

GENIFY

LECTURE 17

03-September-2025

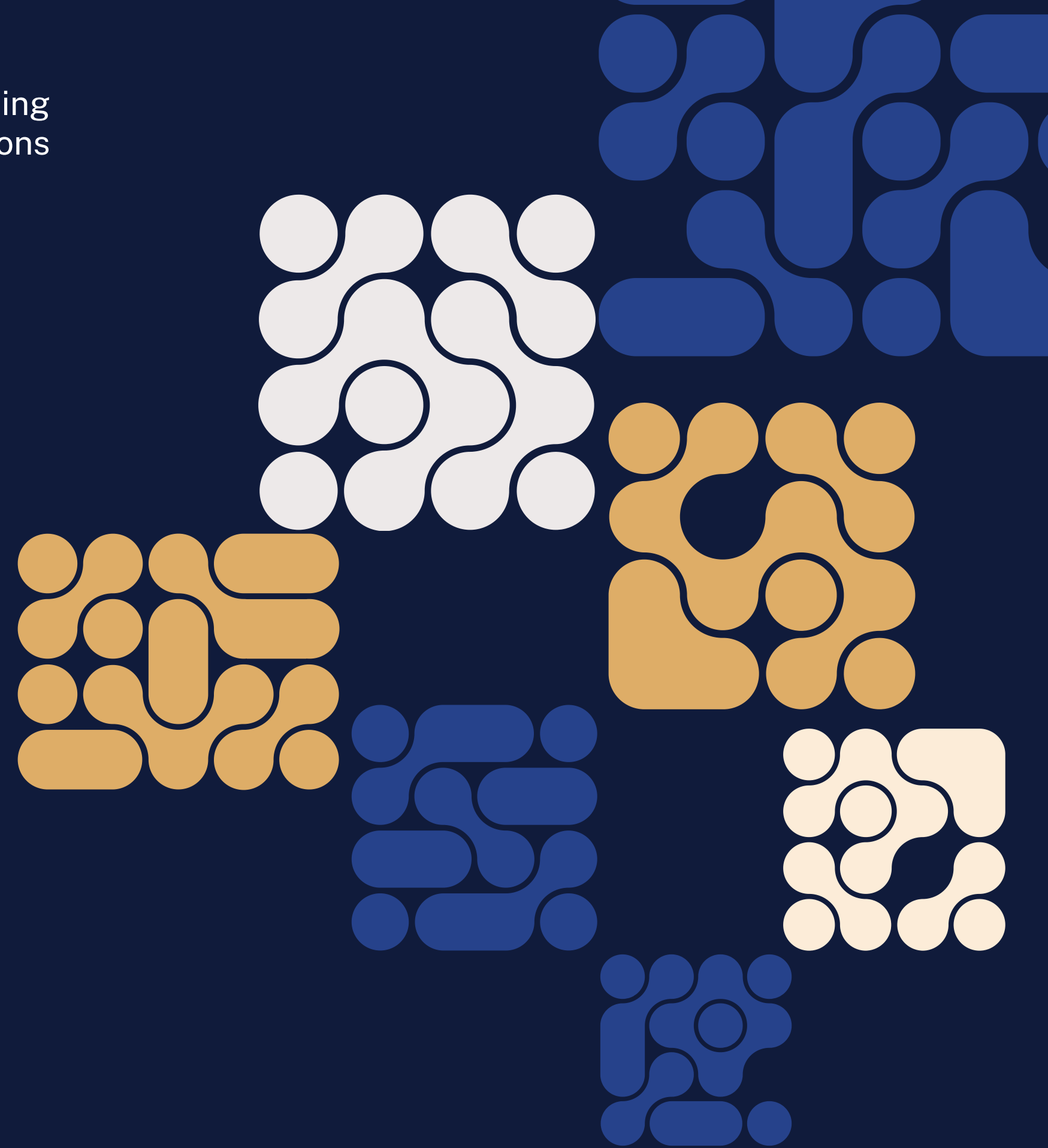
TOPICS COVERED:

RIS

- TRANSFORMERS
 - PREV RECAP
 - DECODER

Presenter
Shahzaib Kashif

Co-Presenter
Shayan Hassan Baig



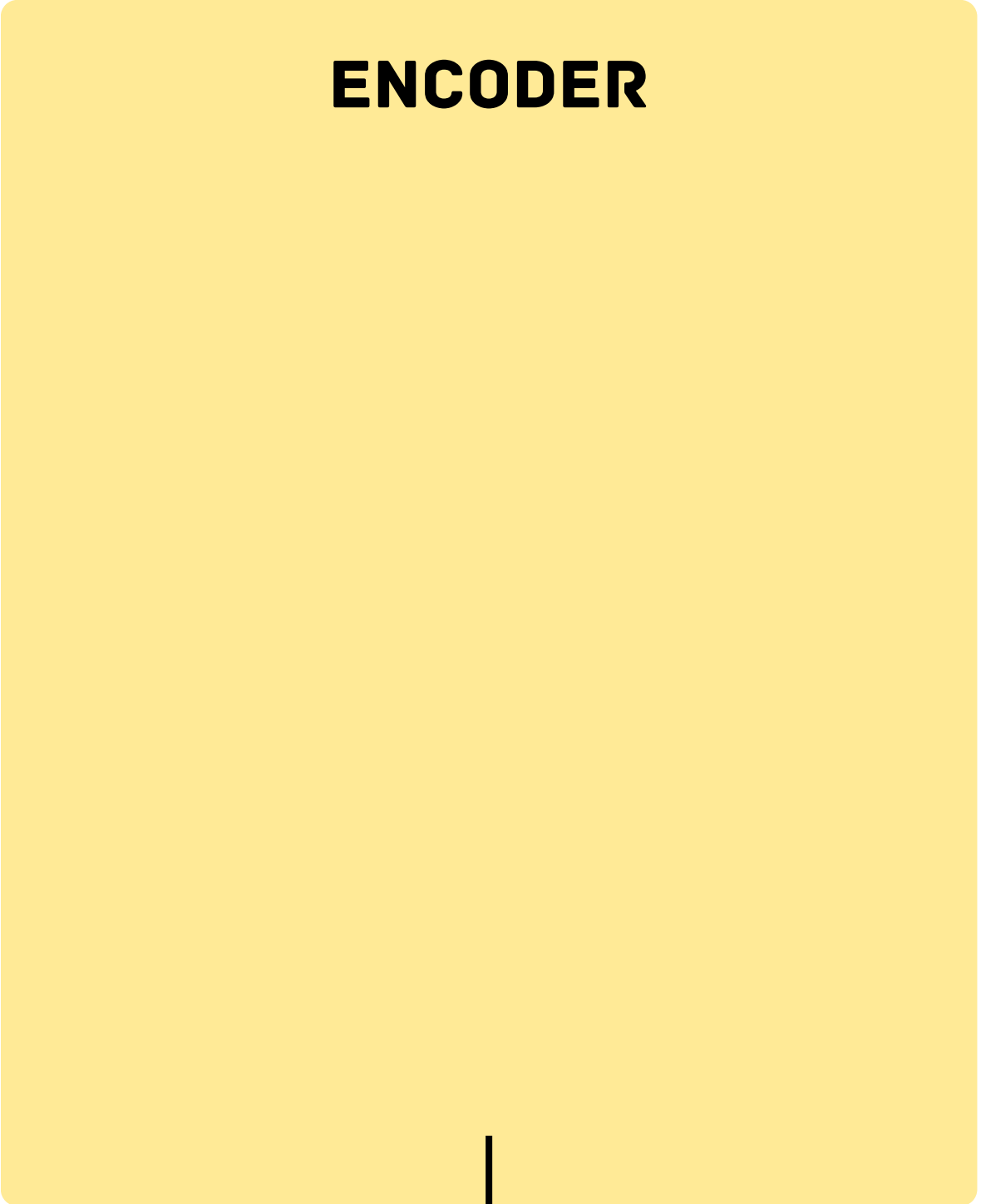


RECAP

ENCODER

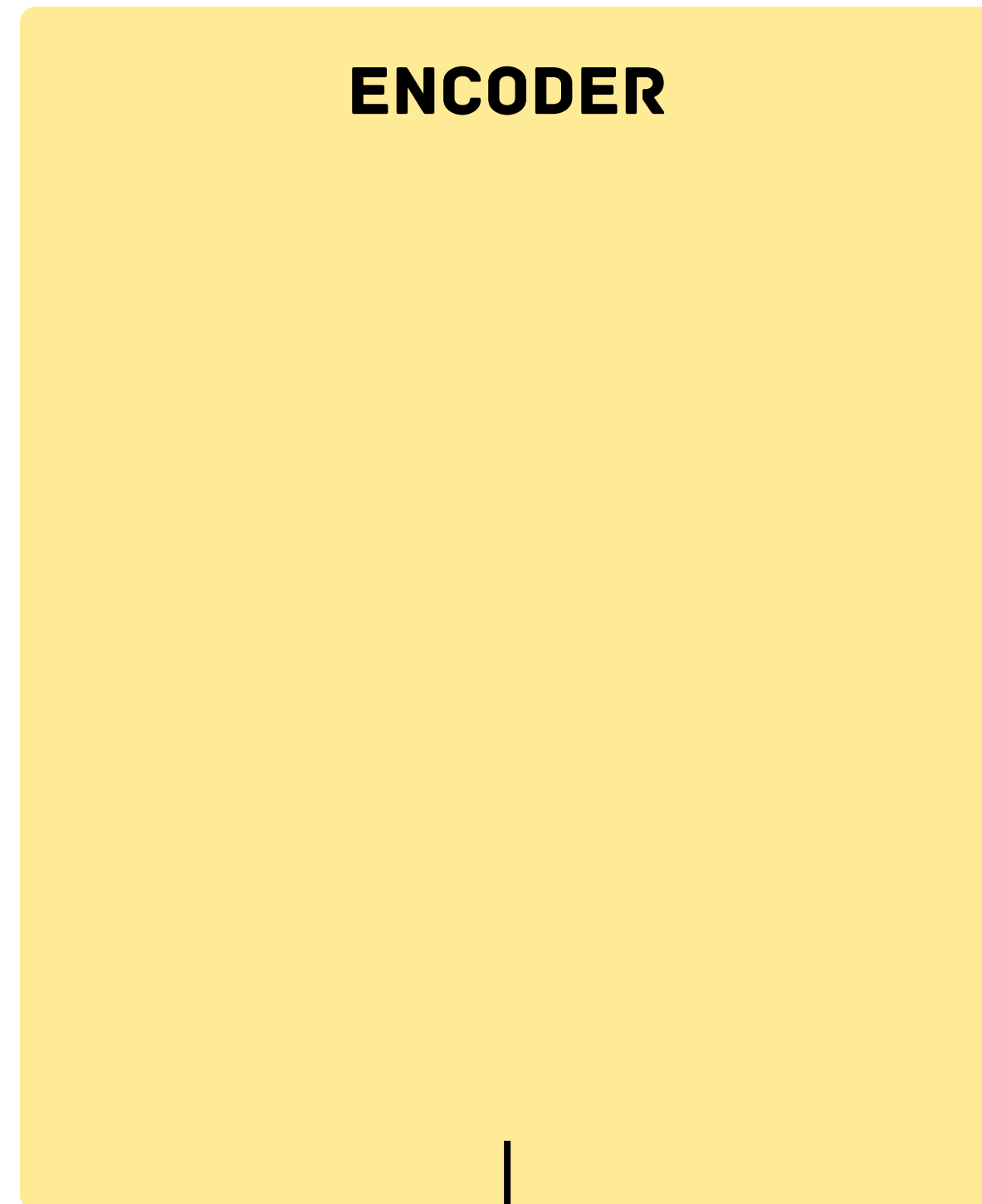
ENCODER

Input



Word Embeddings

King	0.12	0.43	-1.1	0.3	0.4	0.2
Queen	0.22	0.36	-0.9	0.29	0.42	-1.3
Man	-1.5	-0.8	1.0	0.8	0.62	0.23
Woman	-1.3	-0.65	0.98	0.76	0.60	-1.2

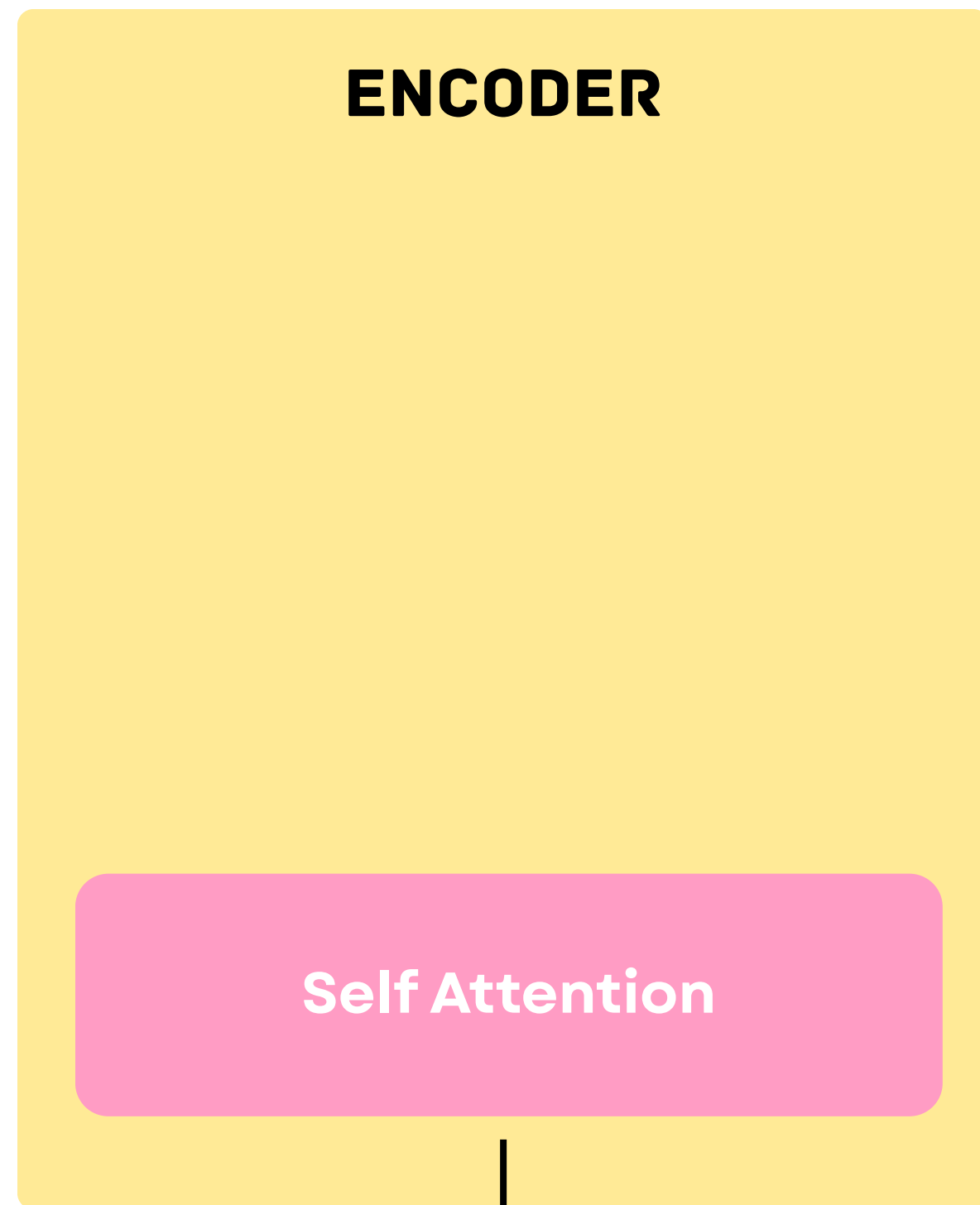


Input

Word Embeddings

King	0.12	0.43	-1.1	0.3	0.4	0.2
Queen	0.22	0.36	-0.9	0.29	0.42	-1.3
Man	-1.5	-0.8	1.0	0.8	0.62	0.23
Woman	-1.3	-0.65	0.98	0.76	0.60	-1.2

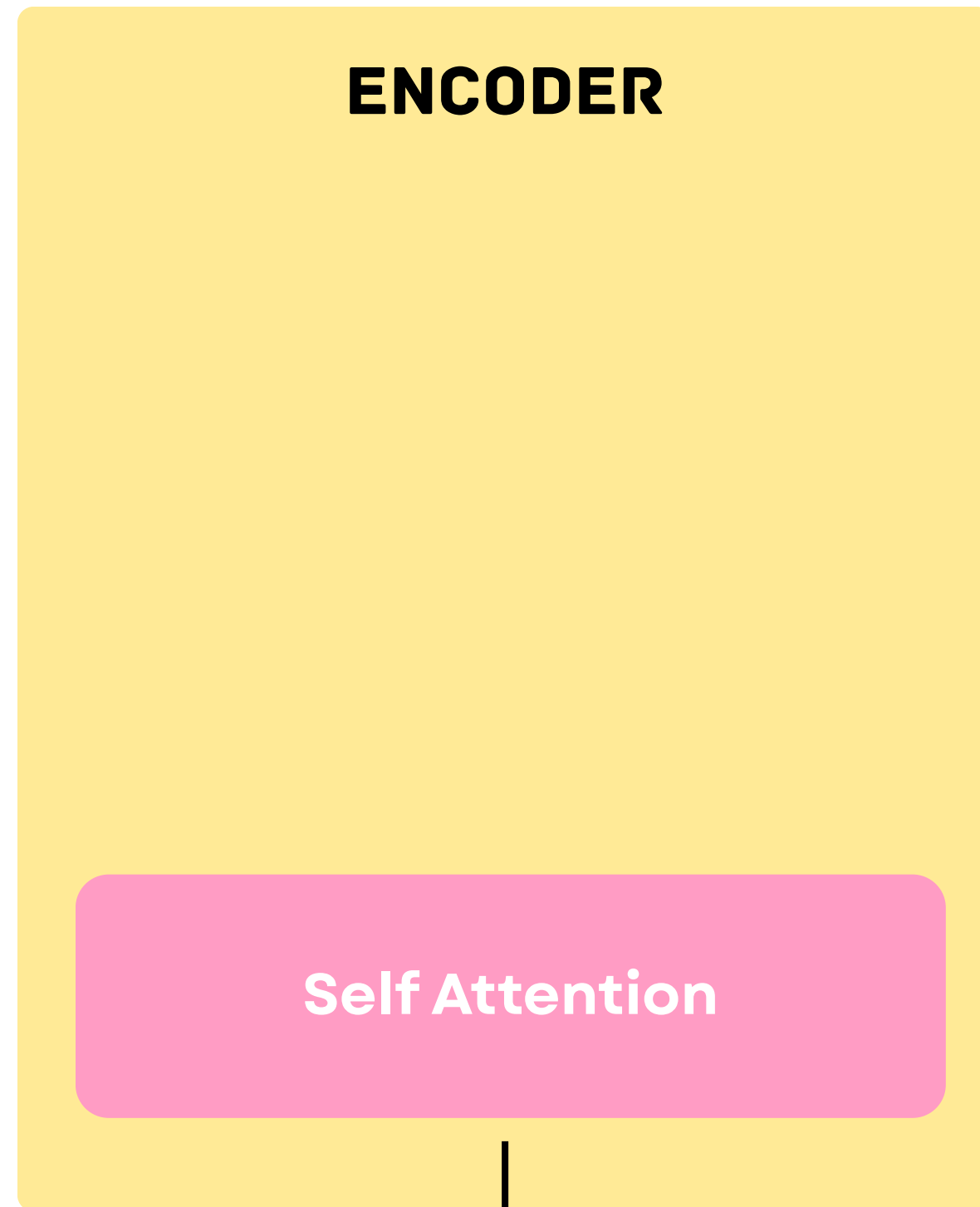
Q: Is there a problem here?



ENCODER

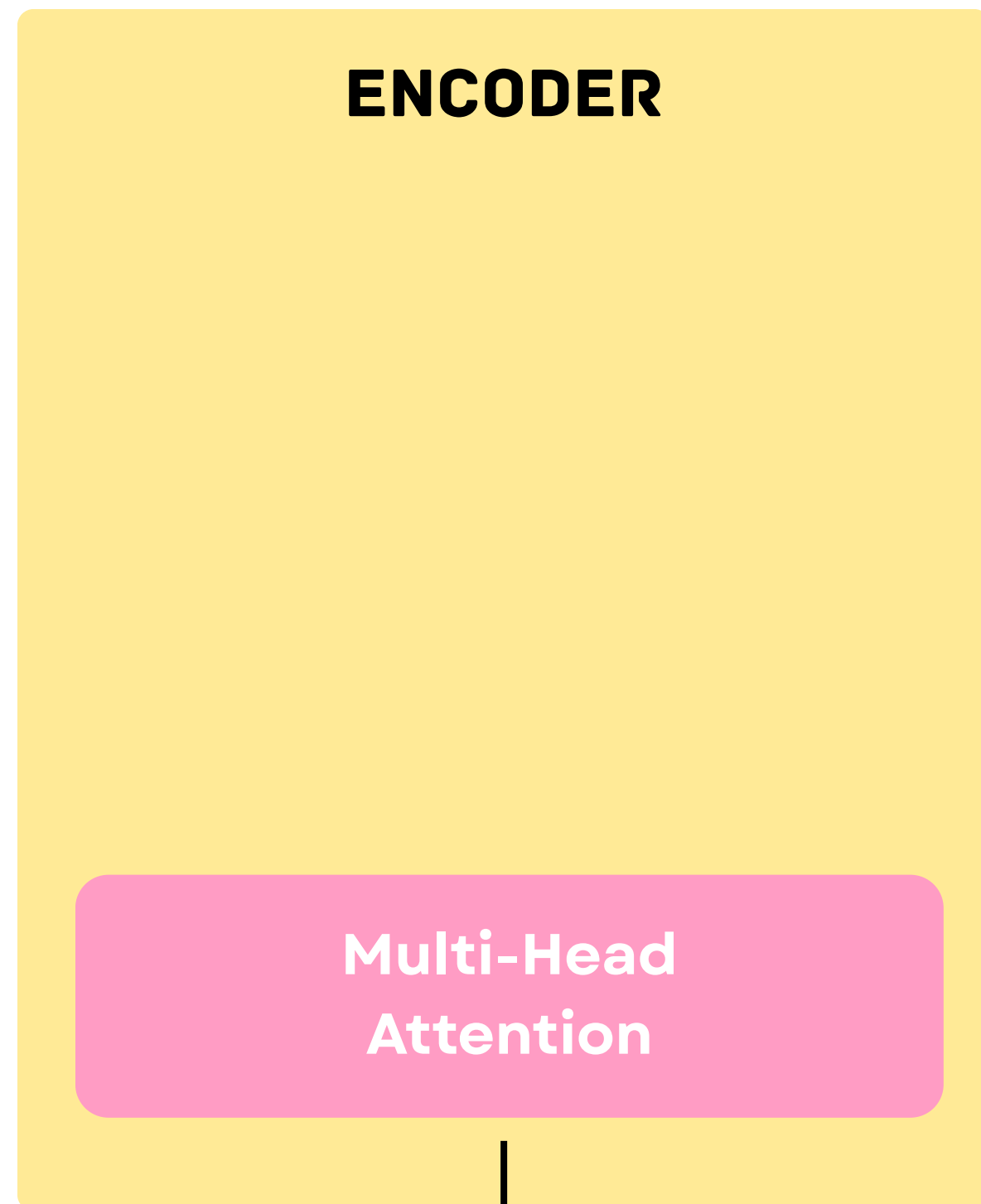
Self Attention

Input



Input

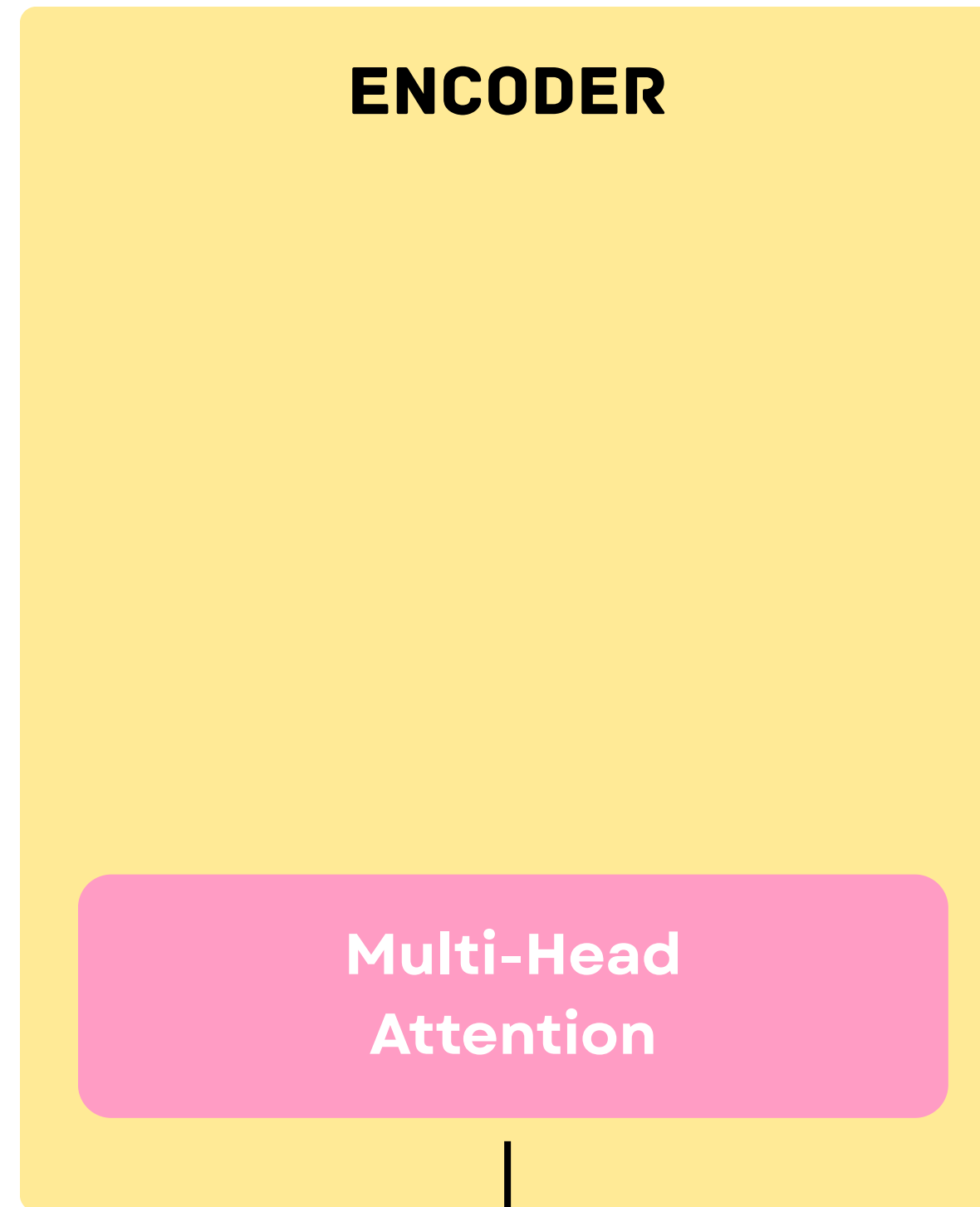
Q: Is there a problem here?



ENCODER

**Multi-Head
Attention**

Input

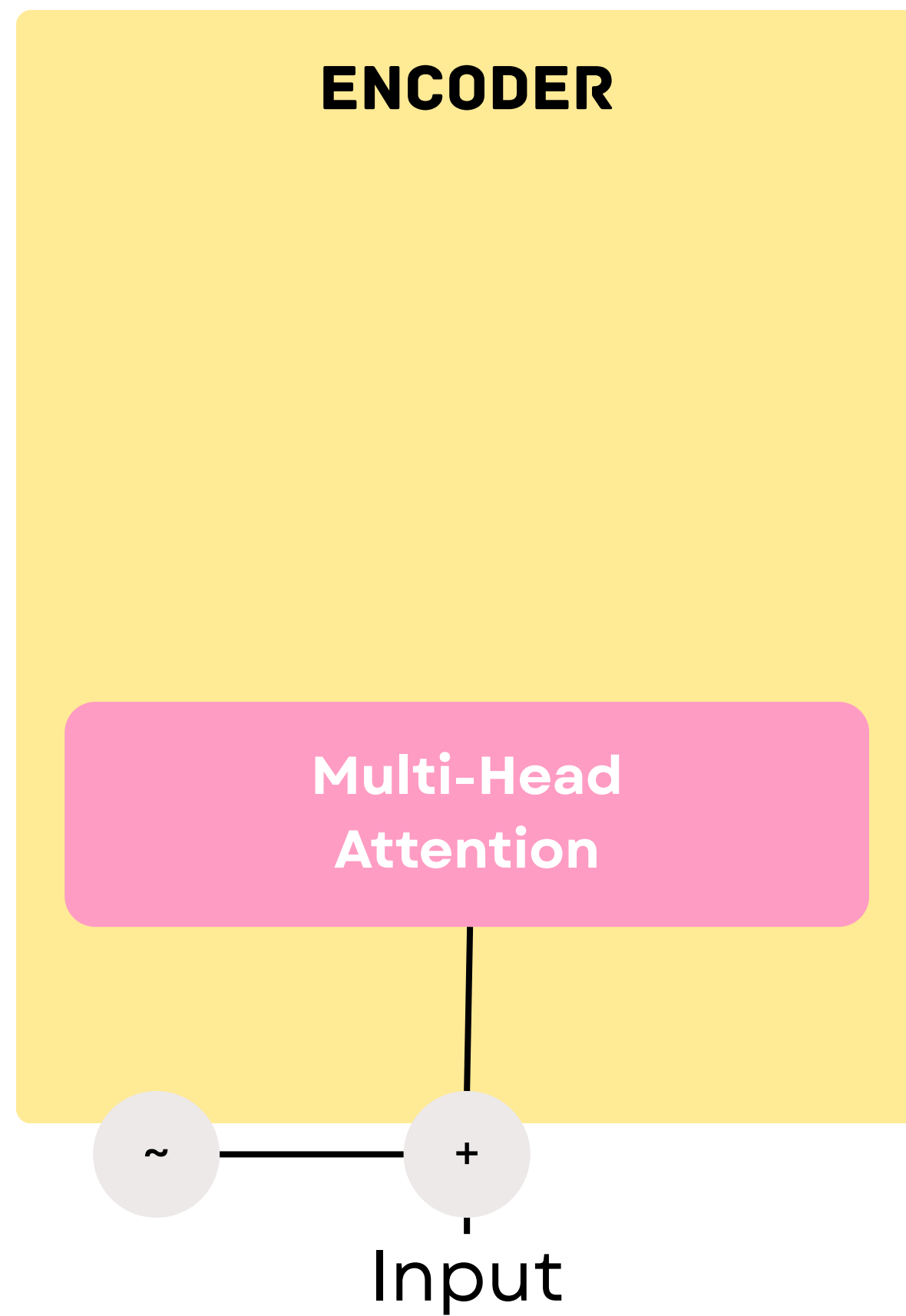


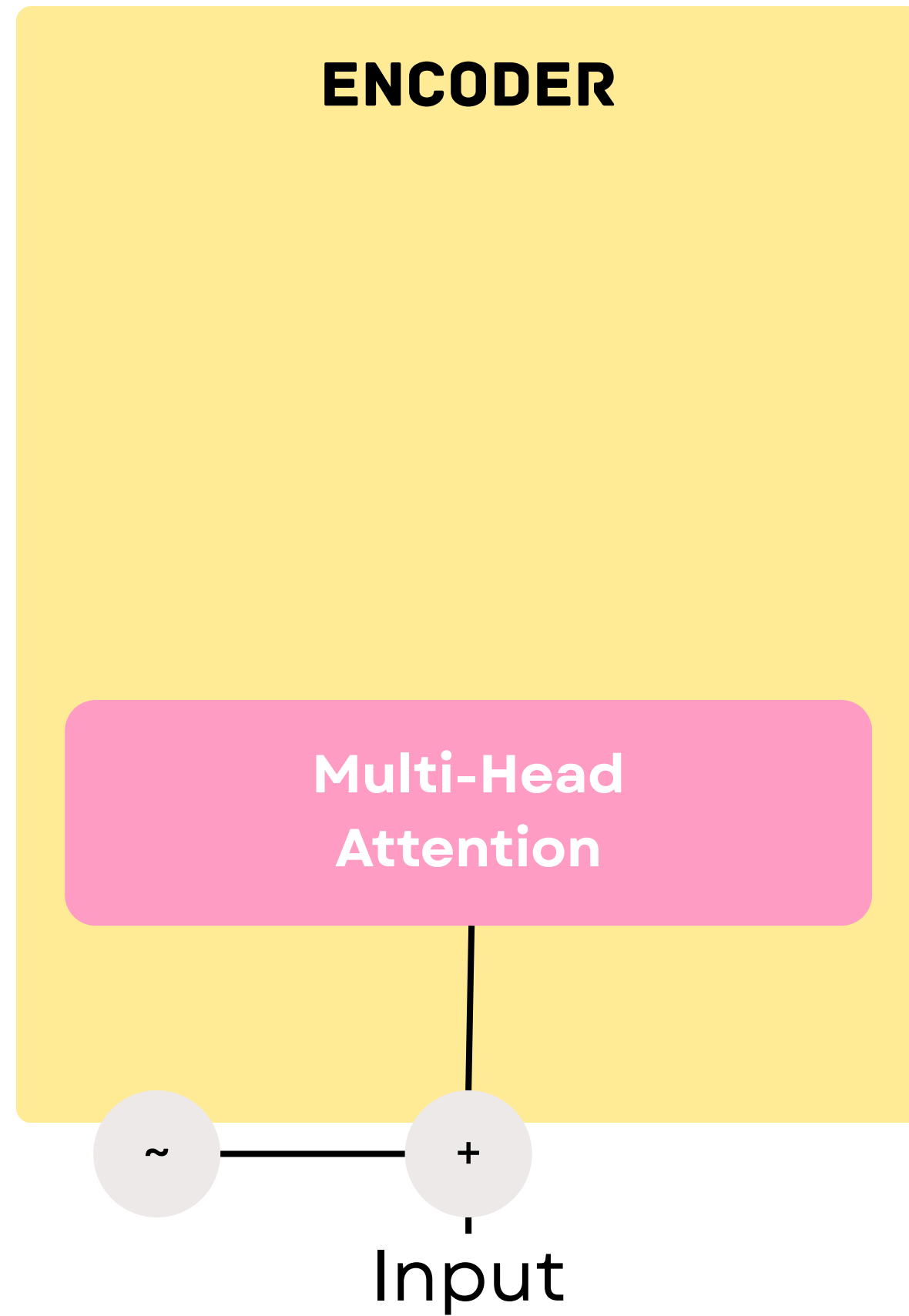
ENCODER

**Multi-Head
Attention**

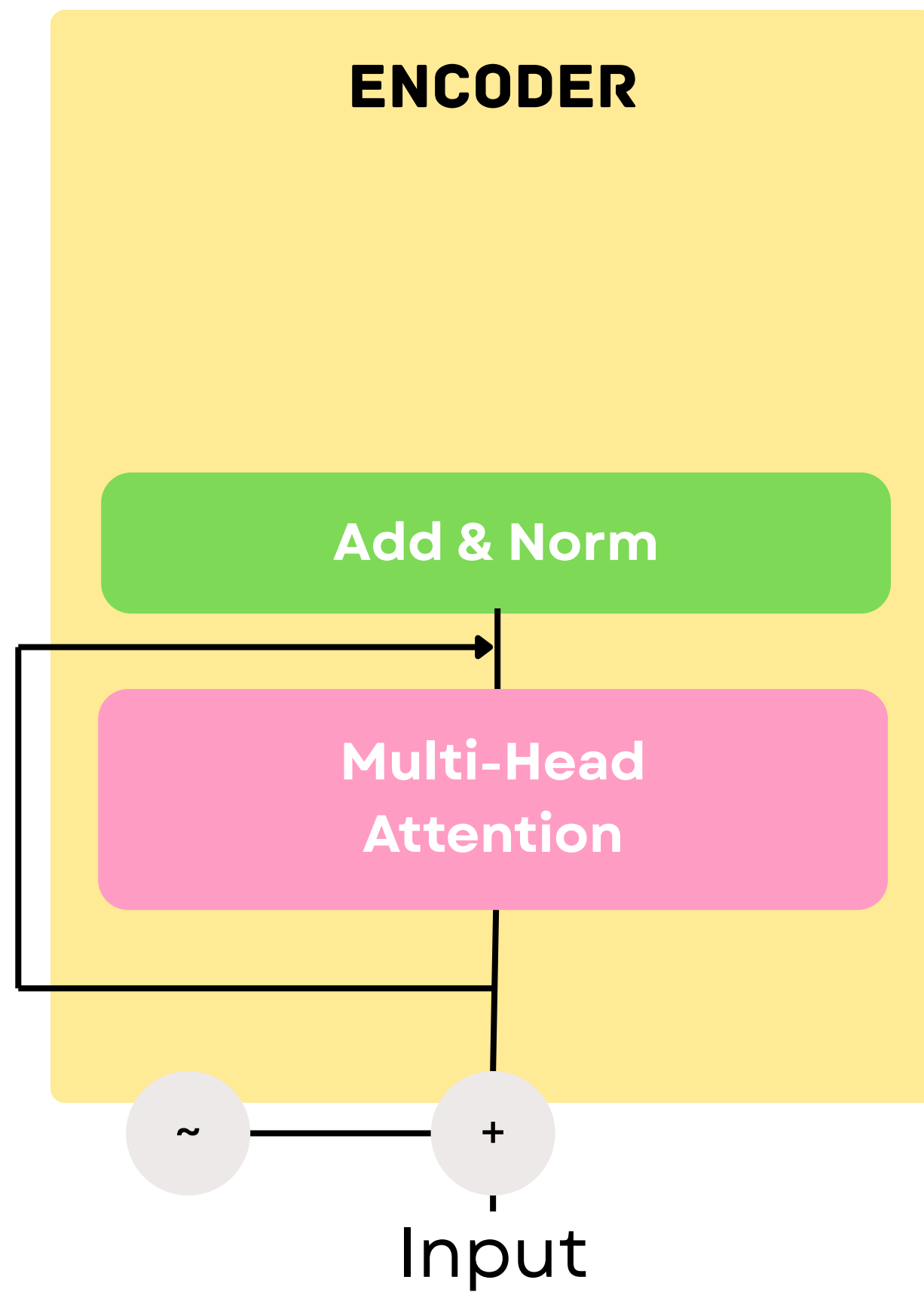
Input

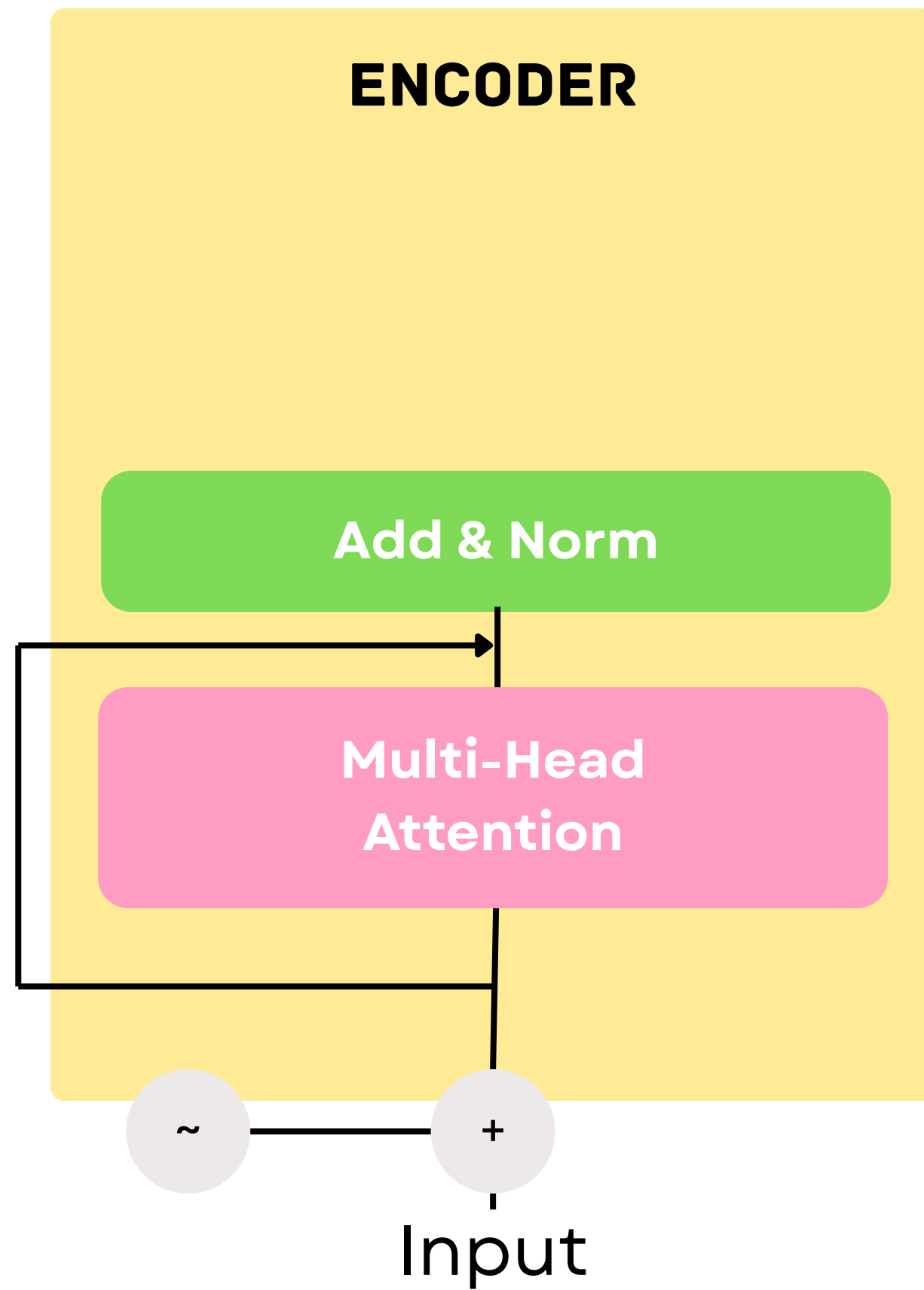
Q: Is there a problem here?



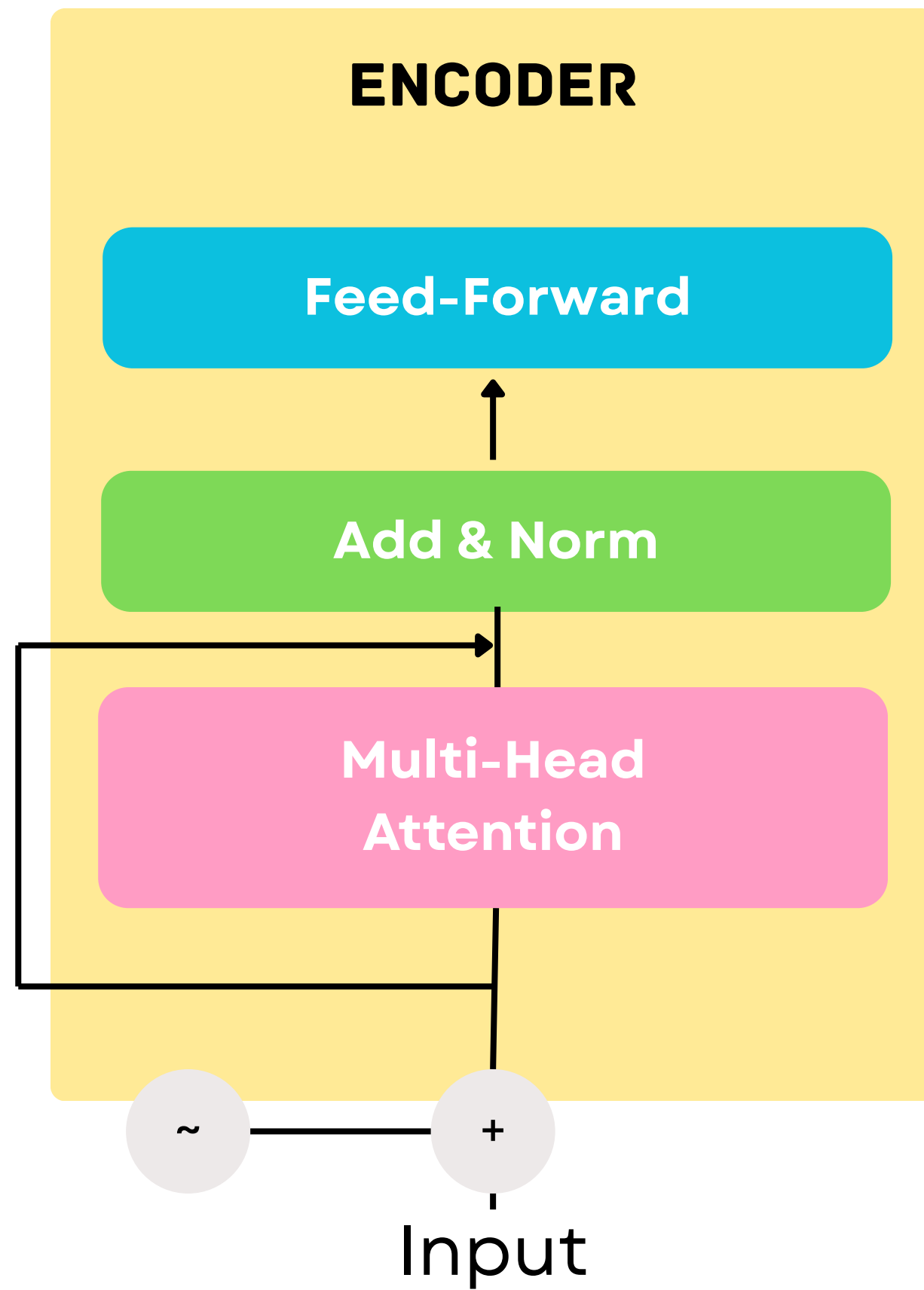


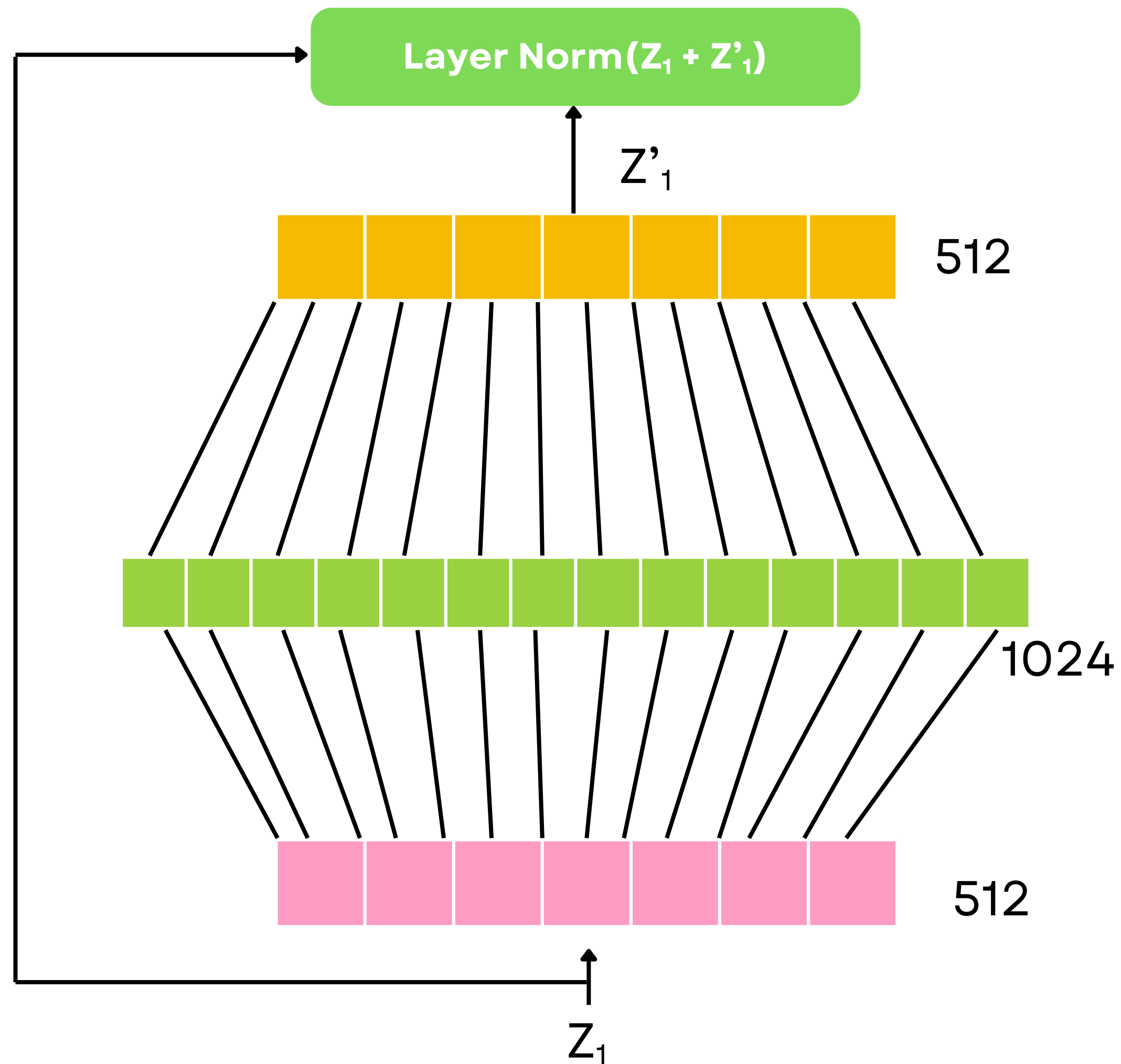
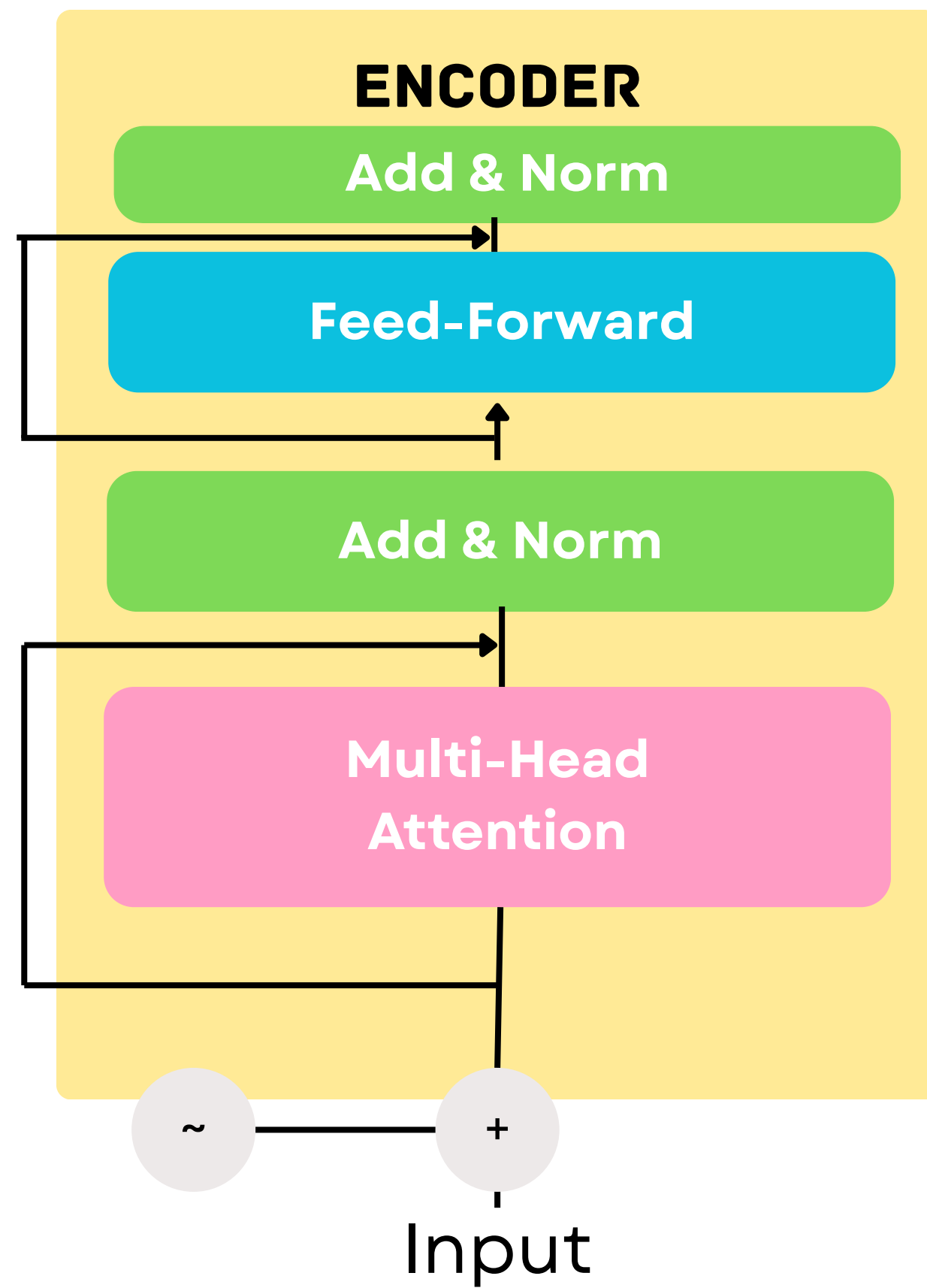
Q: Is there a problem here?



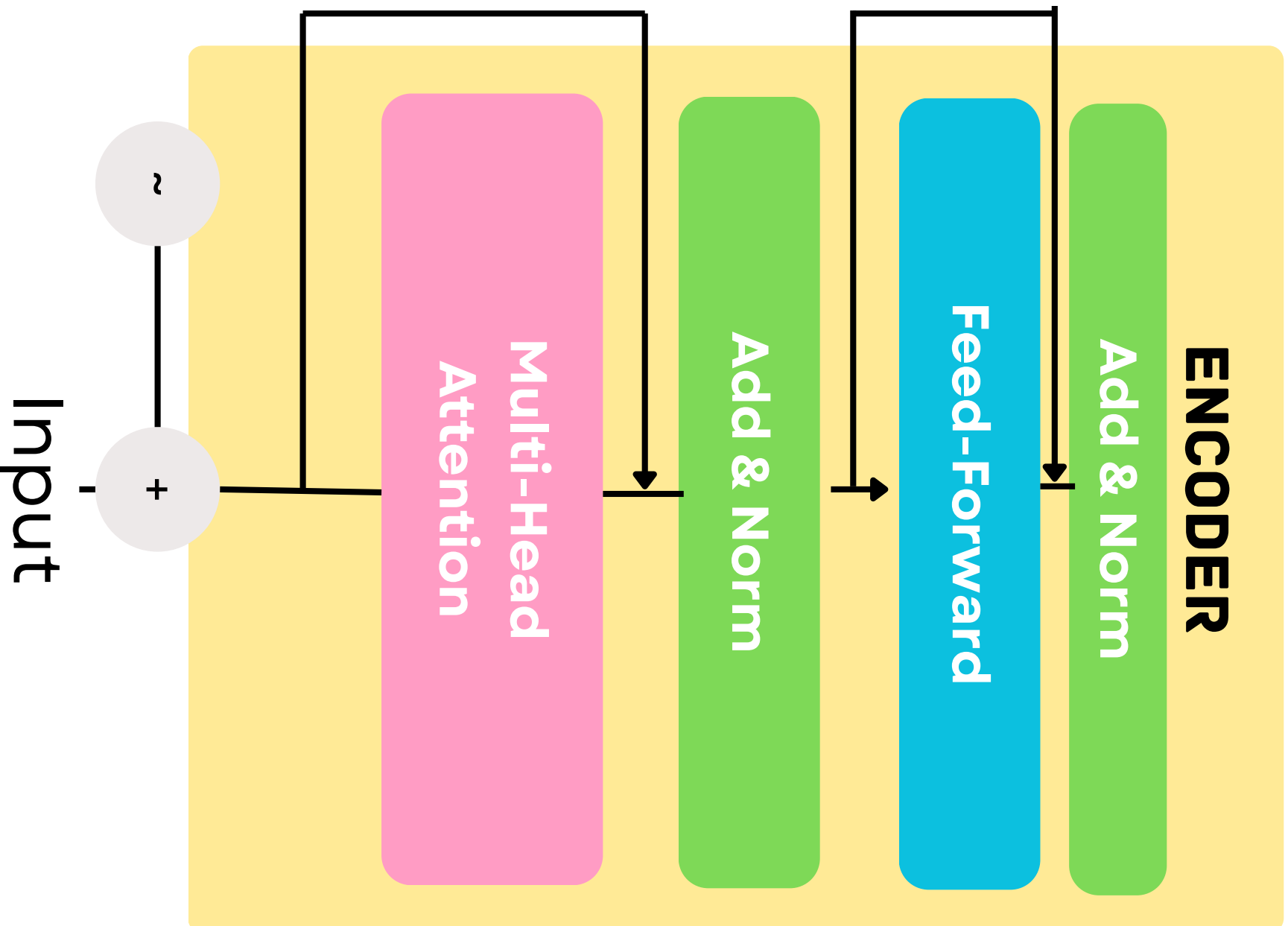
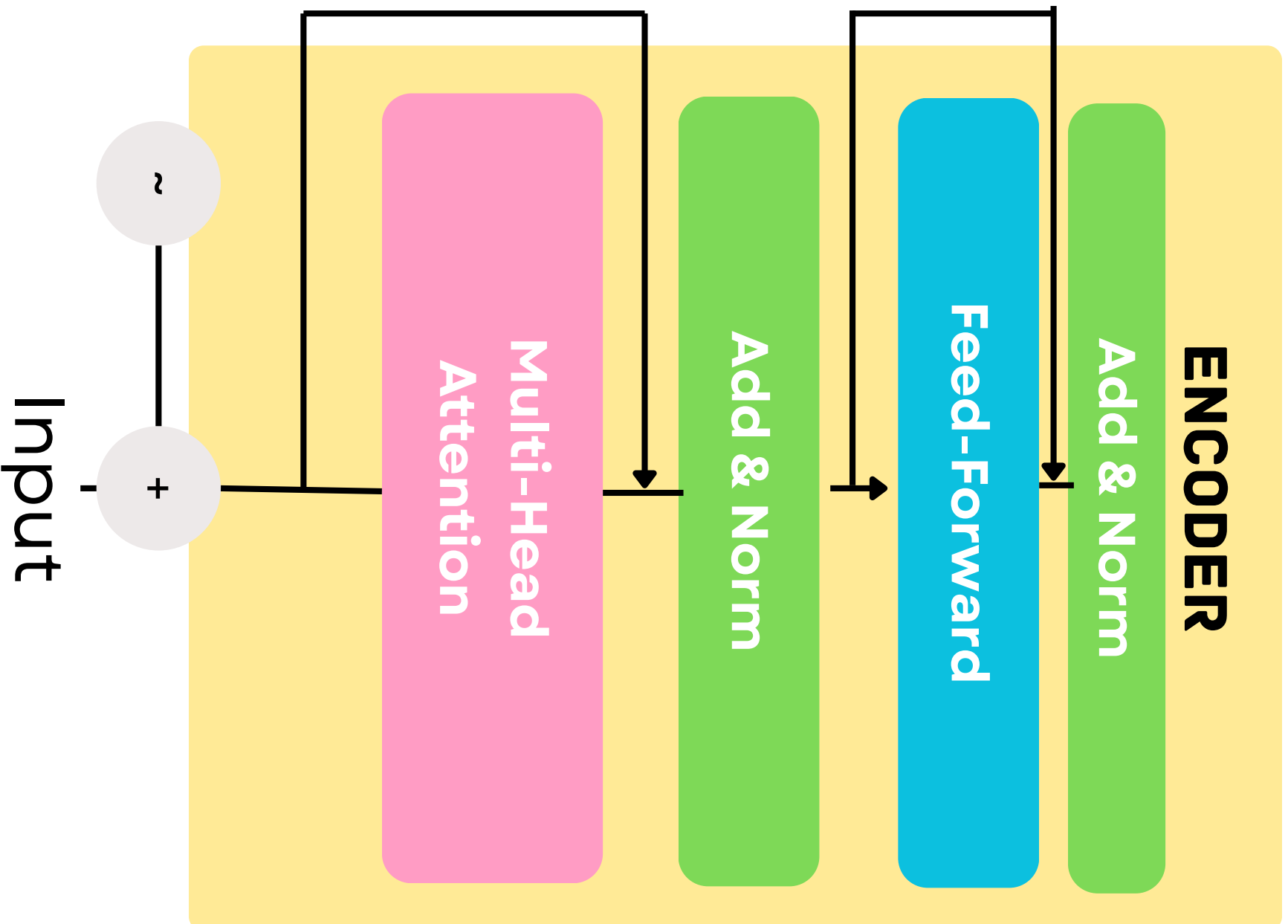


Q: Is there a problem here?





...

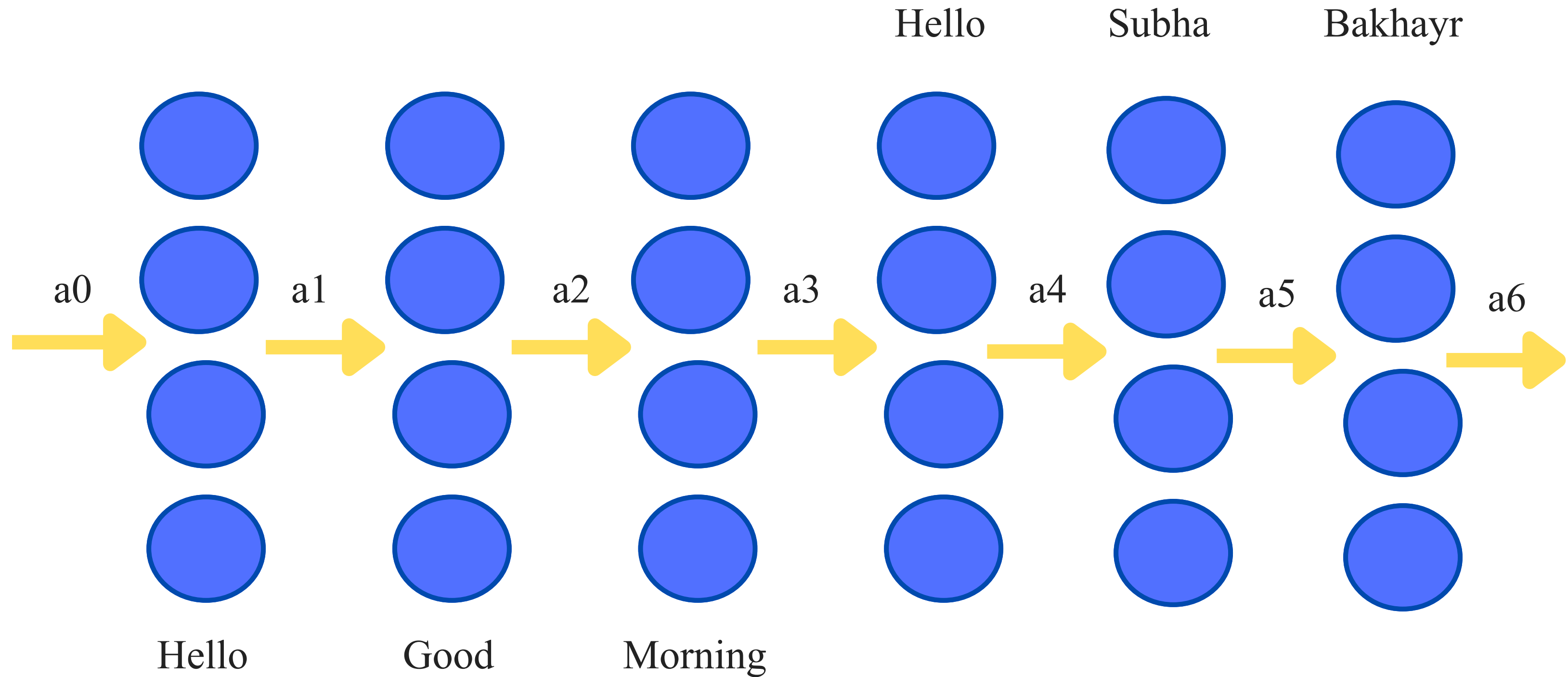




DECODER

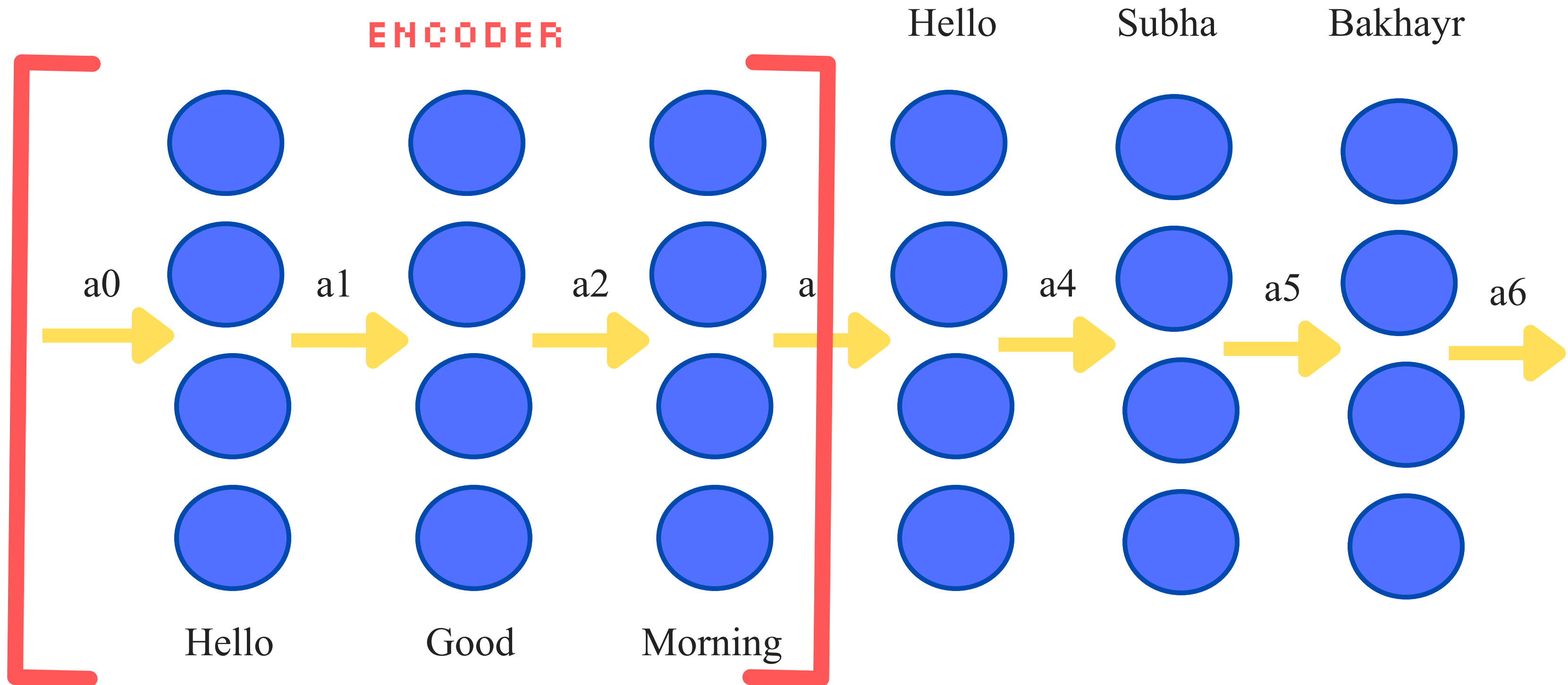
RNN

Language Translation



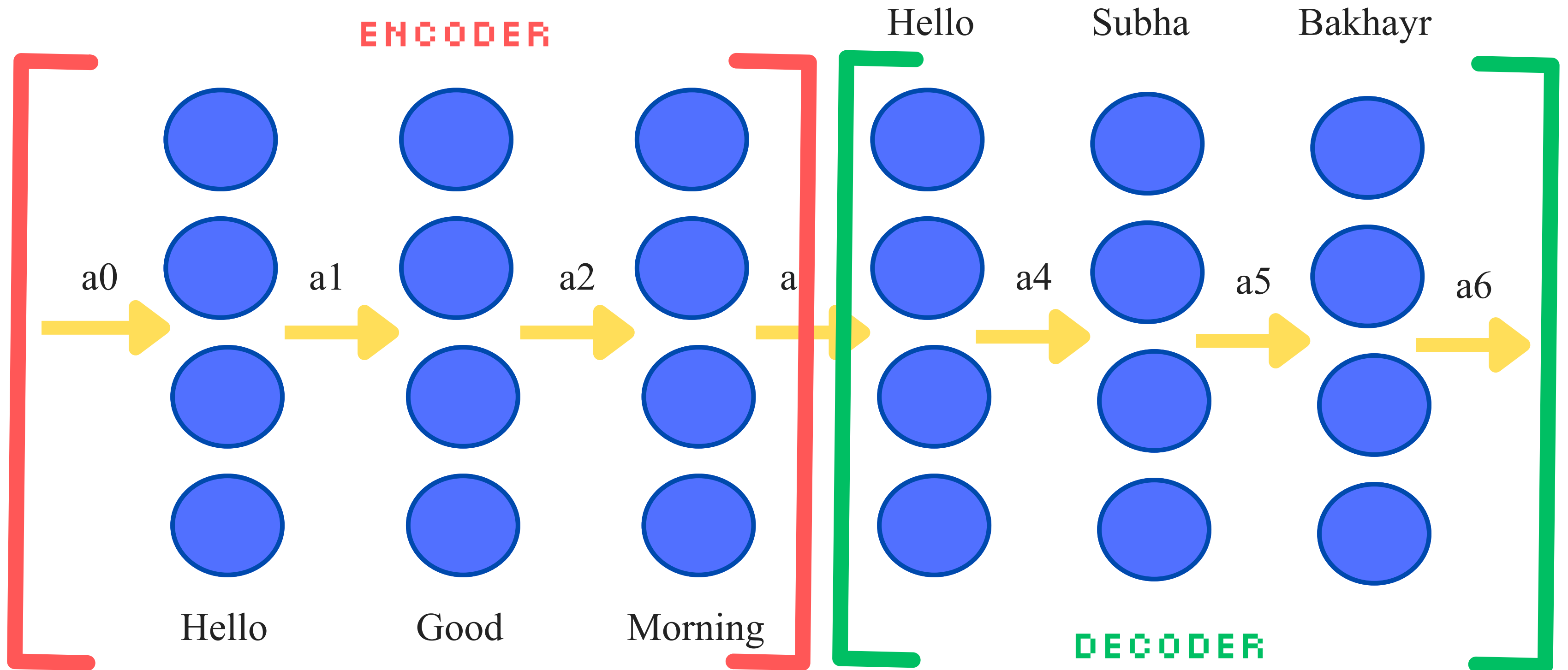
RNN

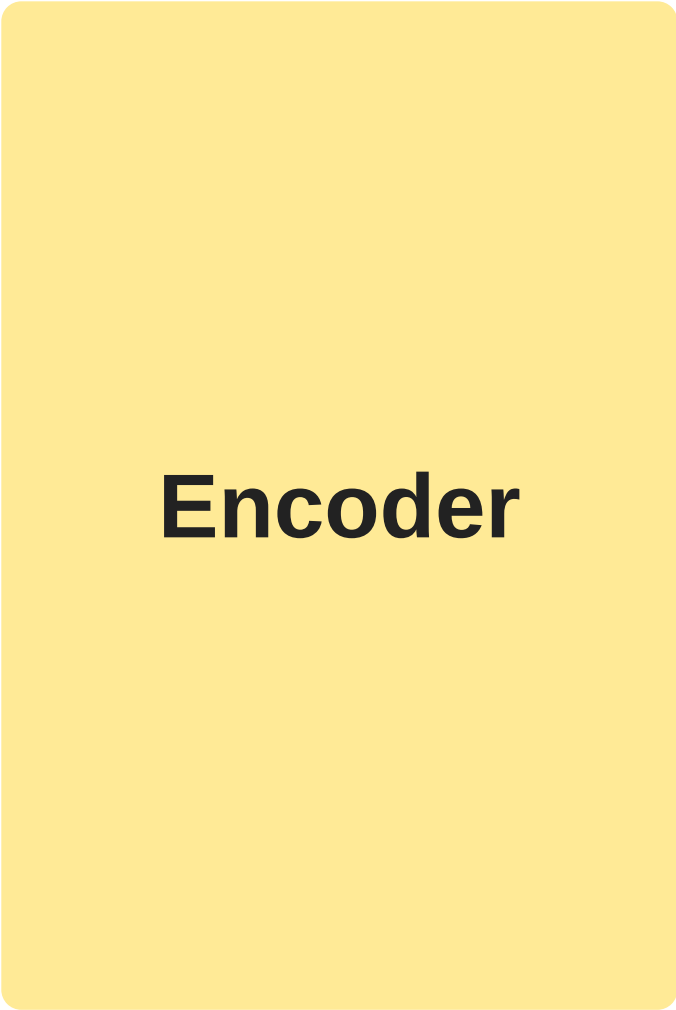
Language Translation



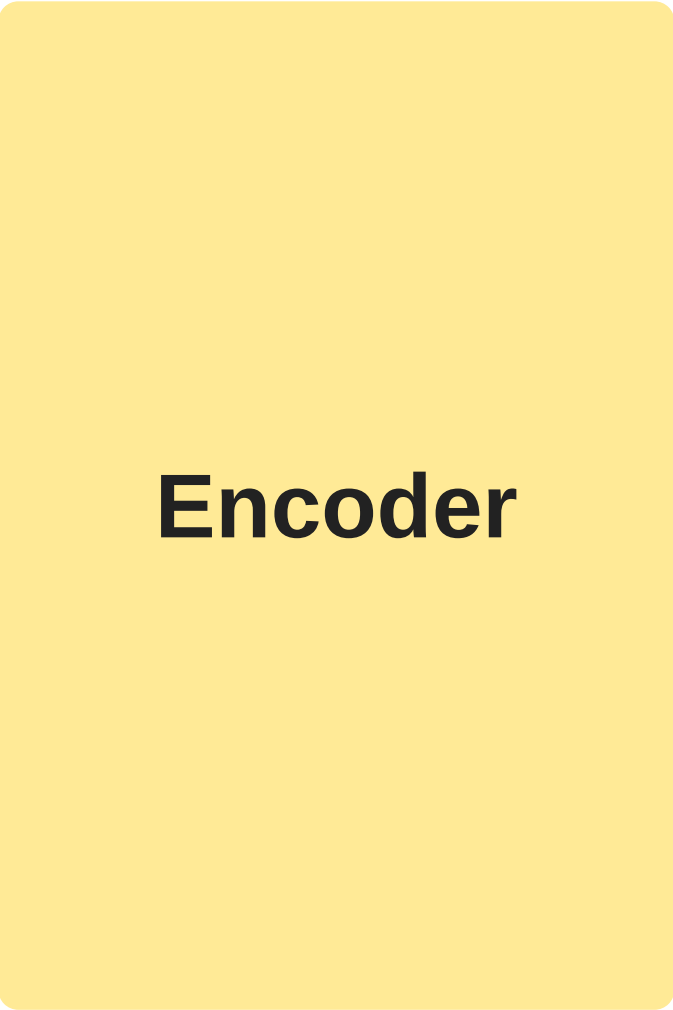
RNN

Language Translation





Encoder



Encoder

A diagram showing a yellow rectangular box with rounded corners, labeled "Encoder". Below the box, the text "E_hello E_good E_morning" is displayed.

E_{hello}

E_{good}

E_{morning}

E'_{hello} E'_{good} E'_{morning}

Encoder

E_{hello} E_{good} E_{morning}

E'_{hello} E'_{good} E'_{morning}

Encoder

Decoder

E_{hello} E_{good} E_{morning}

E'_{hello} E'_{good} E'_{morning}

Encoder

Decoder

↑
<start>

E_{hello} E_{good} E_{morning}

E'_{hello} E'_{good} E'_{morning}

Encoder

Hello

Decoder

<start>

E_{hello} E_{good} E_{morning}

E'_{hello} E'_{good} E'_{morning}

Encoder

Hello

Decoder

Decoder

<start>

<start> Hello

E_{hello} E_{good} E_{morning}

E'_{hello} E'_{good} E'_{morning}

Encoder

Hello

Subha

Decoder

Decoder

<start>

<start> Hello

E_{hello} E_{good} E_{morning}

E'_{hello} E'_{good} E'_{morning}

Encoder

Hello

Decoder

<start>

Subha

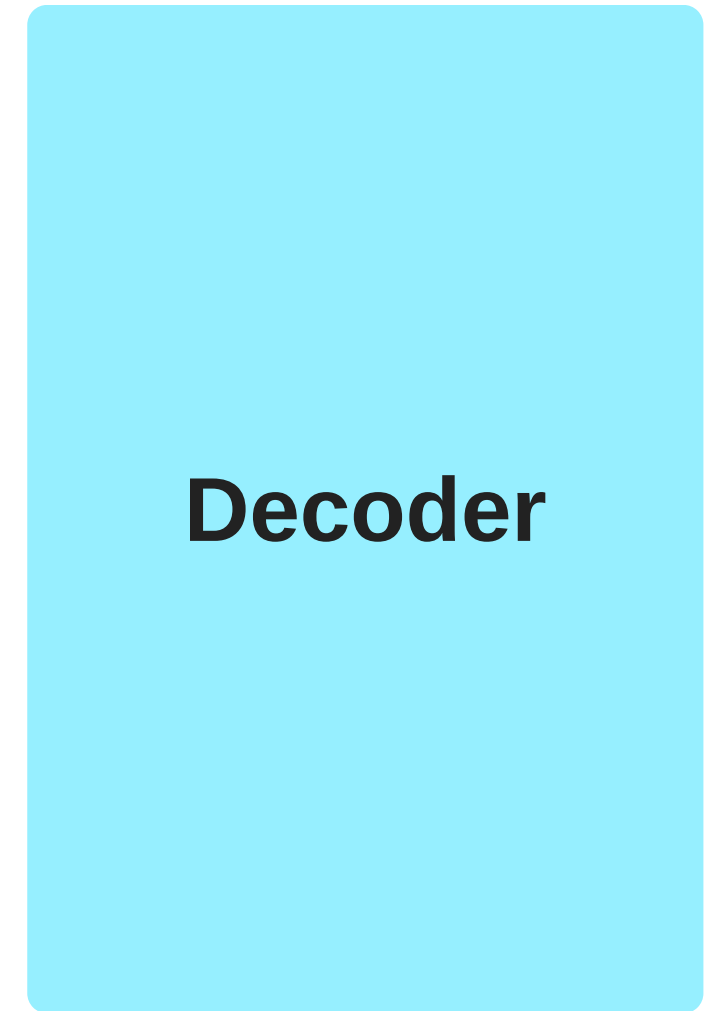
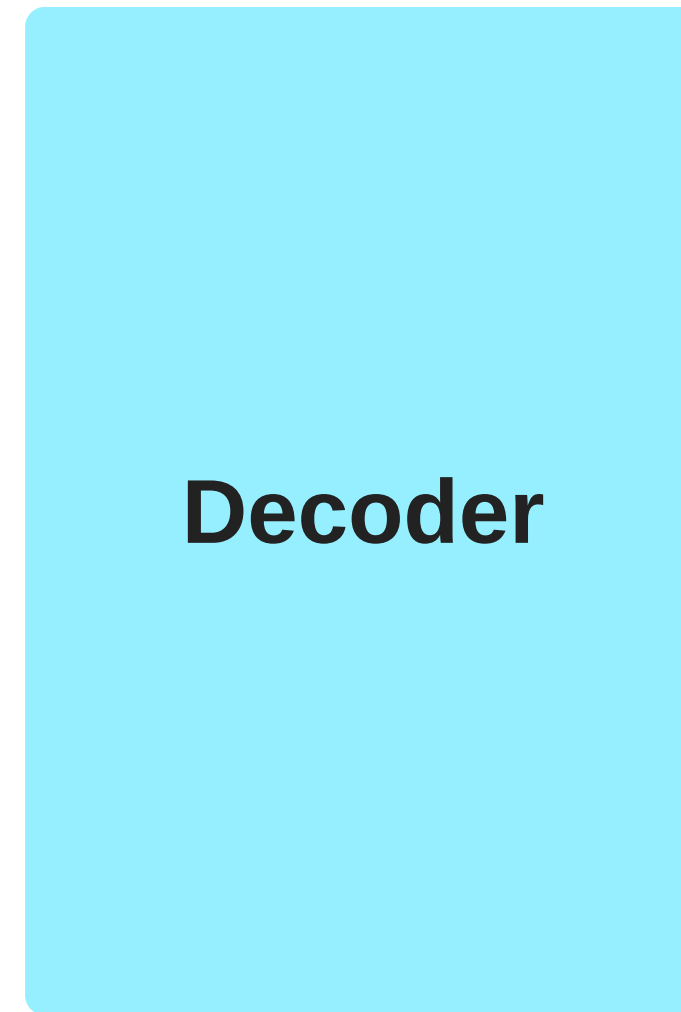
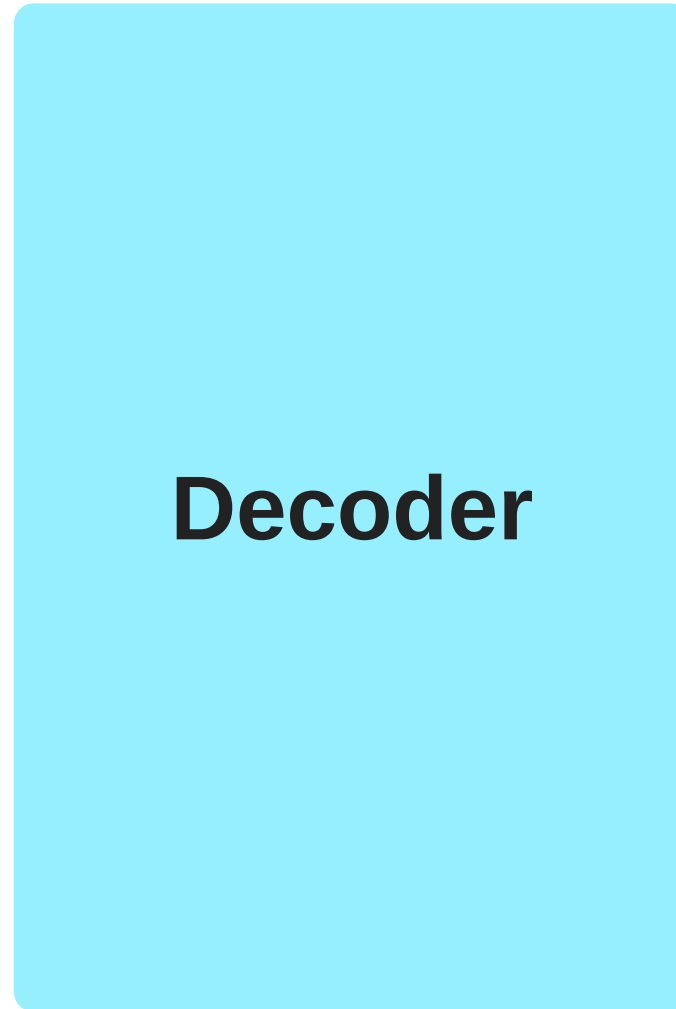
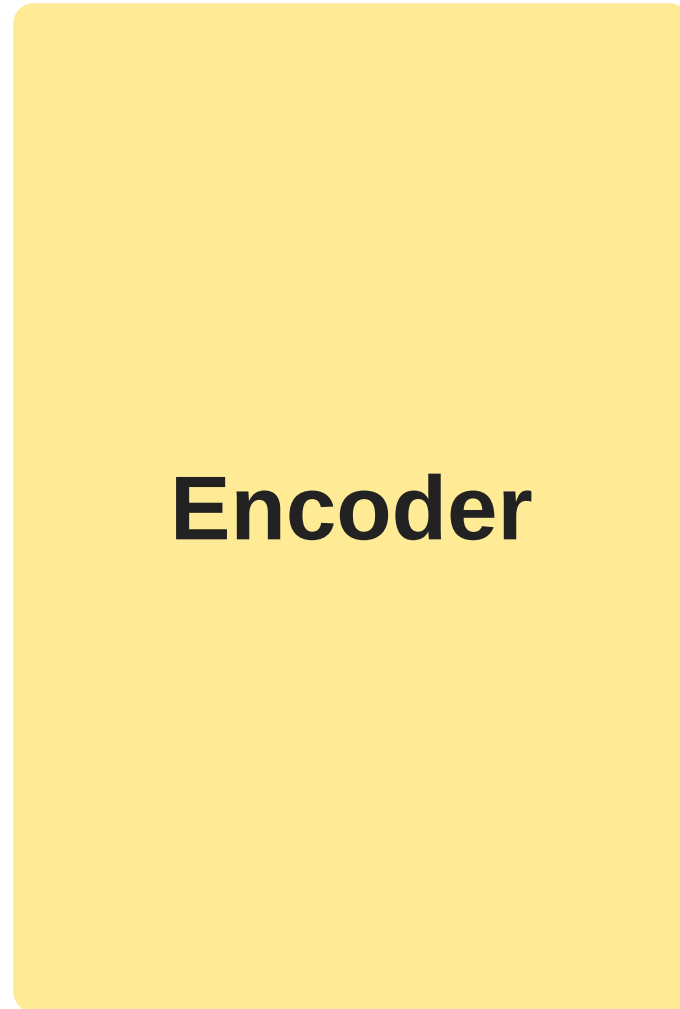
Decoder

<start> Hello

Decoder

<start> Hello Subha

E_{hello} E_{good} E_{morning}



E'_{hello} E'_{good} E'_{morning}

Encoder

Hello

Subha

Bakhayr

Decoder

Decoder

Decoder

<start>

<start> Hello

<start> Hello Subha

E_{hello} E_{good} E_{morning}

E'_{hello} E'_{good} E'_{morning}

Encoder

Hello

Subha

Bakhayr

<end>

Decoder

Decoder

Decoder

...

<start>

<start> Hello

<start> Hello Subha

E_{hello} E_{good} E_{morning}

Inference Time

E'_{hello} E'_{good} E'_{morning}

Encoder

Hello

Subha

Bakhayr

<end>

Decoder

Decoder

Decoder

...

<start>

<start> Hello

<start> Hello Subha

E_{hello}

E_{good}

E_{morning}

Inference Time

Sequential

Q: What about Training? Is it also Sequential? Does it not make process slow?

E'_{hello} E'_{good} E'_{morning}

Encoder

E_{hello} E_{good} E_{morning}

E'_{hello} E'_{good} E'_{morning}

Encoder

Decoder

↑
<start>

E_{hello} E_{good} E_{morning}

E'_{hello} E'_{good} E'_{morning}

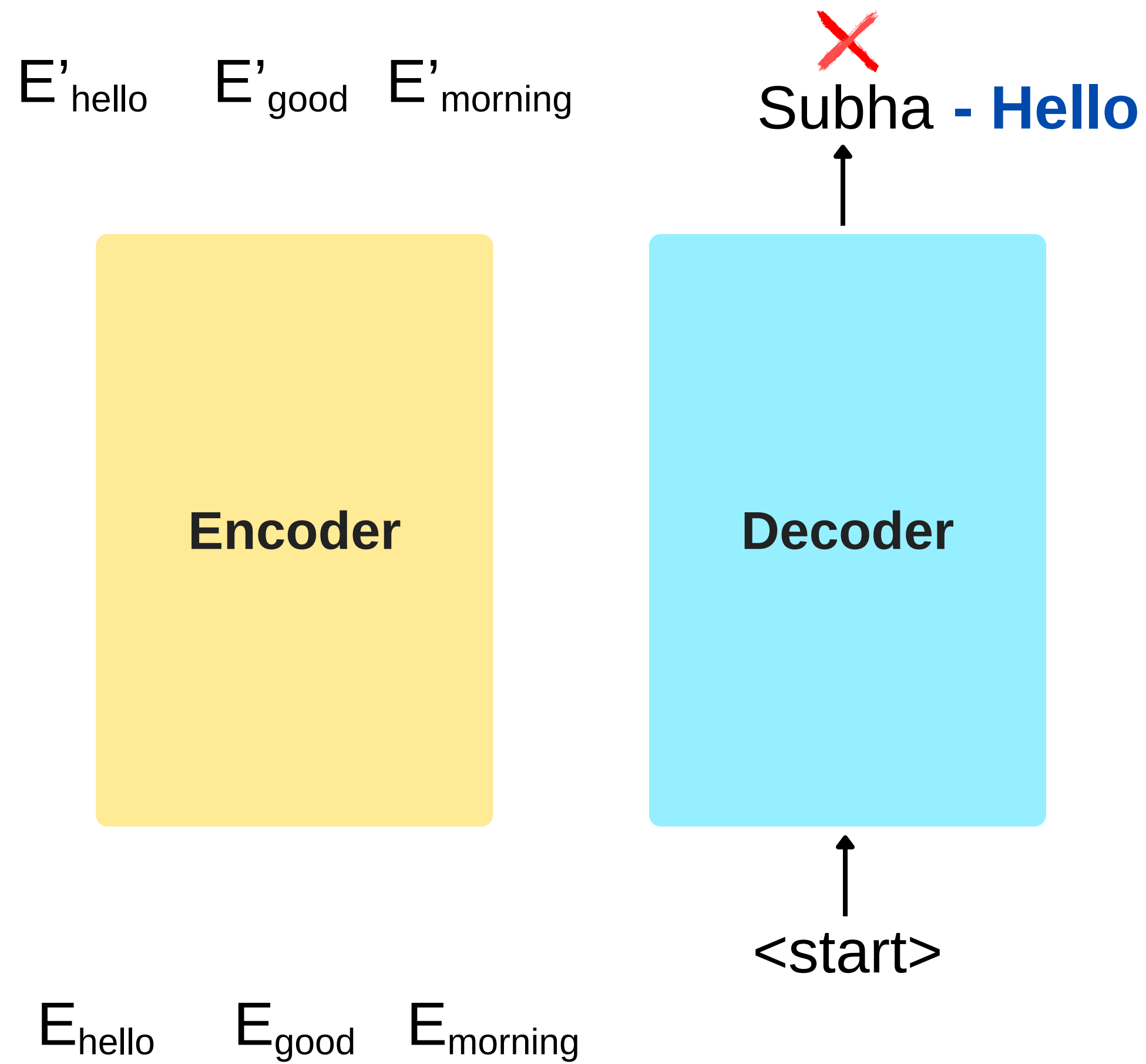
Subha

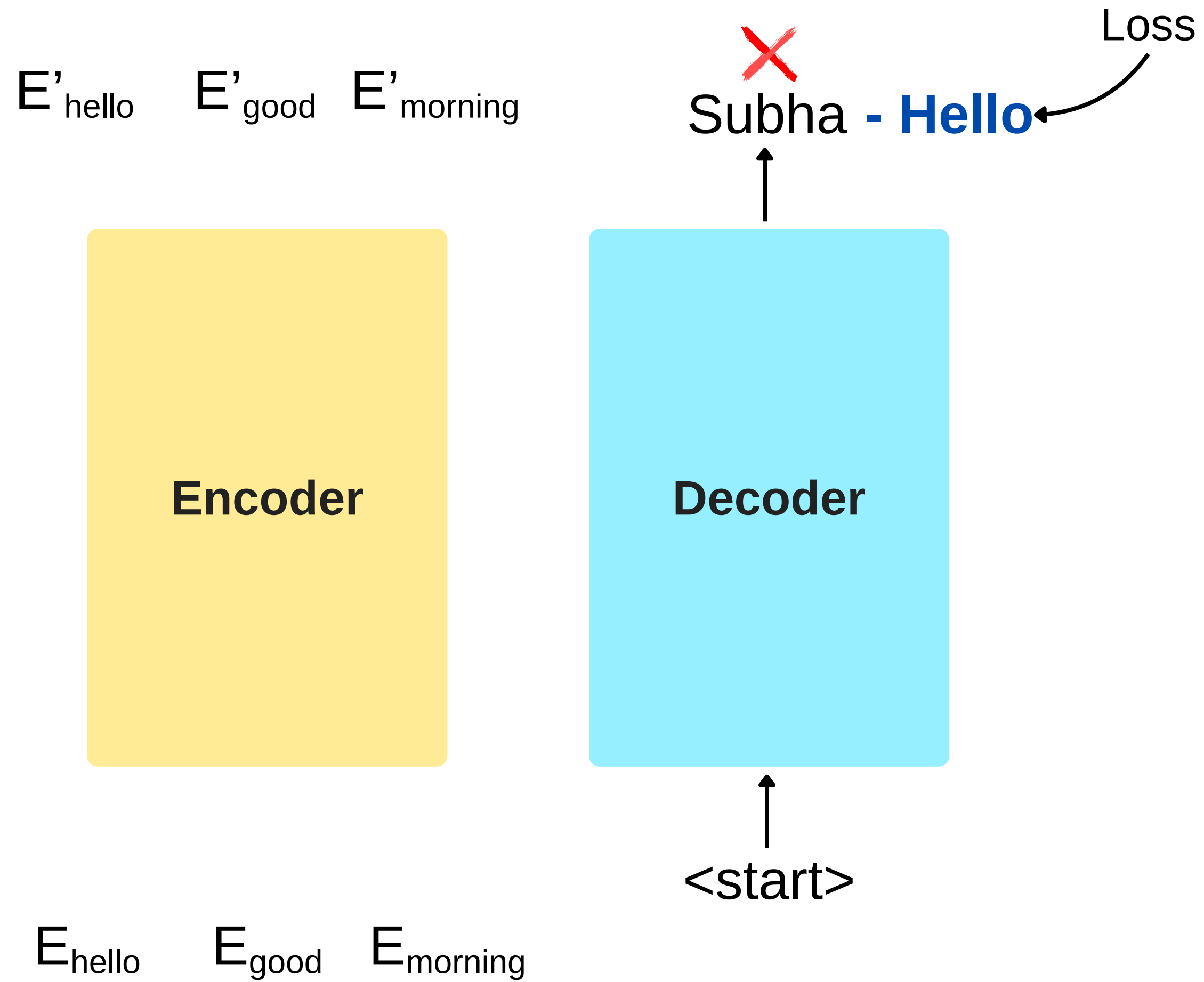
Encoder

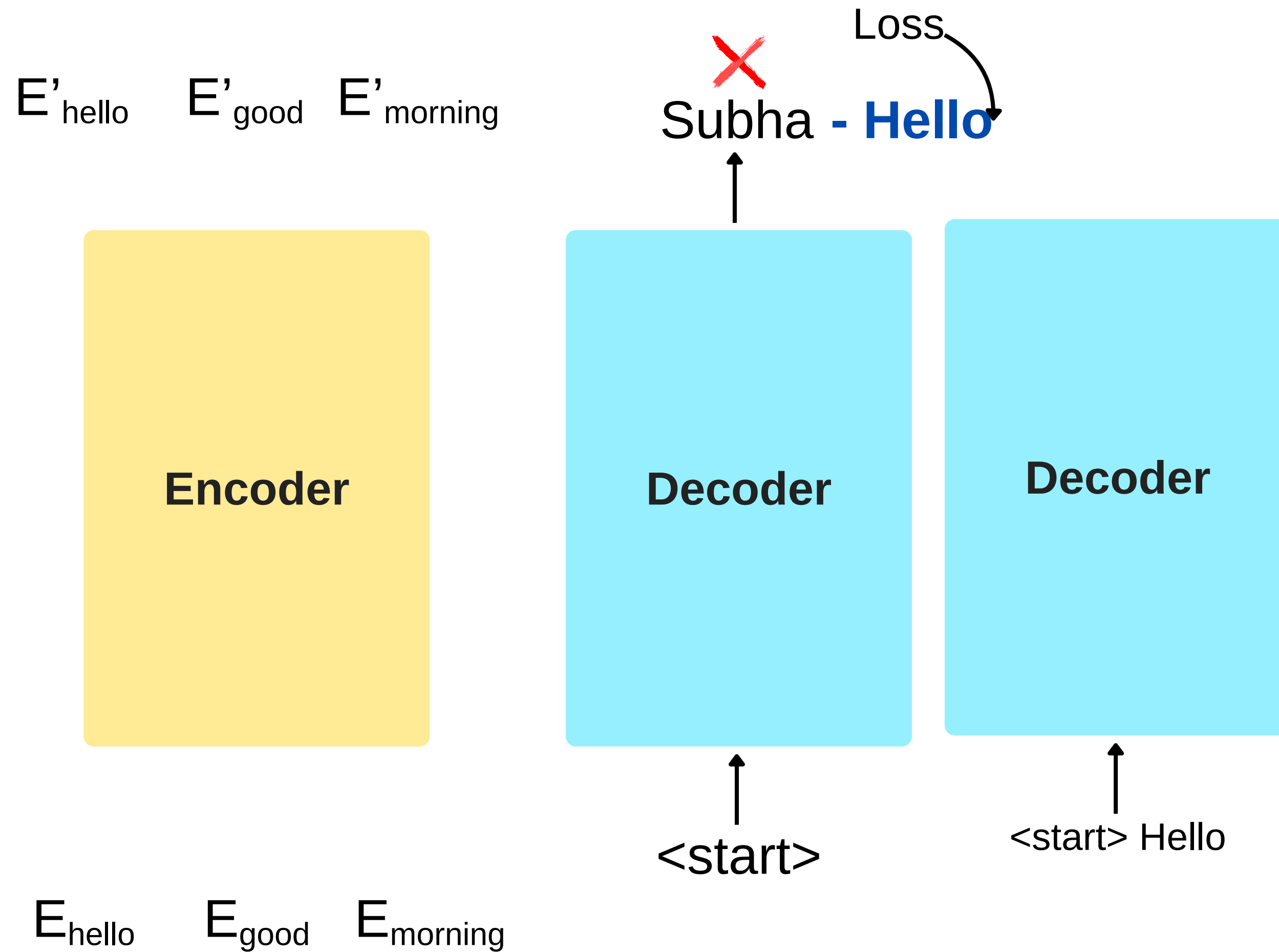
Decoder

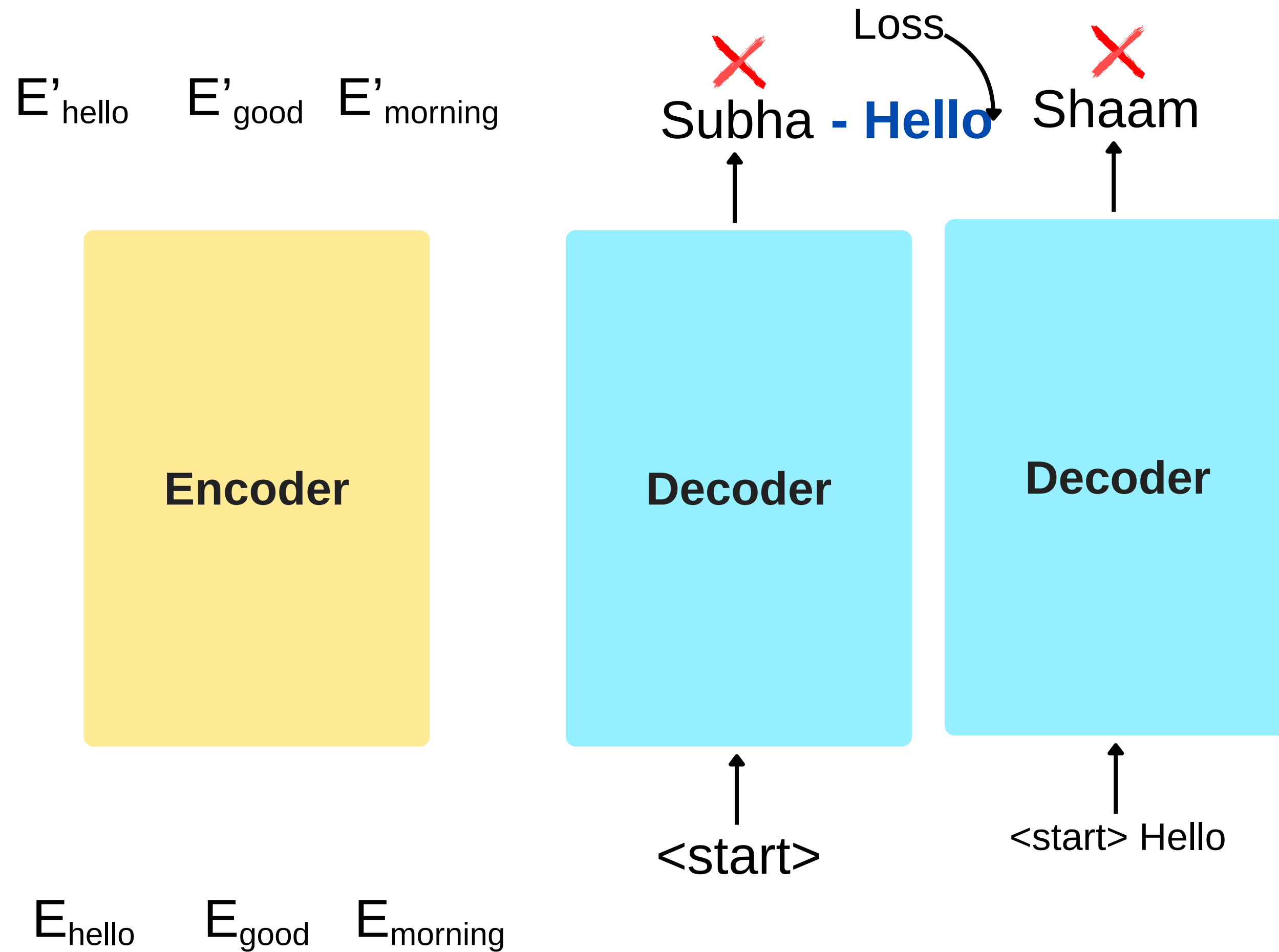
E_{hello} E_{good} E_{morning}

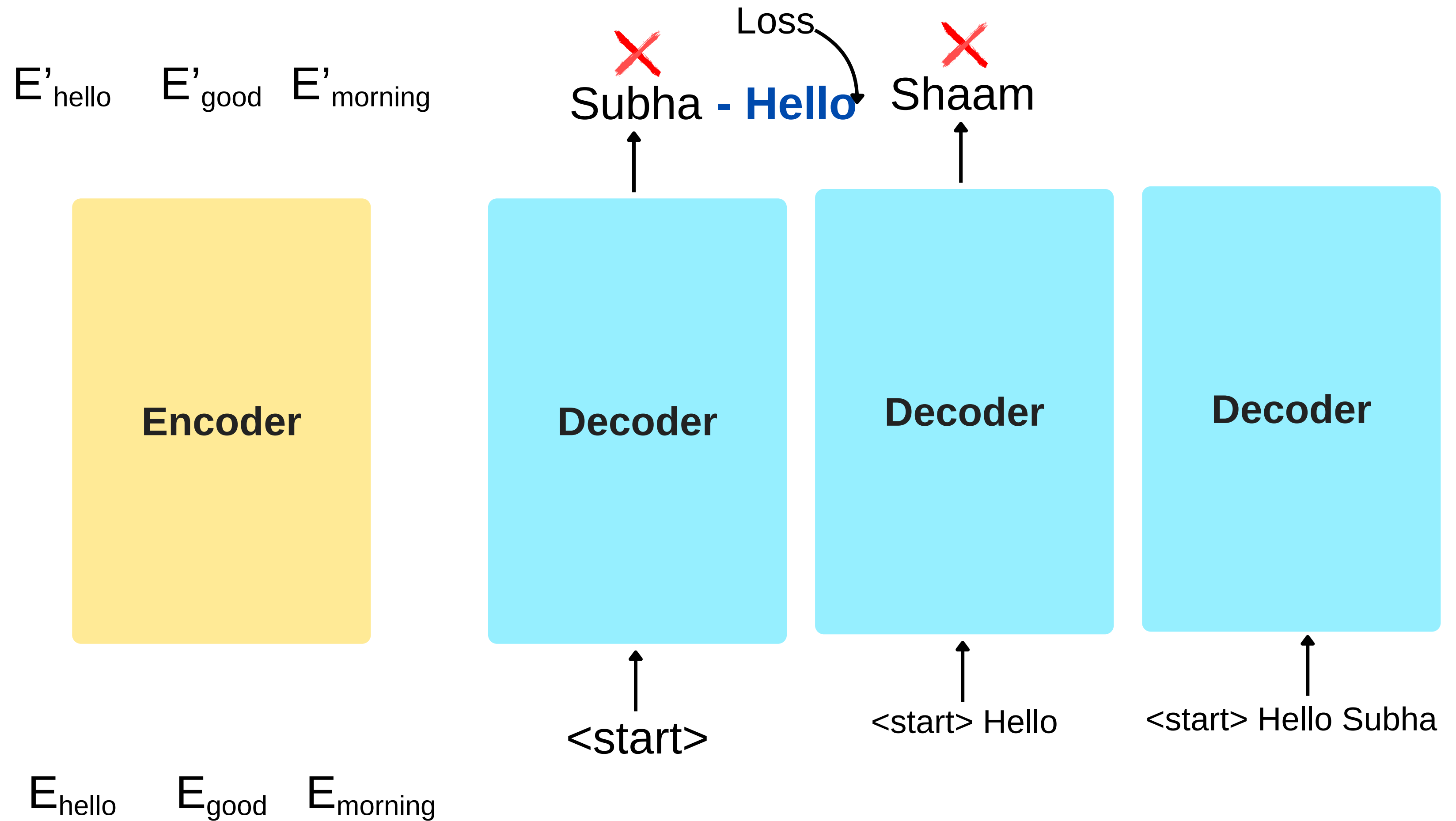
<start>

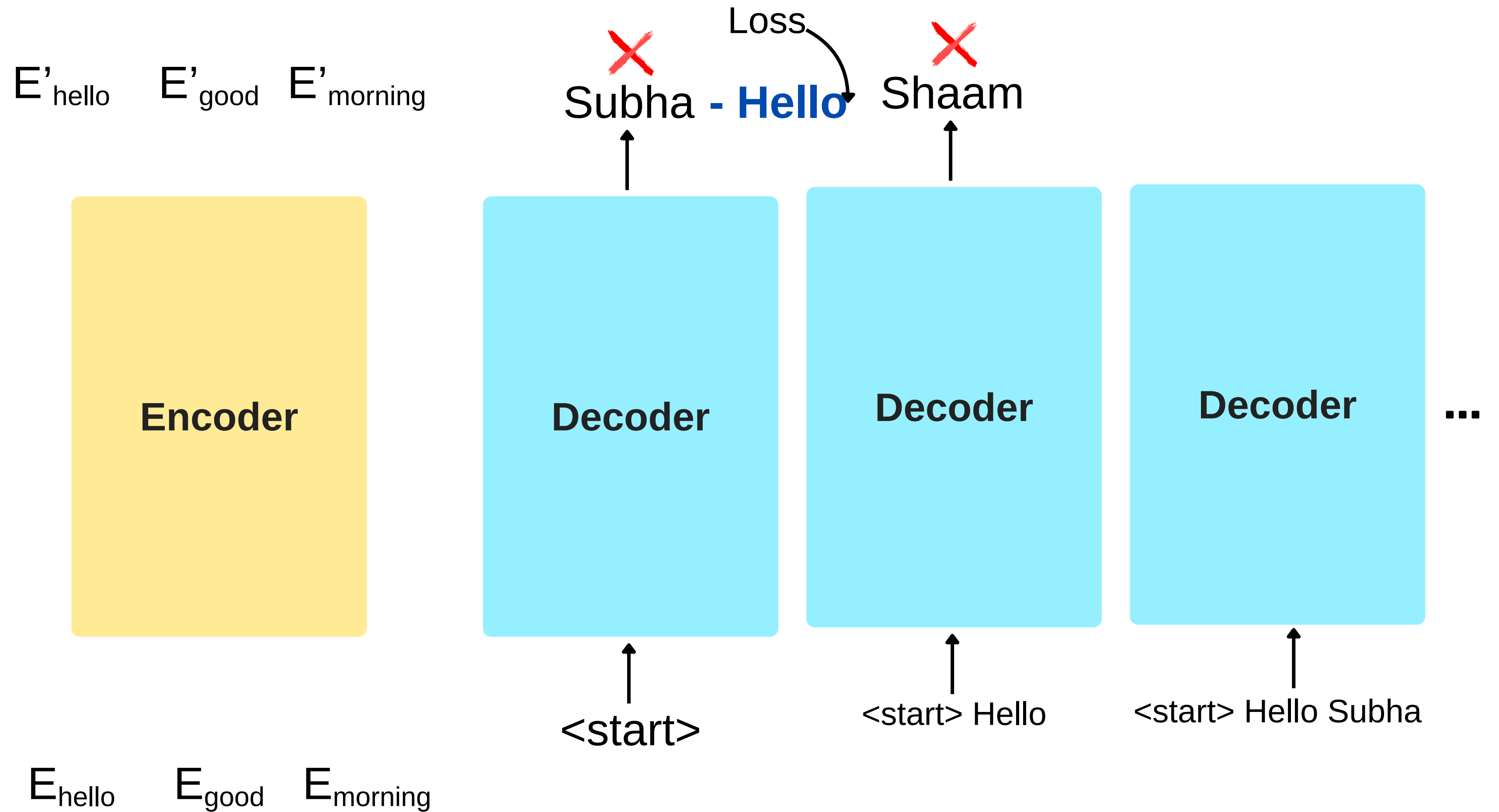


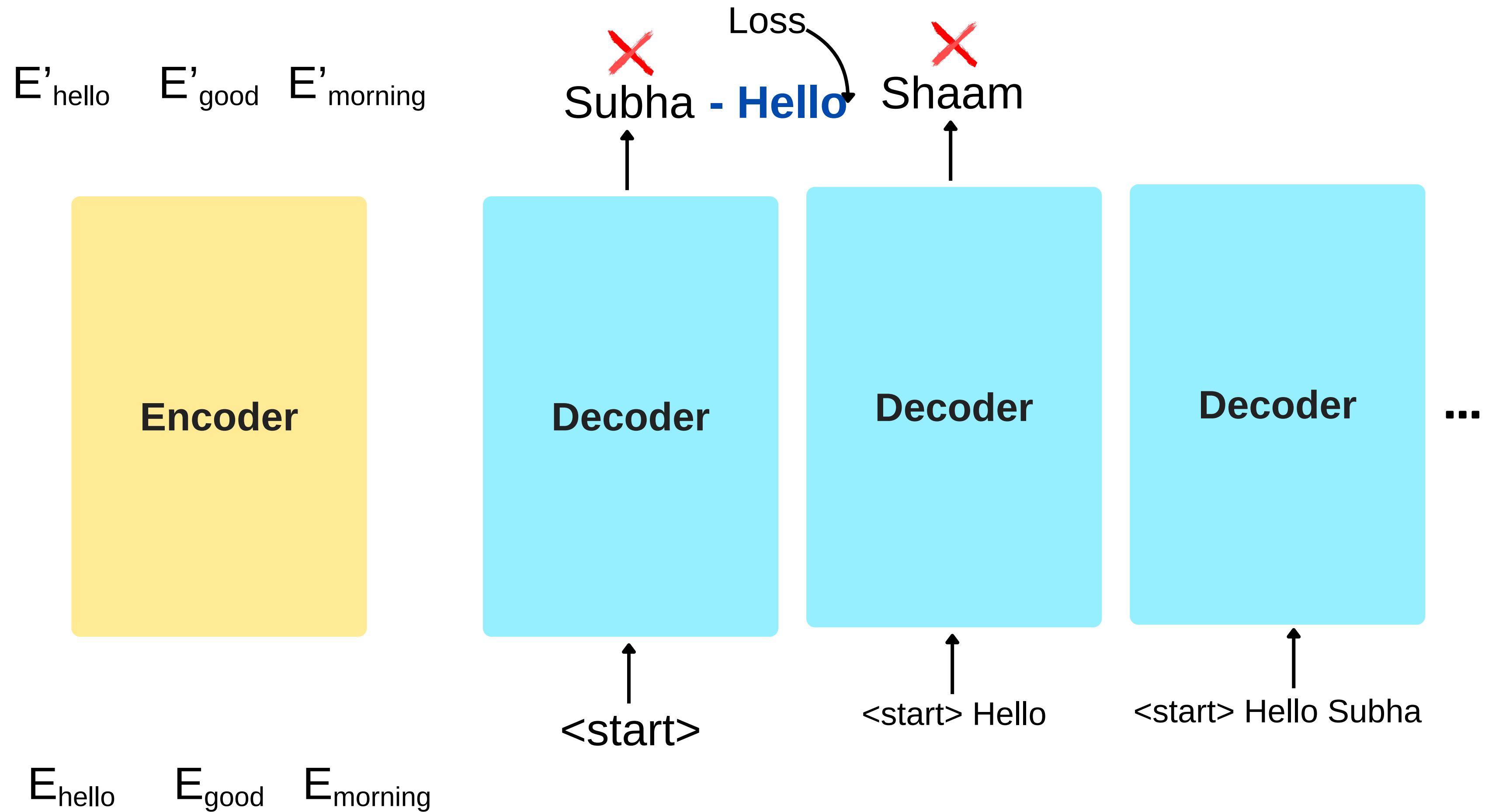






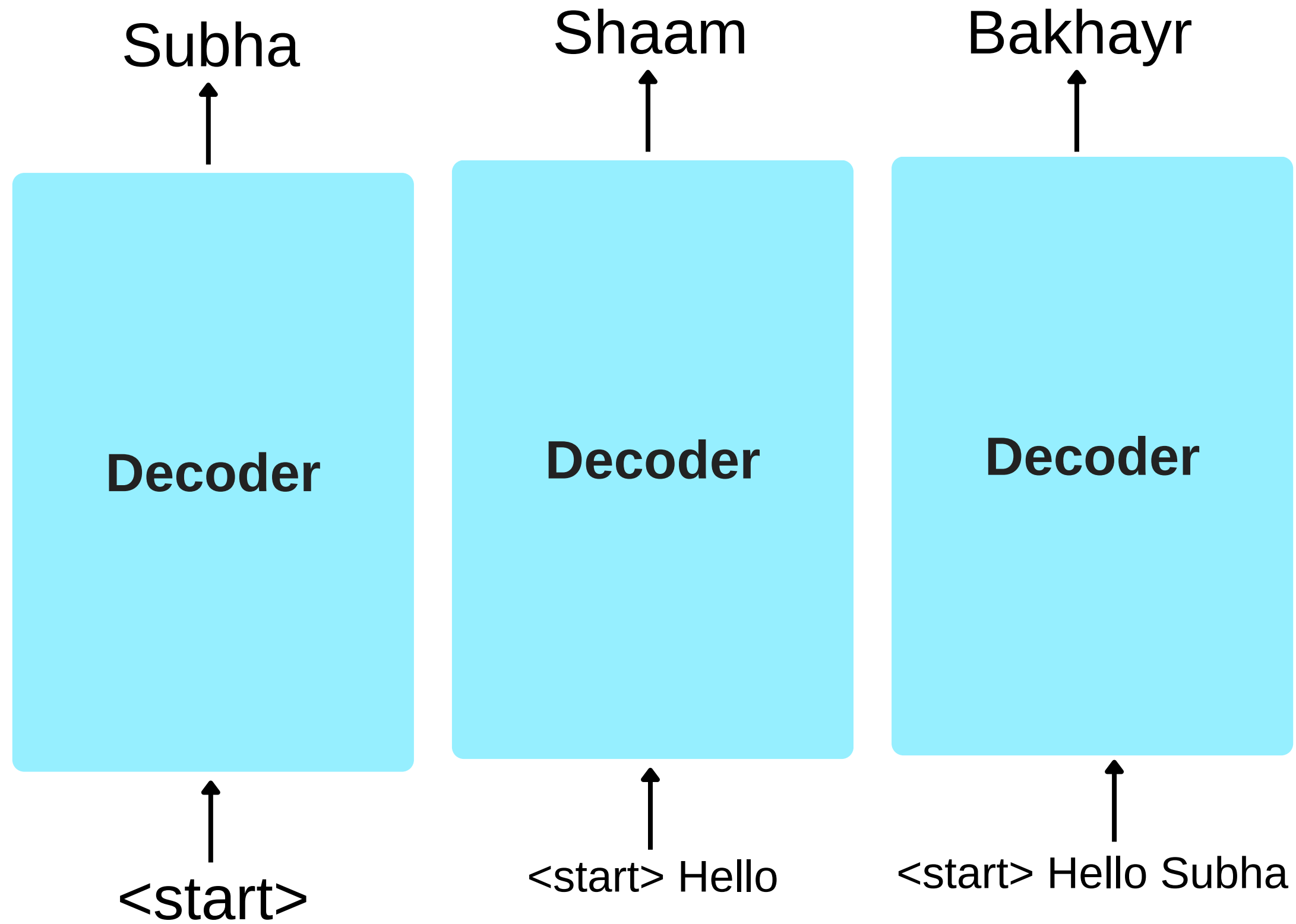


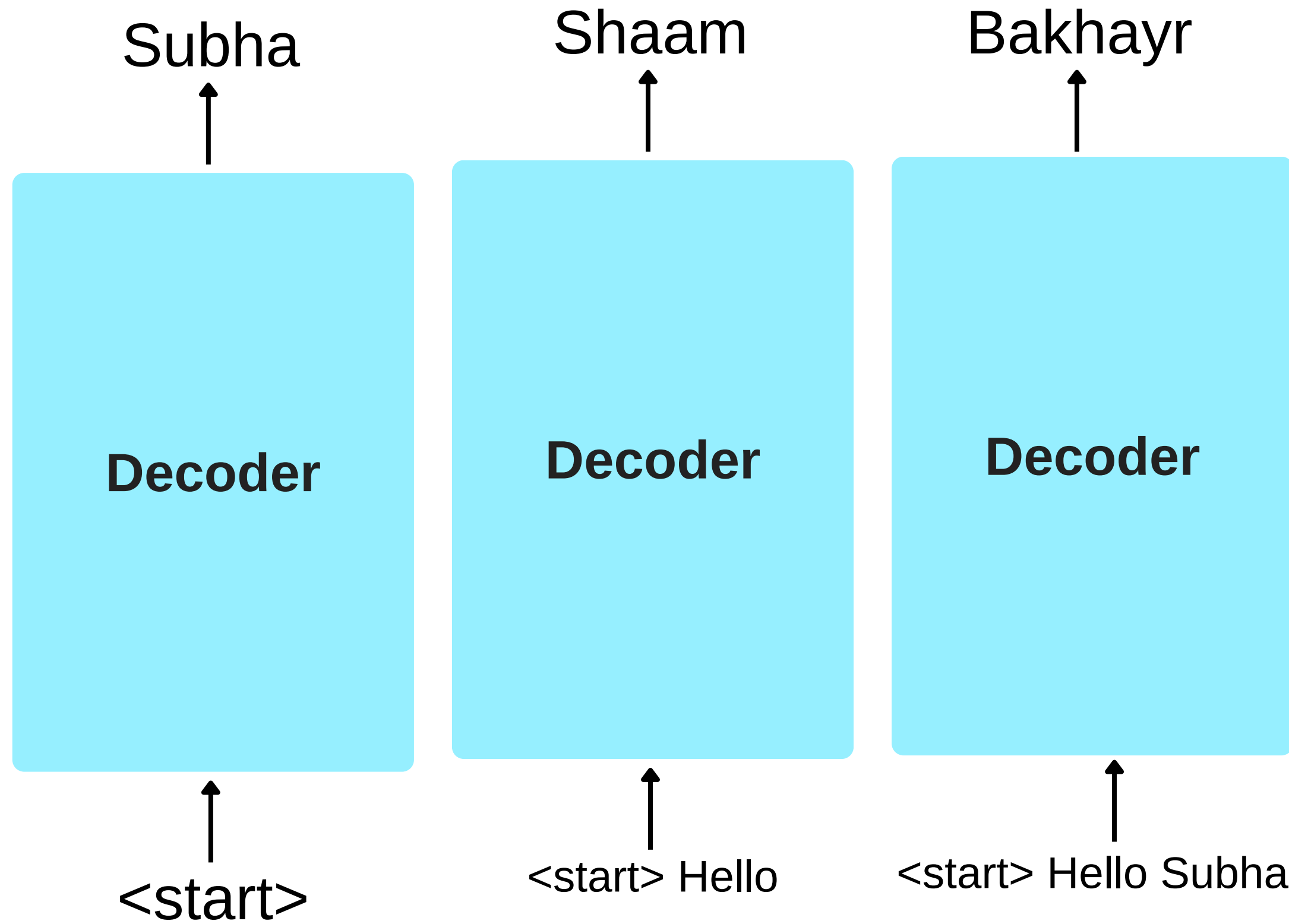




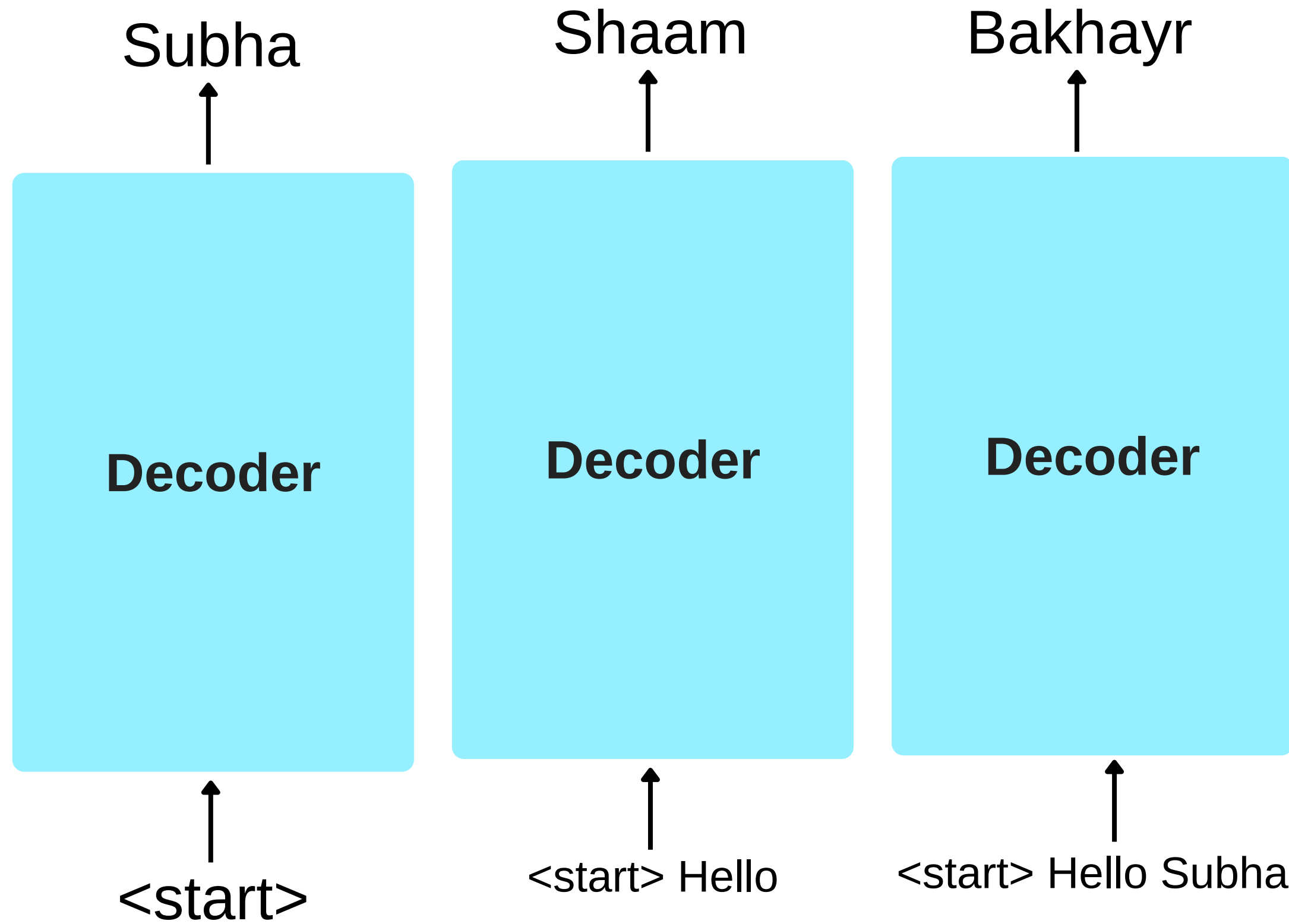
Q: Can I train the model in parallel?

Multiple instances





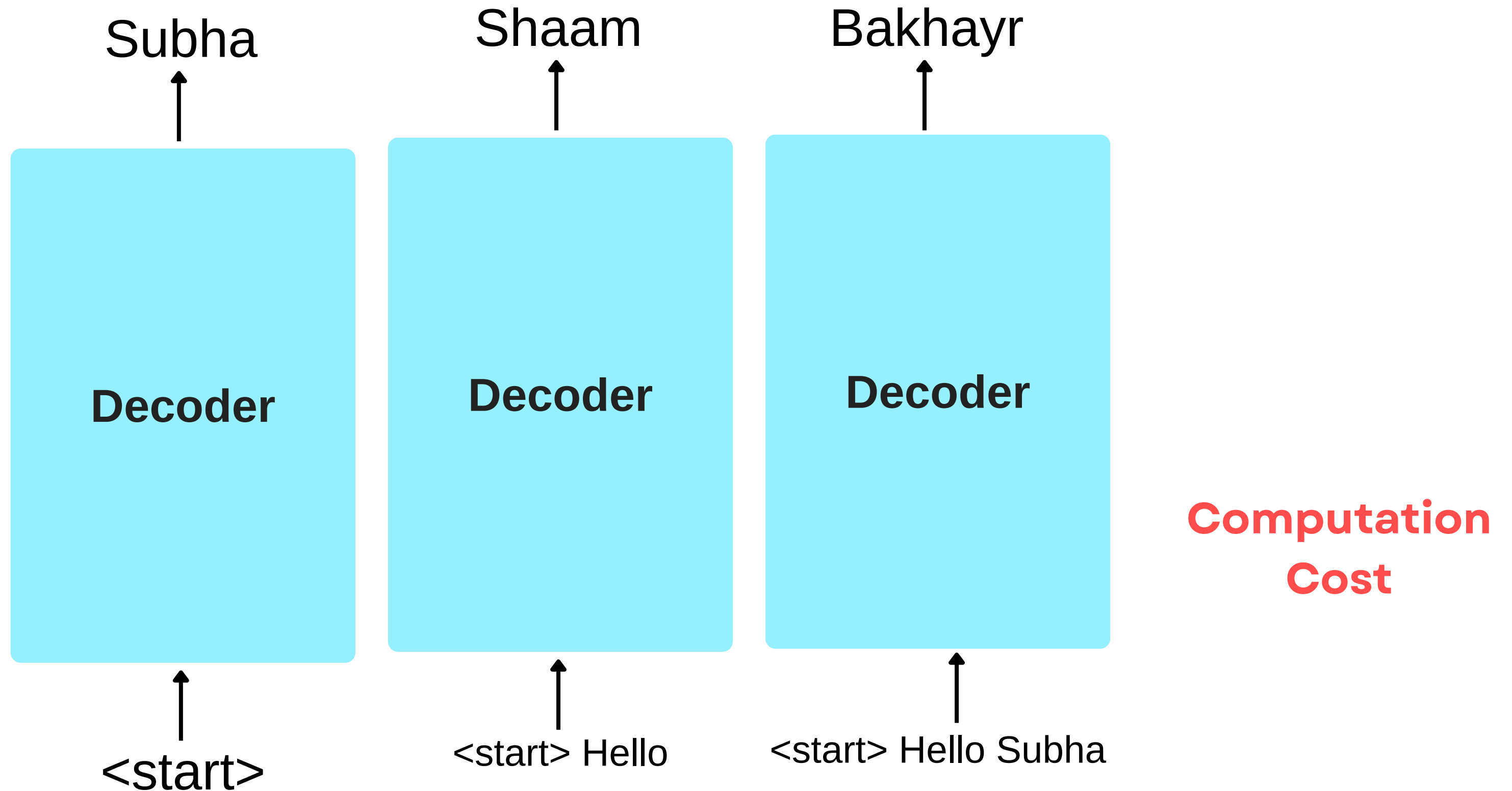
Q: What if I have 100000 words?



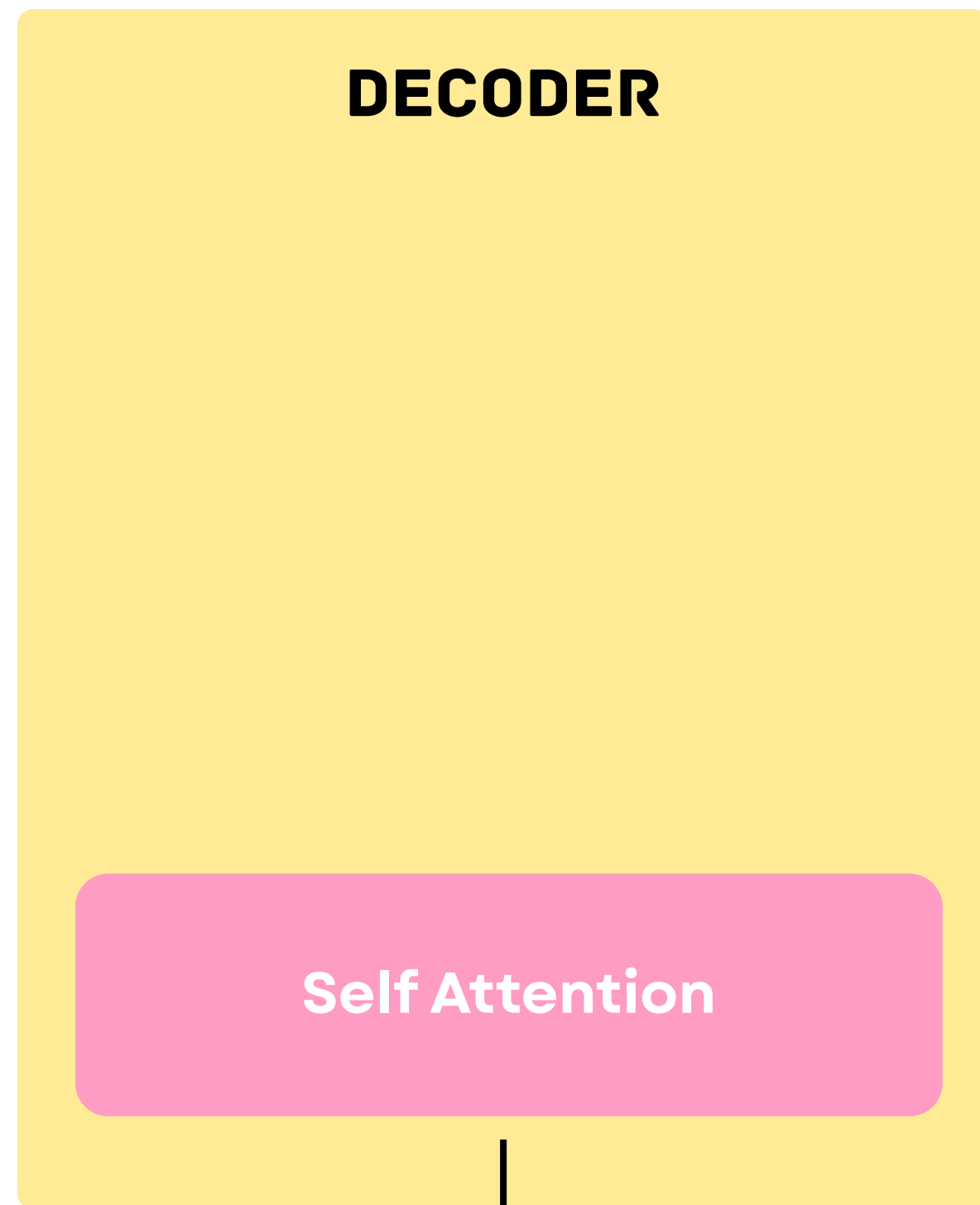
**Computation
Cost**

Q: What if I have 100000 words?

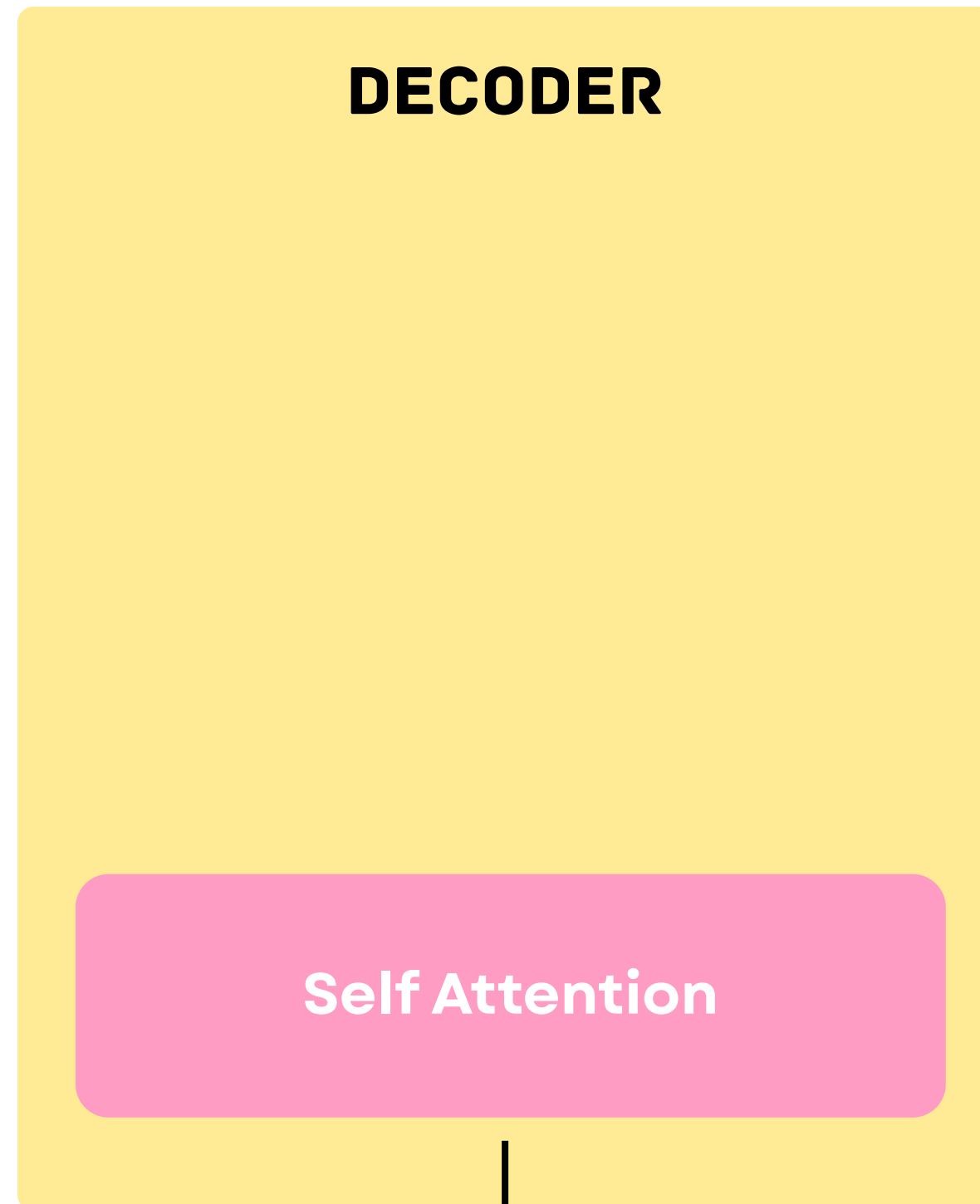
Q: What if we pass all the words altogether?



Q: What if I have 100000 words?



Input

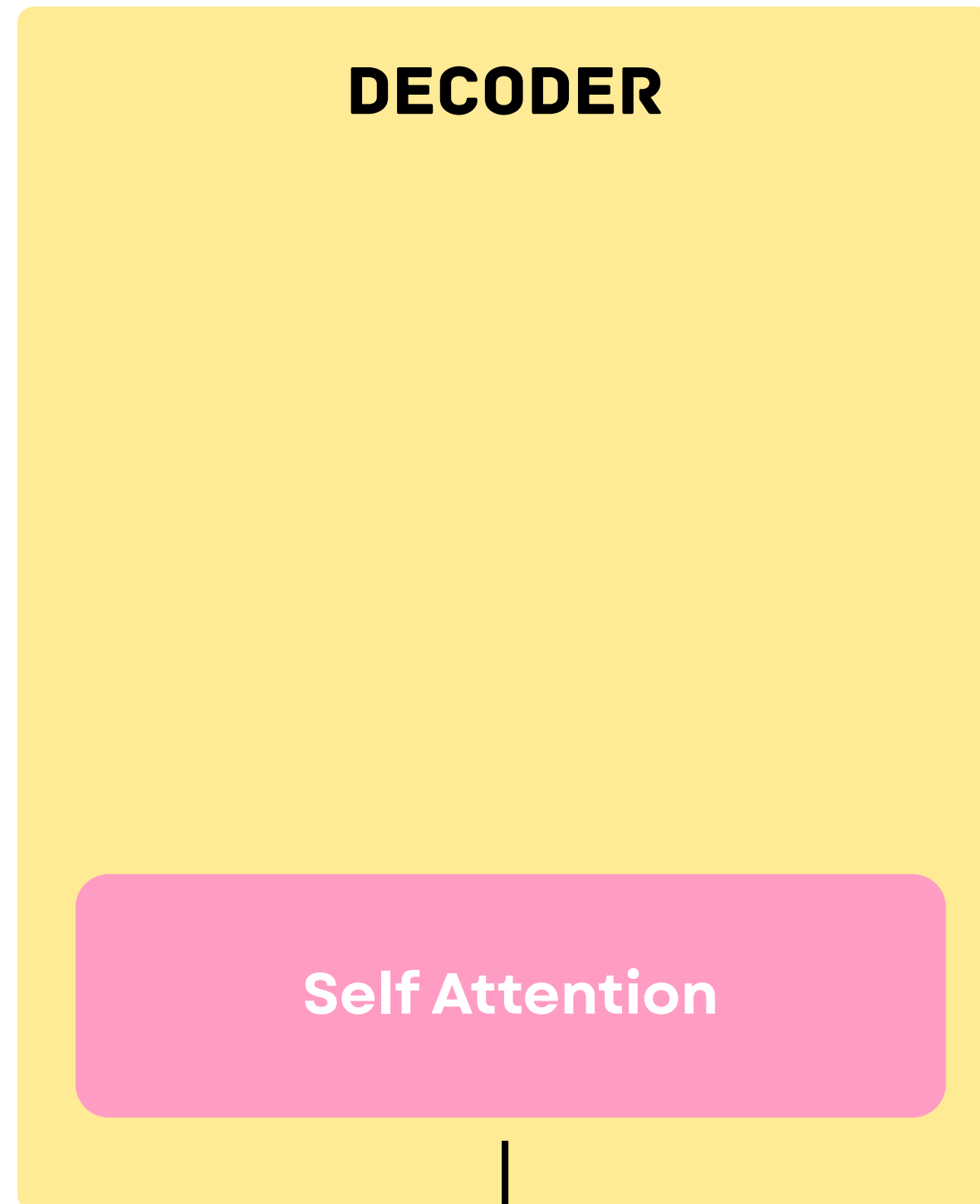


Input

E'_{love} E'_{apple} E'_{phones}



E_{love} E_{apple} E_{phones}



Input

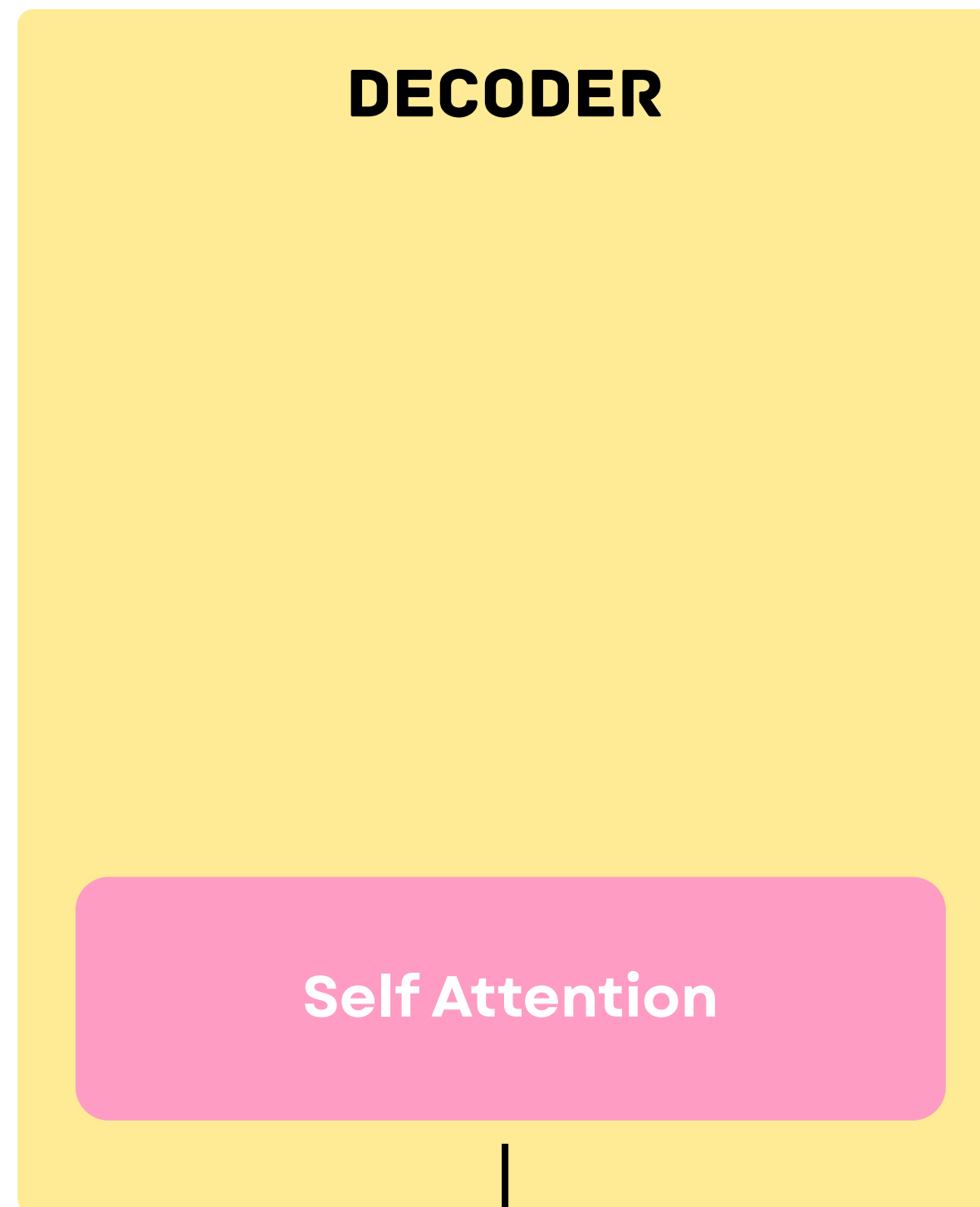
E'_{love} E'_{apple} E'_{phones}



E_{love}

E_{apple}

E_{phones}



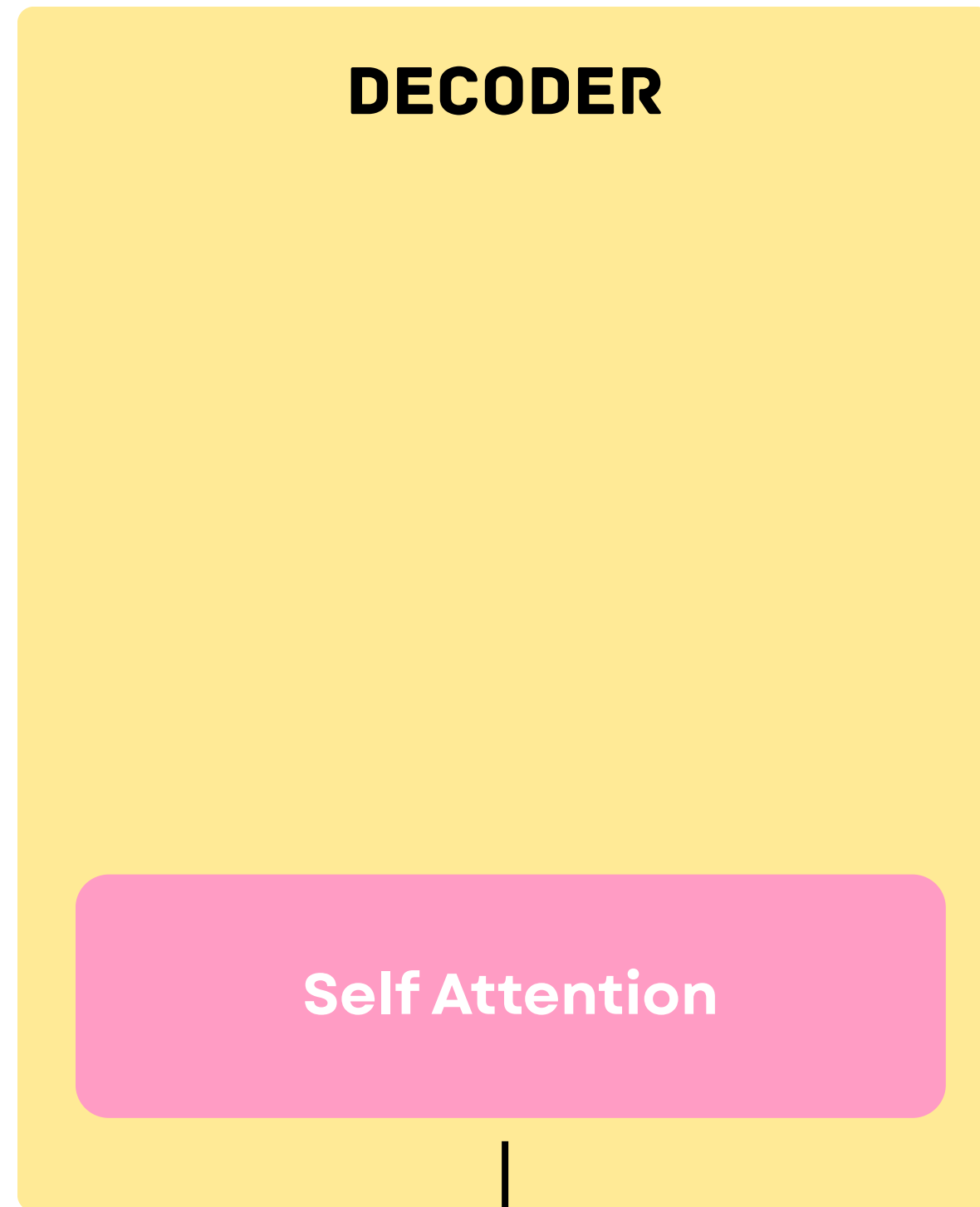
Input

E'_{love} E'_{apple} E'_{phones}

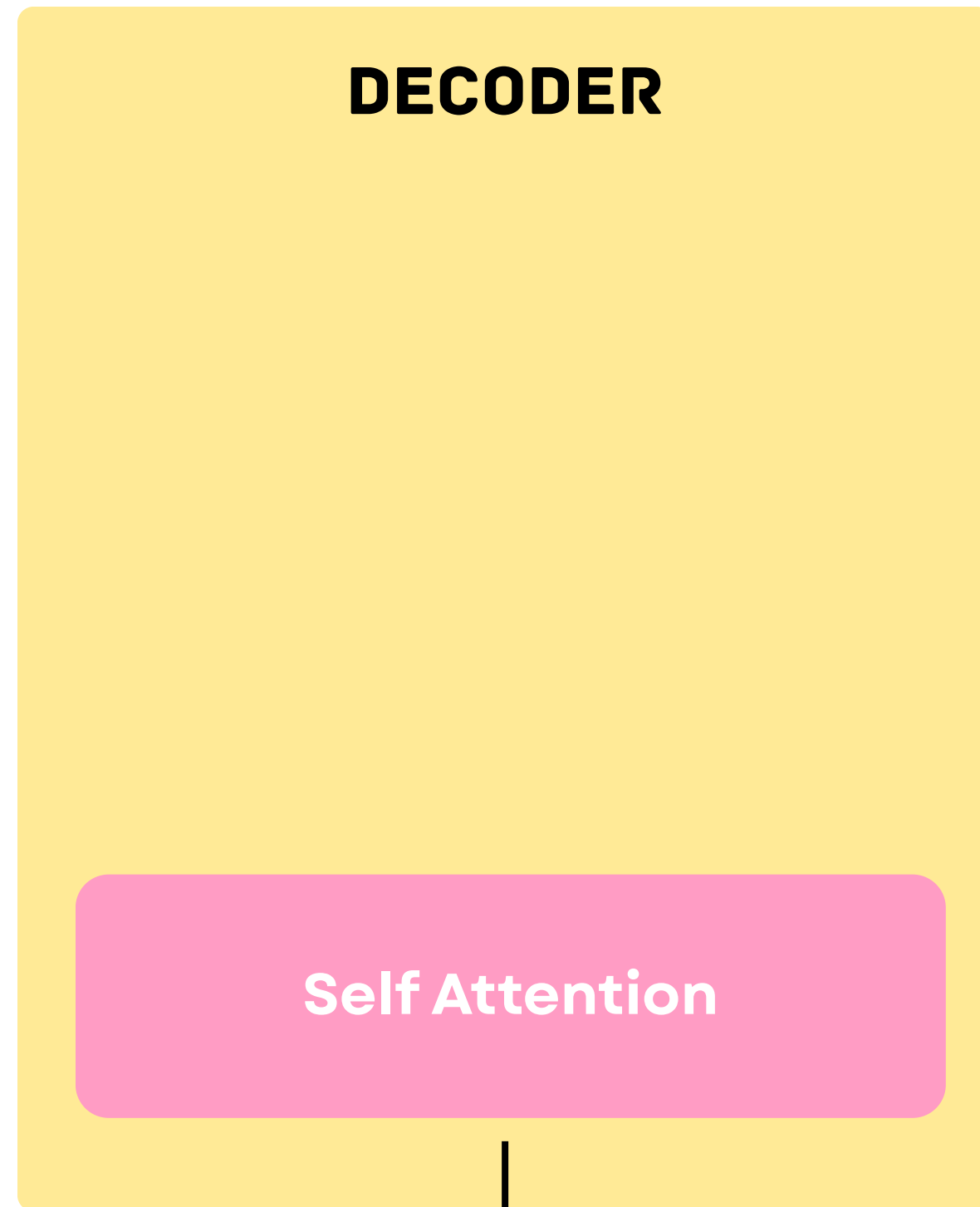


E_{love} E_{apple} E_{phones}

DATA LEAKAGE

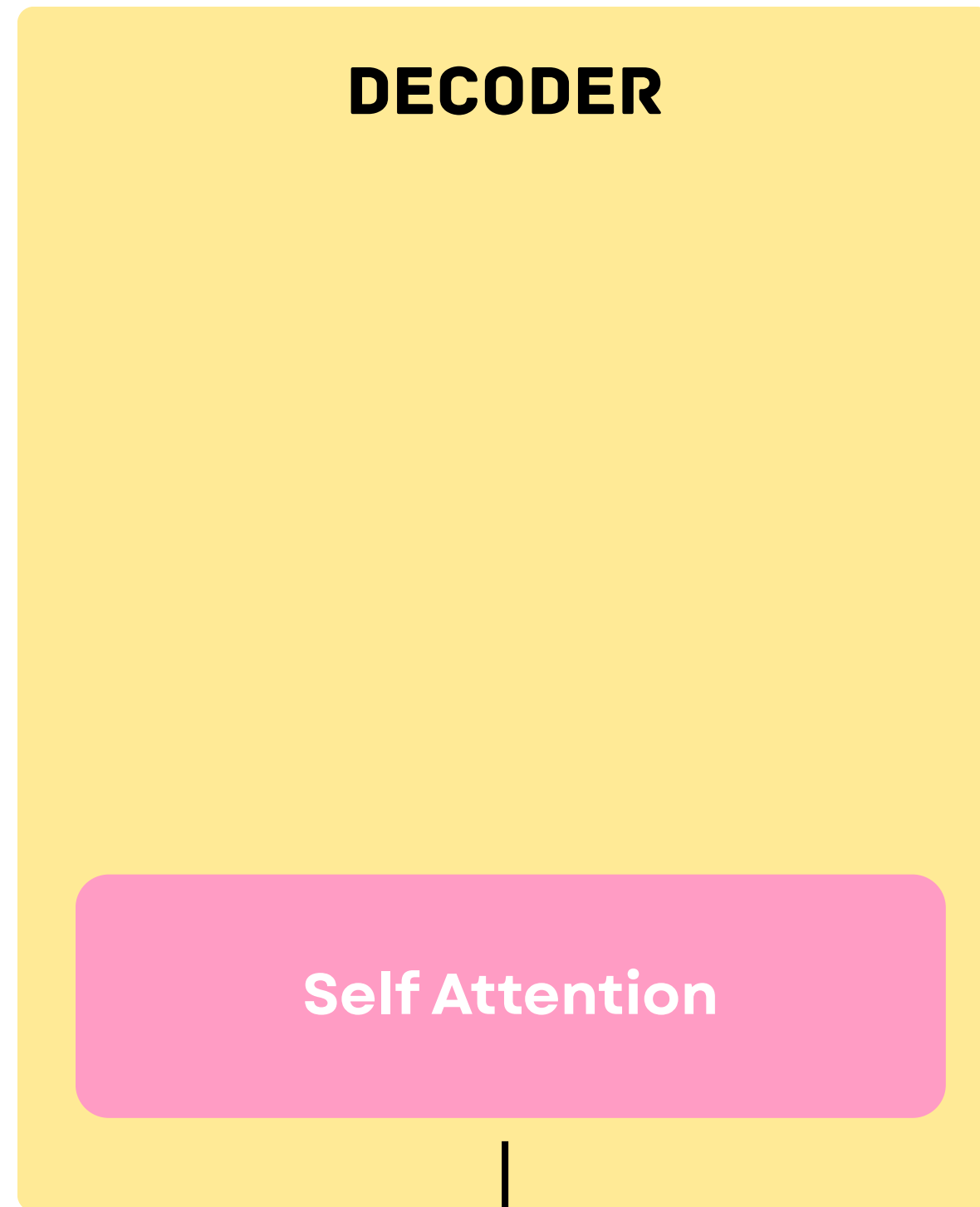


- Sequentially



Input

- Sequentially → More Time

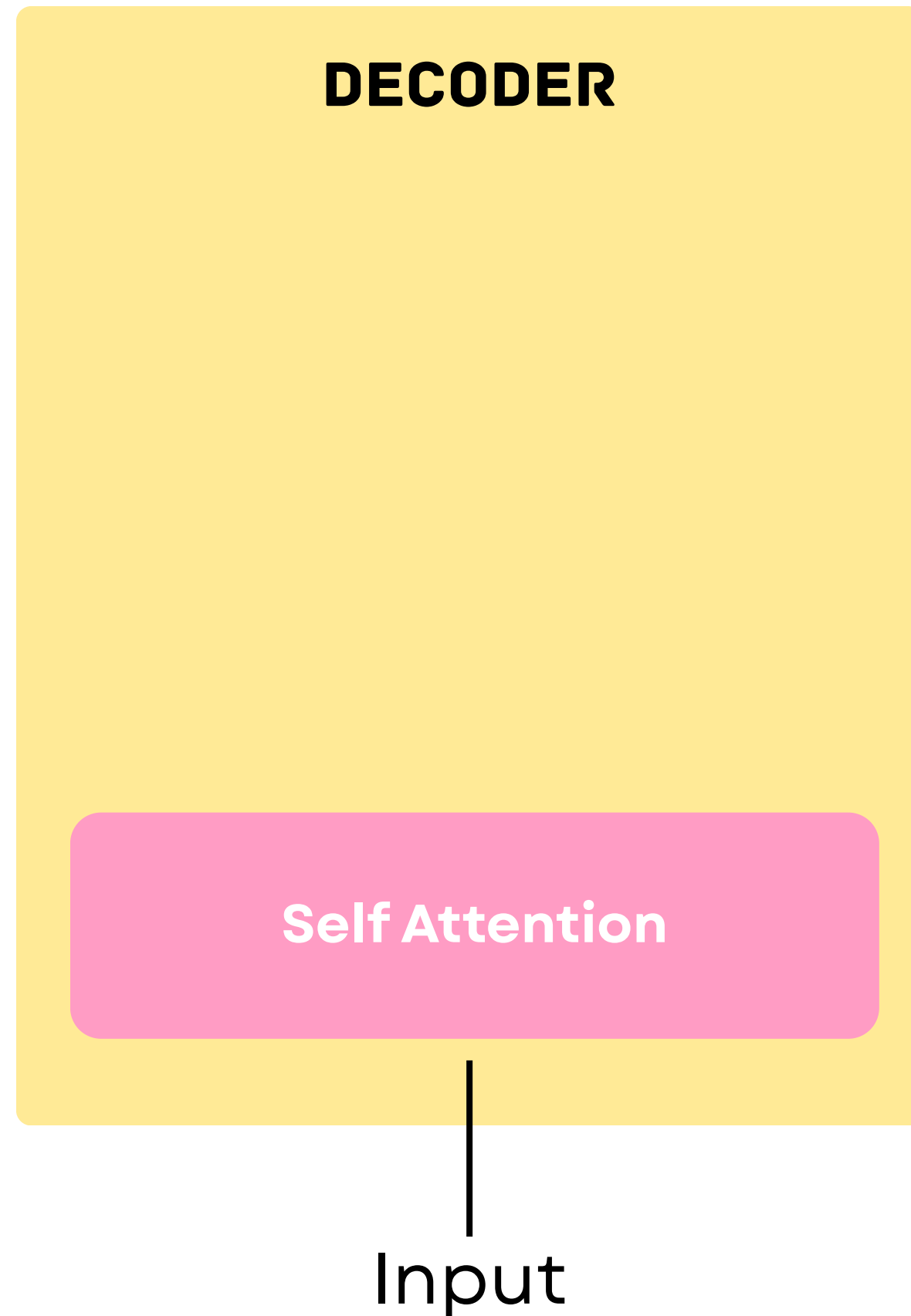


DECODER

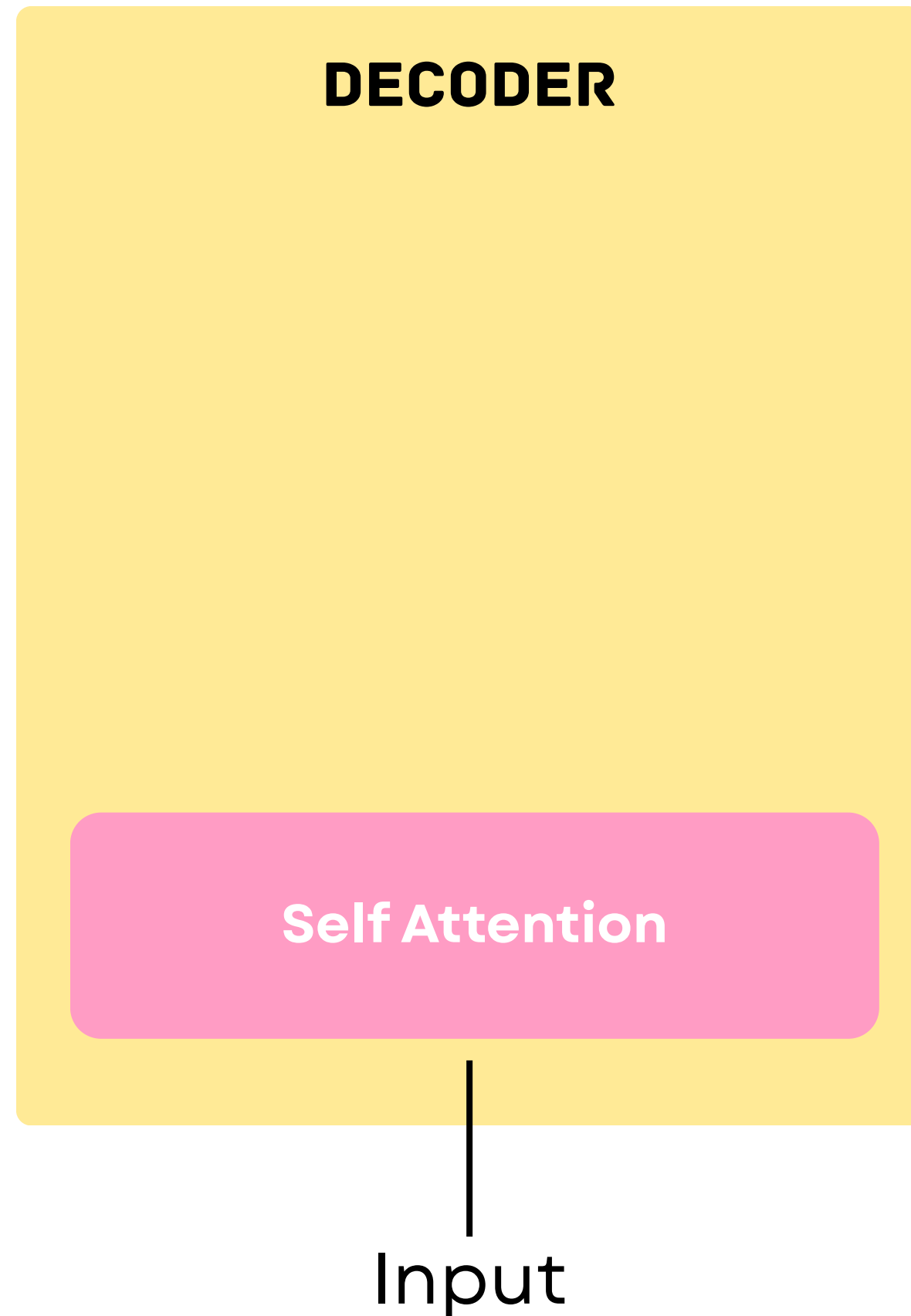
Self Attention

Input

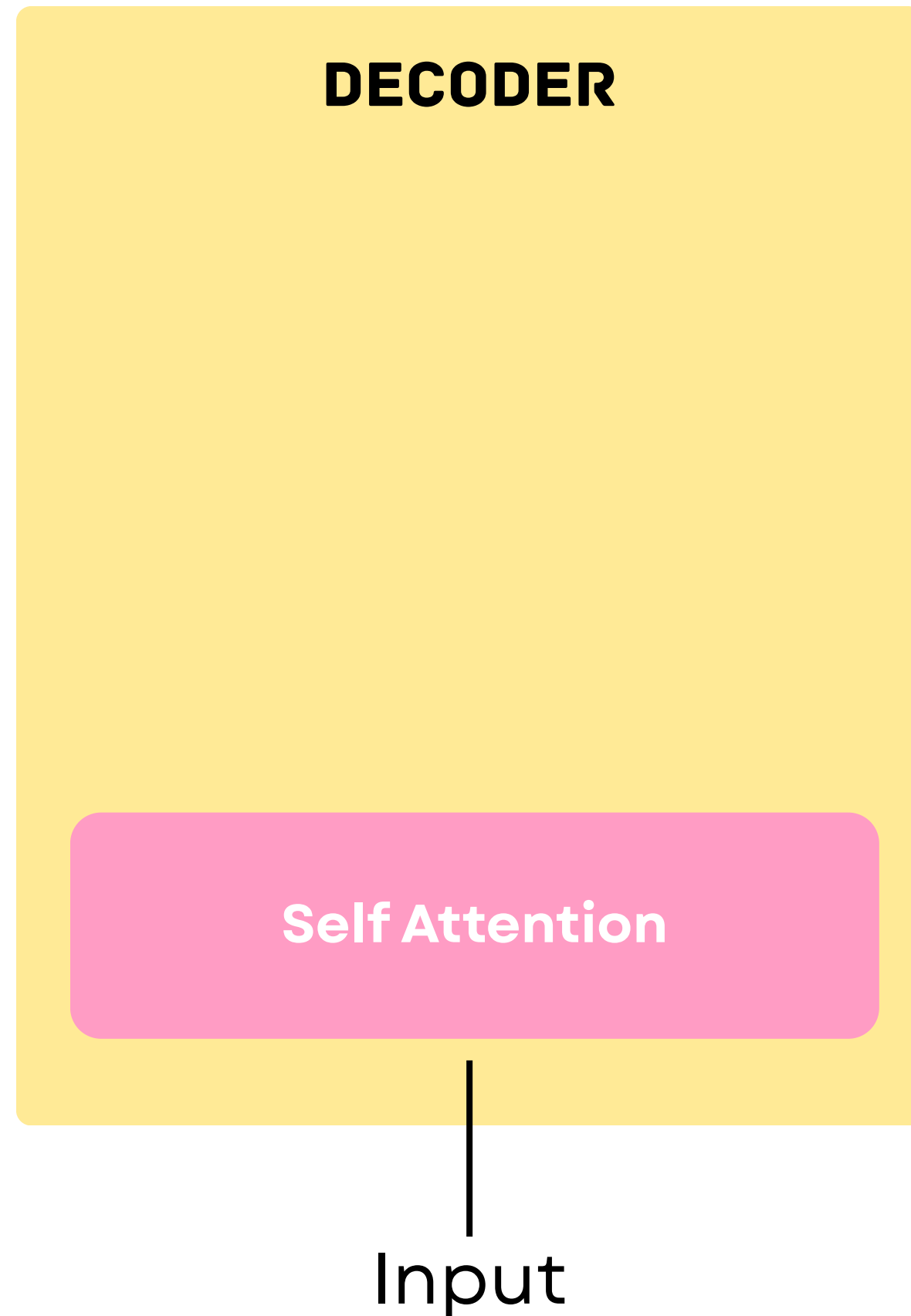
- Sequentially → More Time
- Multiple Decode Instances



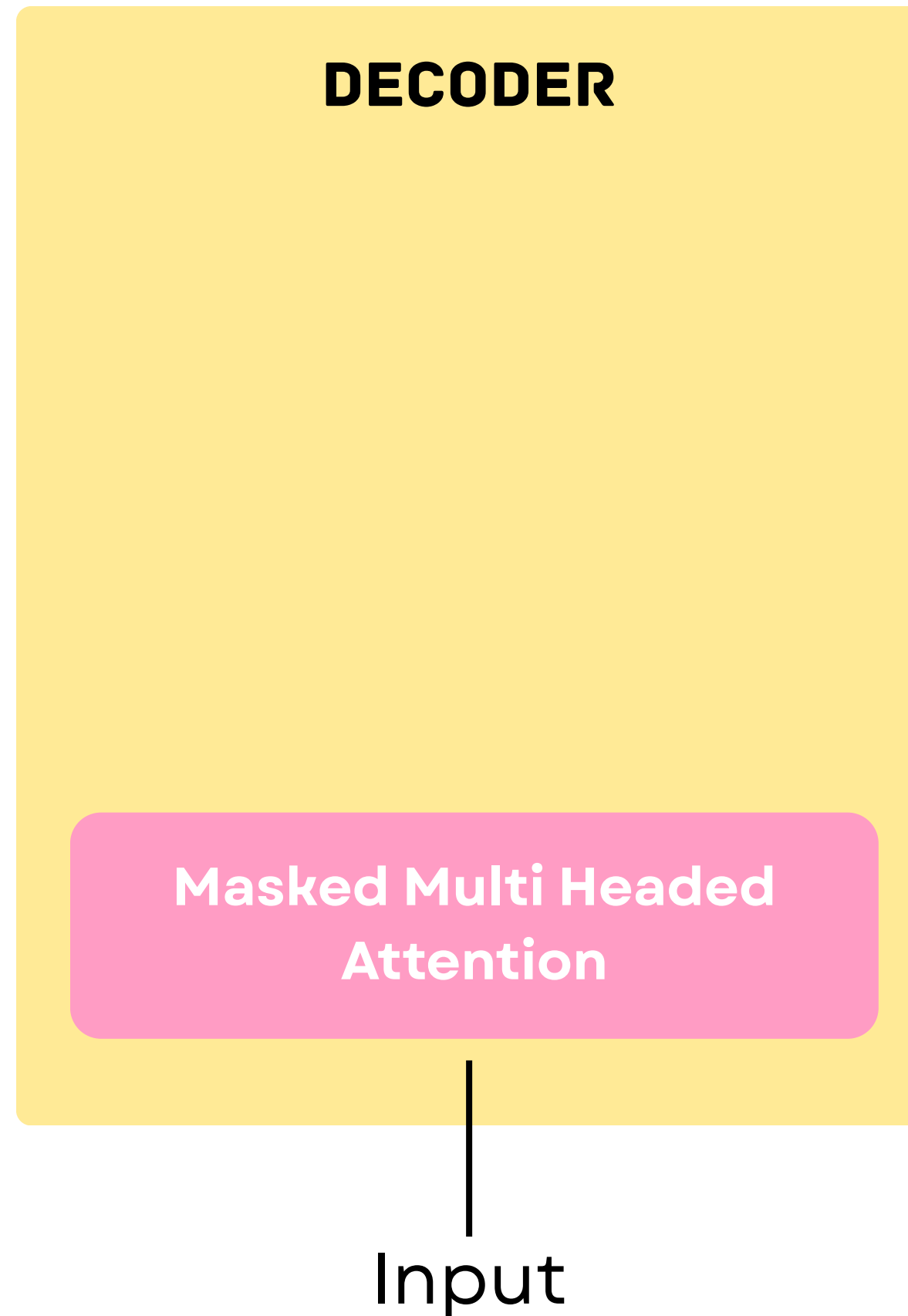
- Sequentially → More Time
- Multiple Decode Instances → More Computation



- Sequentially → More Time
- Multiple Decode Instances → More Computation
- Passing entire target sequence at once



- Sequentially → More Time
- Multiple Decode Instances → More Computation
- Passing entire target sequence at once → Data Leakage



- Sequentially → More Time
- Multiple Decode Instances → More Computation
- Passing entire target sequence at once → Data Leakage

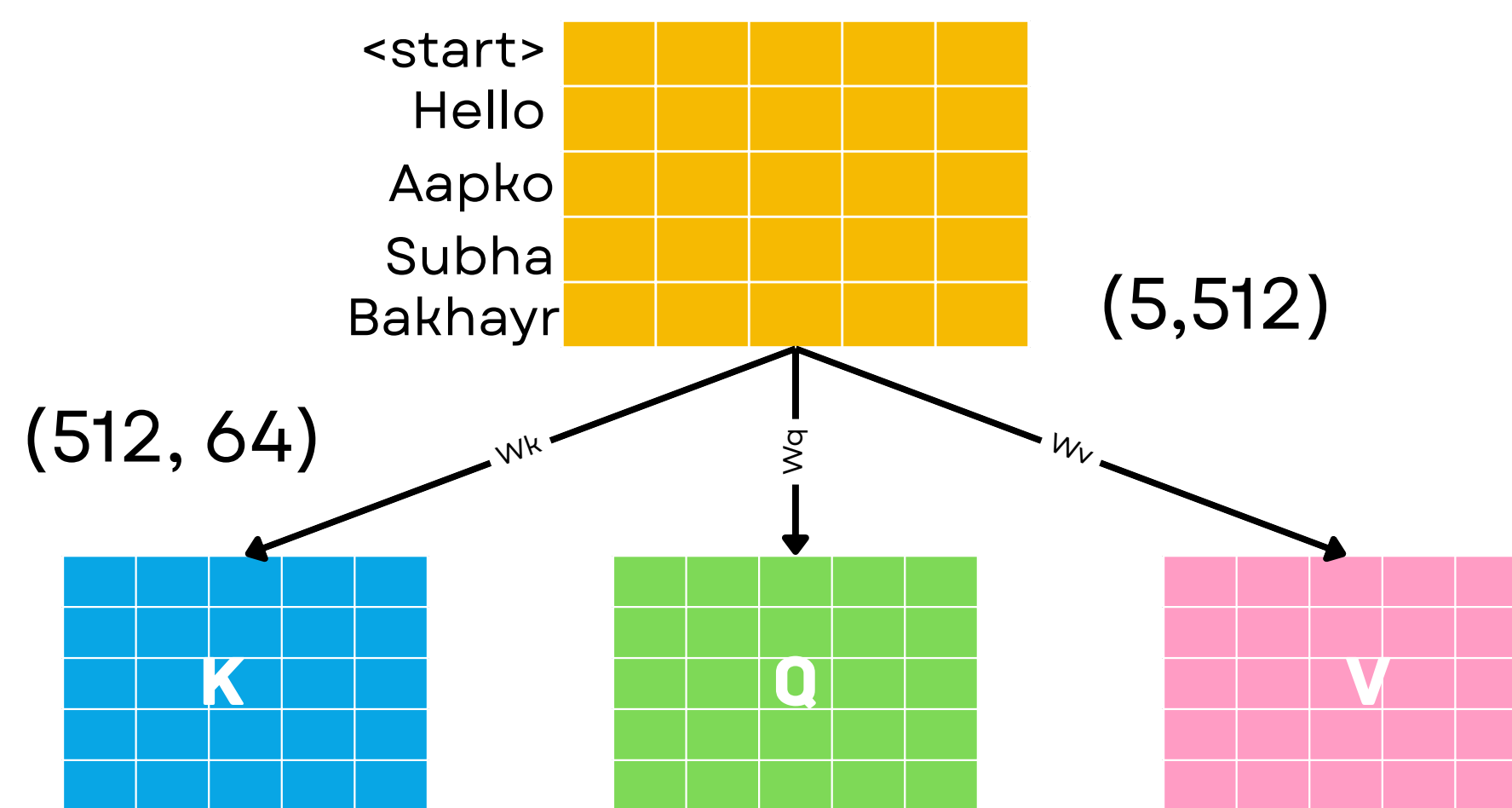


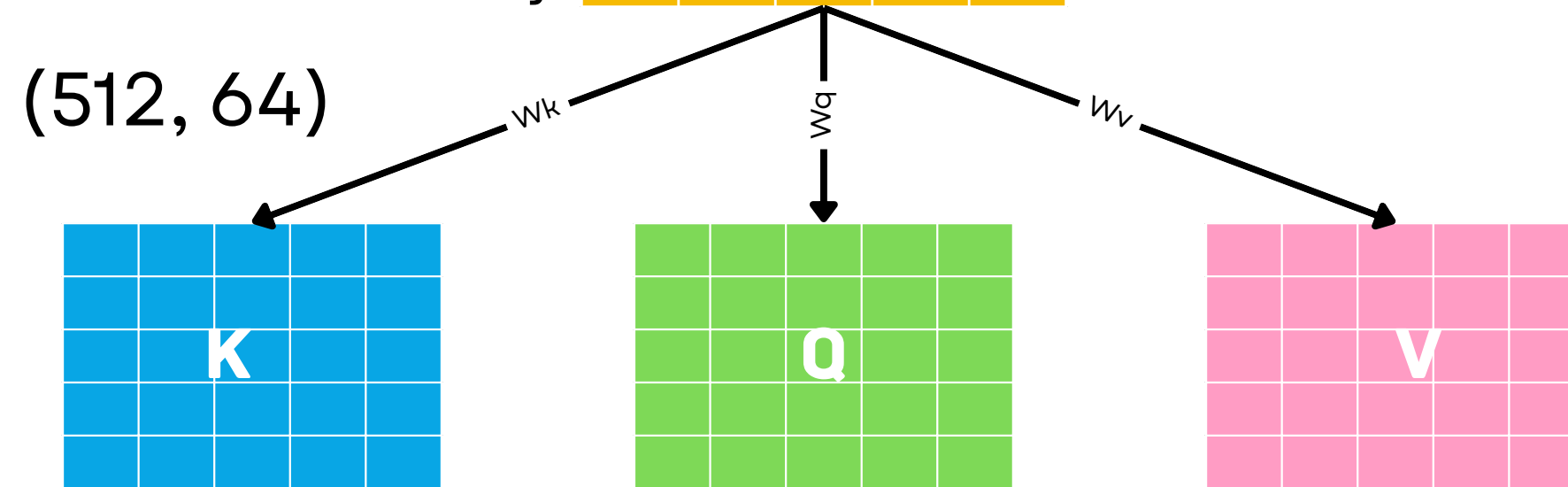
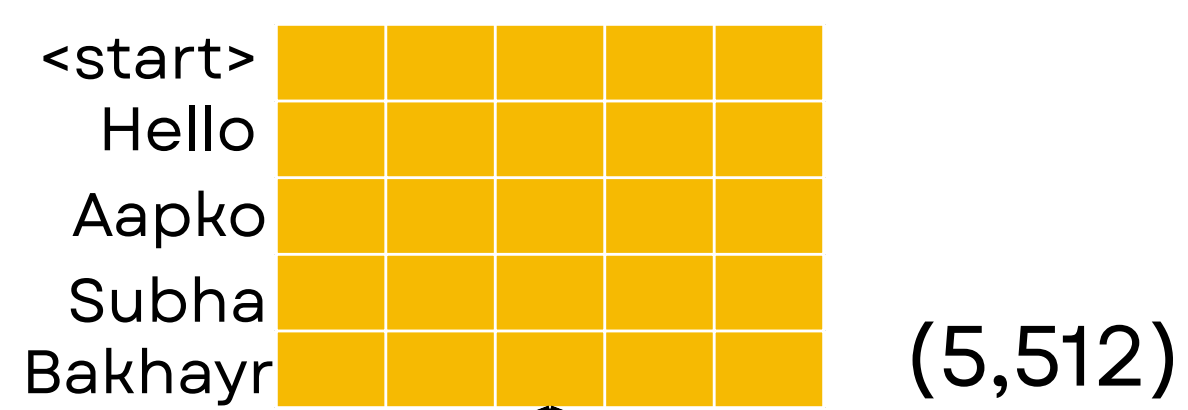
- Sequentially → More Time
- Multiple Decode Instances → More Computation
- Passing entire target sequence at once → Data Leakage

We will pass entire target sequence without
Data Leakage

<start>					
Hello					
Aapko					
Subha					
Bakhayr					

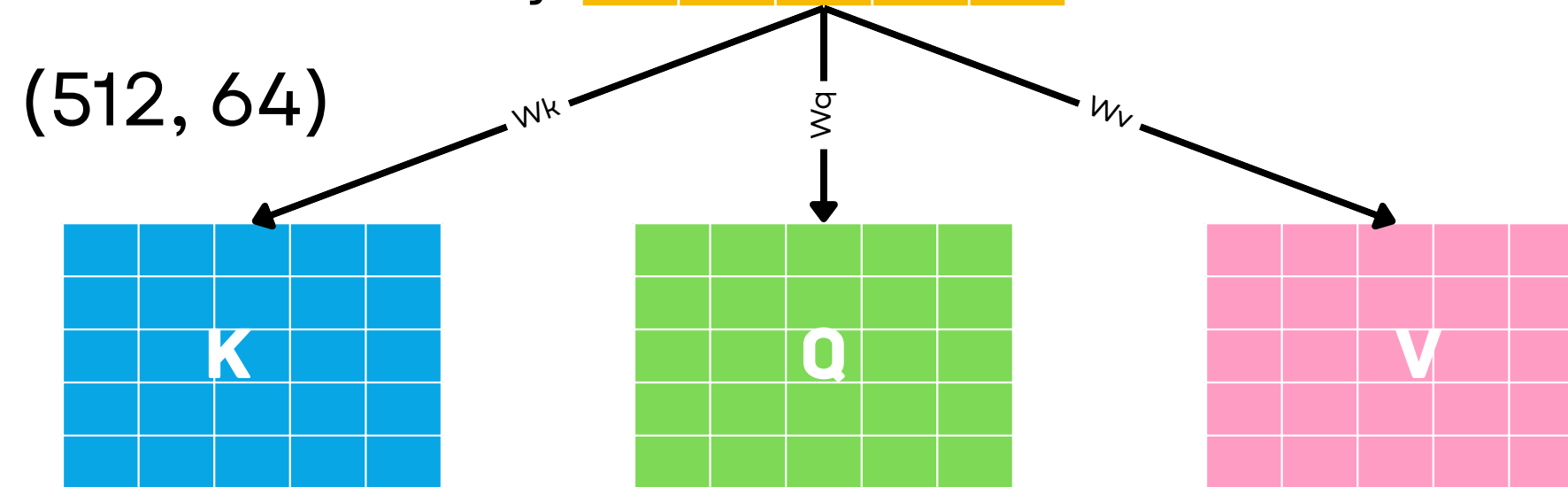
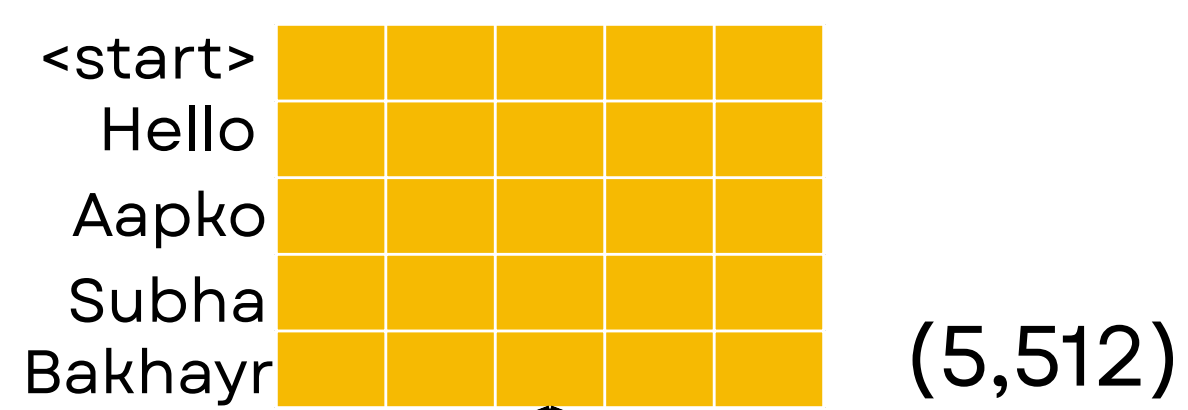
<start>						
Hello						
Aapko						
Subha						
Bakhayr						(5,512)





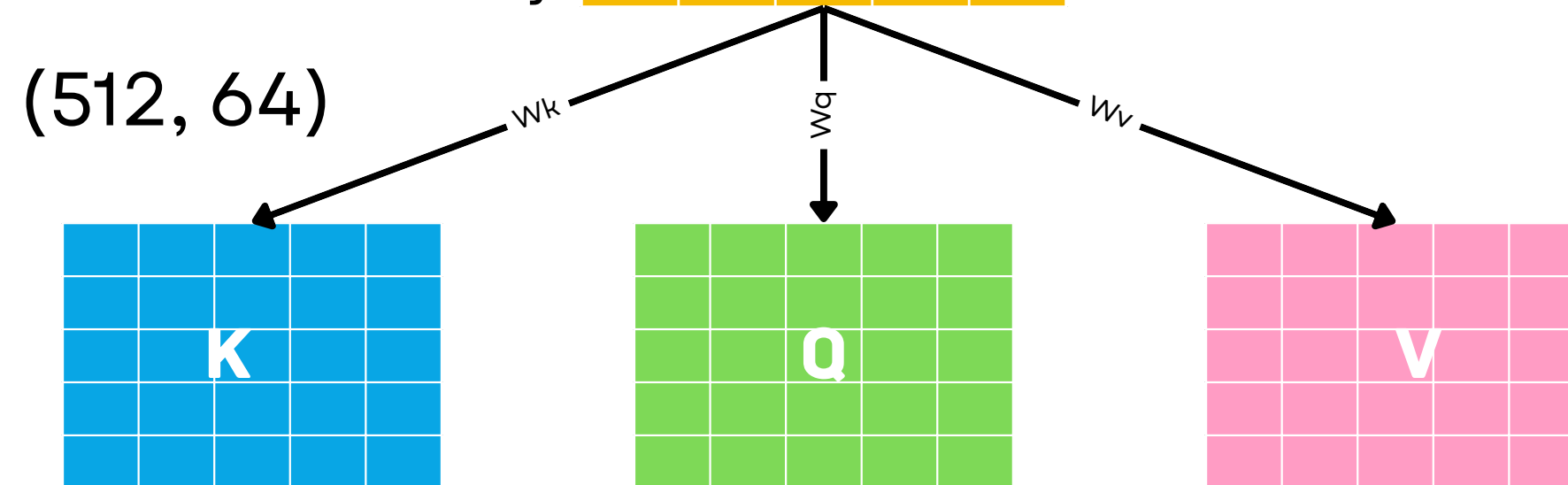
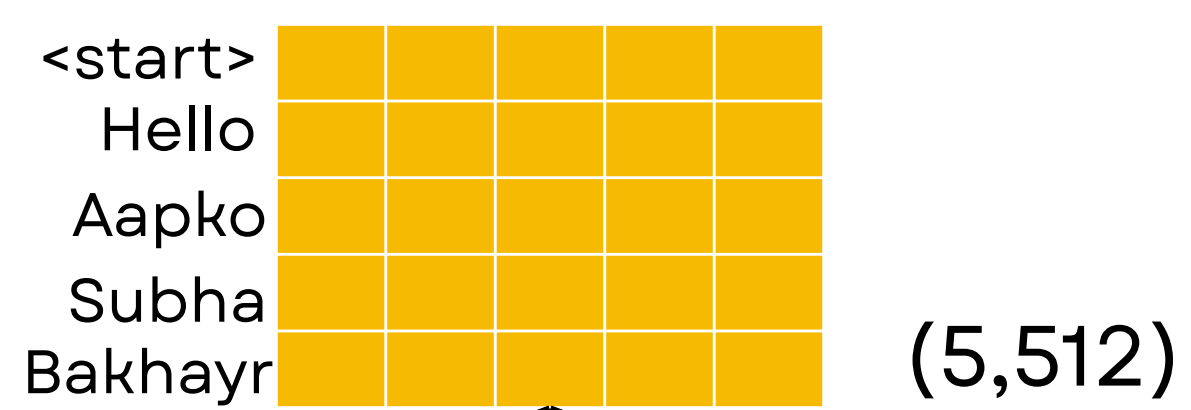
$\left(\begin{array}{c} \text{5x5 grid of green squares} \\ \text{5x5 grid of blue squares} \end{array} \right) \times$

$Q \times K^T$



(512, 64)

$(Q \times K^T)$



(512, 64)

(64, 512)

$\left(\begin{matrix} \text{Green Matrix} & \times & \text{Blue Matrix} \end{matrix} \right)$

$Q \times K^T$

<start>					
Hello					
Aapko					
Subha					
Bakhayr					

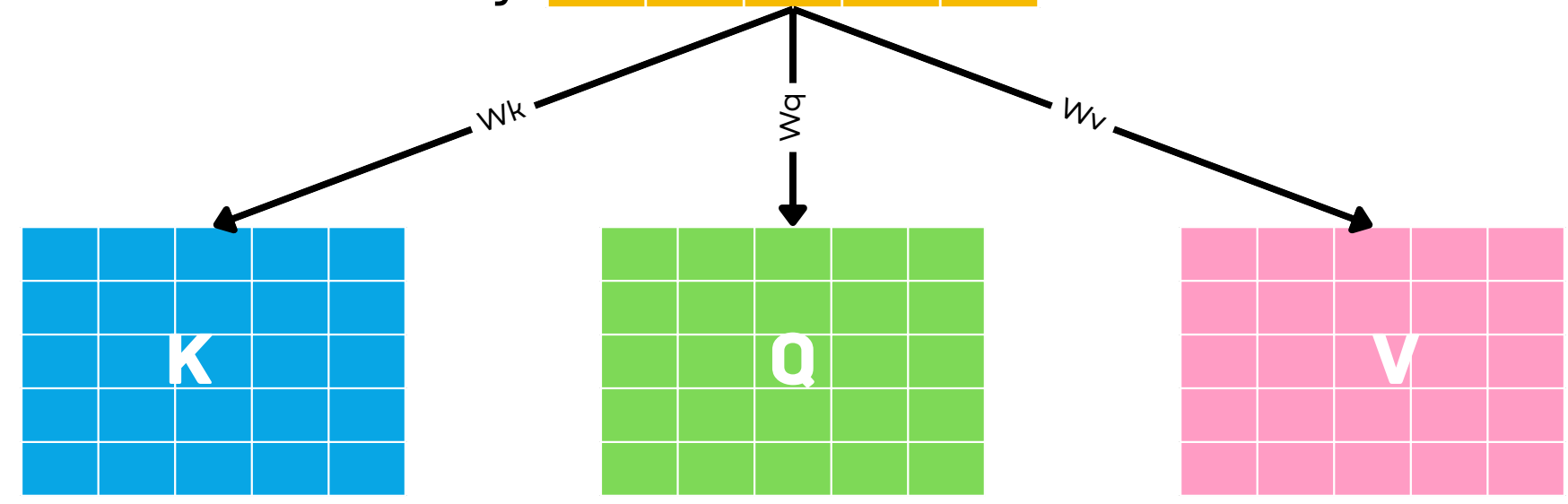
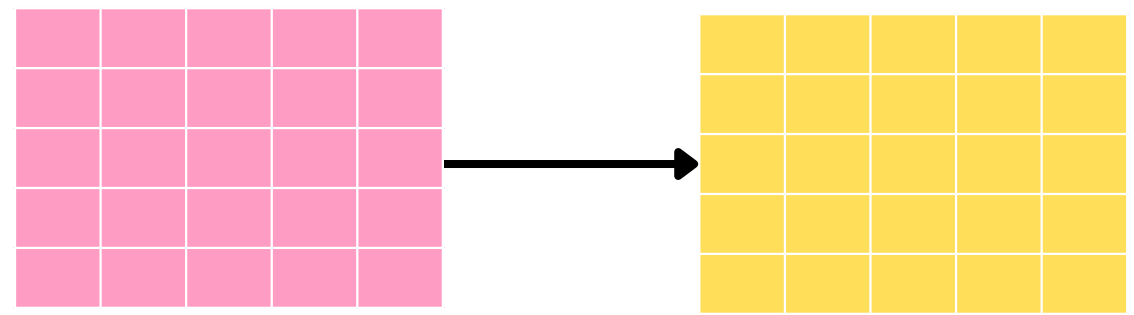


Diagram illustrating the matrix multiplication $Q \times K^T$. A green matrix Q (5x5) is multiplied by a blue matrix K^T (5x5) to produce a result matrix of size 5x5.

	$K_{<start>}$	K_{hello}	K_{aapko}	K_{subha}	$K_{bakhayr}$
$Q_{<start>}$	V11	V12	V13	V14	V15
Q_{hello}	V21	V22	V23	V24	V25
Q_{aapko}	V31	V32	V33	V34	V35
Q_{subha}	V41	V42	V43	V44	V45
$Q_{bakhayr}$	V51	V52	V53	V54	V55

Scale
Softmax

X11	X12	X13	X14	X15
X21	X22	X23	X24	X25
X31	X32	X33	X34	X35
X41	X42	X43	X44	X45
X51	X52	X53	X54	X55



$\text{Softmax}(Q \times K^T) \times V$

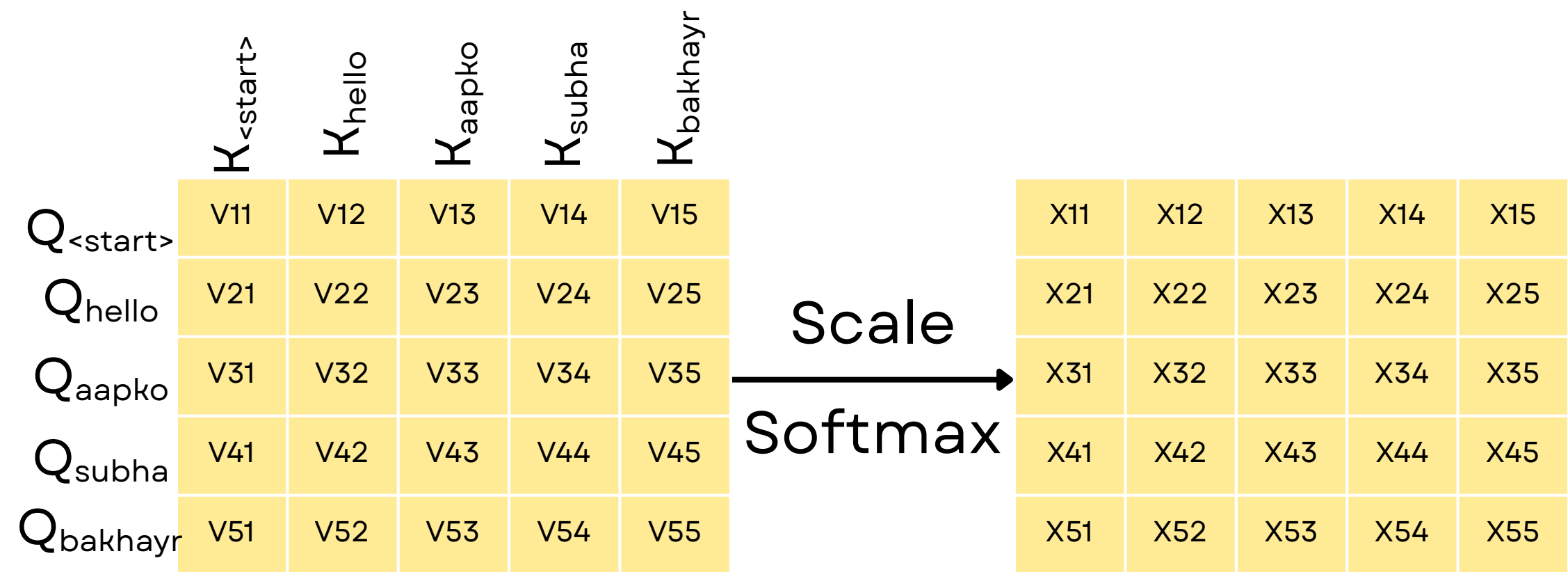
$$E'_{\text{<start>}} = 0.9 \times V_{\text{<start>}} + 0.07 \times V_{\text{hello}} + 0.02 \times V_{\text{aapko}} + 0.01 \times V_{\text{subha}} + 0.01 \times V_{\text{bakhayr}}$$

$$E'_{\text{hello}} = 0.9 \times V_{\text{<start>}} + 0.07 \times V_{\text{hello}} + 0.02 \times V_{\text{aapko}} + 0.01 \times V_{\text{subha}} + 0.01 \times V_{\text{bakhayr}}$$

$$E'_{\text{aapko}} = 0.9 \times V_{\text{<start>}} + 0.07 \times V_{\text{hello}} + 0.02 \times V_{\text{aapko}} + 0.01 \times V_{\text{subha}} + 0.01 \times V_{\text{bakhayr}}$$

$$E'_{\text{subha}} = 0.9 \times V_{\text{<start>}} + 0.07 \times V_{\text{hello}} + 0.02 \times V_{\text{aapko}} + 0.01 \times V_{\text{subha}} + 0.01 \times V_{\text{bakhayr}}$$

$$E'_{\text{bakhayr}} = 0.9 \times V_{\text{<start>}} + 0.07 \times V_{\text{hello}} + 0.02 \times V_{\text{aapko}} + 0.01 \times V_{\text{subha}} + 0.01 \times V_{\text{bakhayr}}$$



$$E'_{\langle \text{start} \rangle} = 0.9 \times V_{\langle \text{start} \rangle} + 0.07 \times V_{\text{hello}} + 0.02 \times V_{\text{aapko}} + 0.01 \times V_{\text{subha}} + 0.01 \times V_{\text{bakhayr}}$$

$$E'_{\text{hello}} = 0.9 \times V_{\langle \text{start} \rangle} + 0.07 \times V_{\text{hello}} + 0.02 \times V_{\text{aapko}} + 0.01 \times V_{\text{subha}} + 0.01 \times V_{\text{bakhayr}}$$

$$E'_{\text{aapko}} = 0.9 \times V_{\langle \text{start} \rangle} + 0.07 \times V_{\text{hello}} + 0.02 \times V_{\text{aapko}} + 0.01 \times V_{\text{subha}} + 0.01 \times V_{\text{bakhayr}}$$

$$E'_{\text{subha}} = 0.9 \times V_{\langle \text{start} \rangle} + 0.07 \times V_{\text{hello}} + 0.02 \times V_{\text{aapko}} + 0.01 \times V_{\text{subha}} + 0.01 \times V_{\text{bakhayr}}$$

$$E'_{\text{bakhayr}} = 0.9 \times V_{\langle \text{start} \rangle} + 0.07 \times V_{\text{hello}} + 0.02 \times V_{\text{aapko}} + 0.01 \times V_{\text{subha}} + 0.01 \times V_{\text{bakhayr}}$$

	$K_{\langle \text{start} \rangle}$	K_{hello}	K_{aapko}	K_{subha}	K_{bakhayr}						
$Q_{\langle \text{start} \rangle}$	V11	V12	V13	V14	V15	Scale → Softmax	X11	X12	X13	X14	X15
Q_{hello}	V21	V22	V23	V24	V25		X21	X22	X23	X24	X25
Q_{aapko}	V31	V32	V33	V34	V35		X31	X32	X33	X34	X35
Q_{subha}	V41	V42	V43	V44	V45		X41	X42	X43	X44	X45
Q_{bakhayr}	V51	V52	V53	V54	V55		X51	X52	X53	X54	X55

$$E'_{\text{<start>}} = 0.9 \times V_{\text{<start>}} + 0.07 \times V_{\text{hello}} + 0.02 \times V_{\text{aapko}} + 0.01 \times V_{\text{subha}} + 0.01 \times V_{\text{bakhayr}}$$

$$E'_{\text{hello}} = 0.9 \times V_{\text{<start>}} + 0.07 \times V_{\text{hello}} + 0.02 \times V_{\text{aapko}} + 0.01 \times V_{\text{subha}} + 0.01 \times V_{\text{bakhayr}}$$

$$E'_{\text{aapko}} = 0.9 \times V_{\text{<start>}} + 0.07 \times V_{\text{hello}} + 0.02 \times V_{\text{aapko}} + 0.01 \times V_{\text{subha}} + 0.01 \times V_{\text{bakhayr}}$$

$$E'_{\text{subha}} = 0.9 \times V_{\text{<start>}} + 0.07 \times V_{\text{hello}} + 0.02 \times V_{\text{aapko}} + 0.01 \times V_{\text{subha}} + 0.01 \times V_{\text{bakhayr}}$$

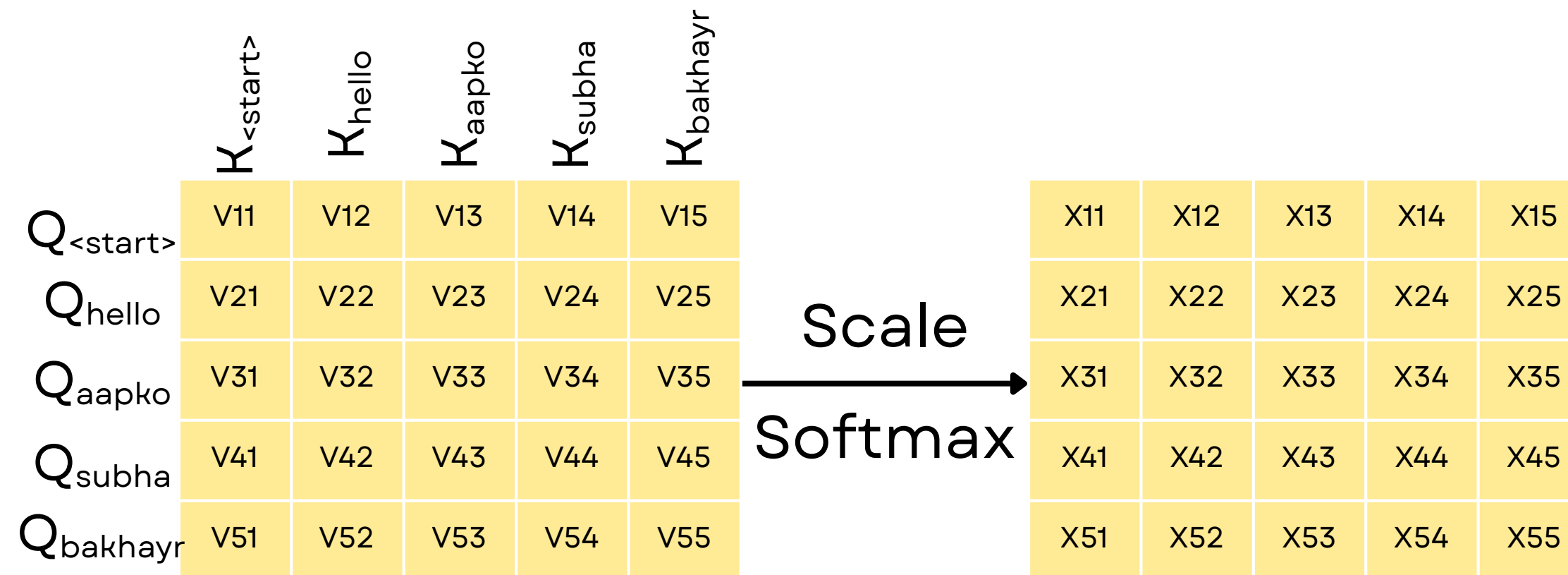
$$E'_{\text{bakhayr}} = 0.9 \times V_{\text{<start>}} + 0.07 \times V_{\text{hello}} + 0.02 \times V_{\text{aapko}} + 0.01 \times V_{\text{subha}} + 0.01 \times V_{\text{bakhayr}}$$

	$K_{\text{<start>}}$	K_{hello}	K_{aapko}	K_{subha}	K_{bakhayr}						
$Q_{\text{<start>}}$	V11	V12	V13	V14	V15	Scale → Softmax	X11	X12	X13	X14	X15
Q_{hello}	V21	V22	V23	V24	V25		X21	X22	X23	X24	X25
Q_{aapko}	V31	V32	V33	V34	V35		X31	X32	X33	X34	X35
Q_{subha}	V41	V42	V43	V44	V45		X41	X42	X43	X44	X45
Q_{bakhayr}	V51	V52	V53	V54	V55		X51	X52	X53	X54	X55

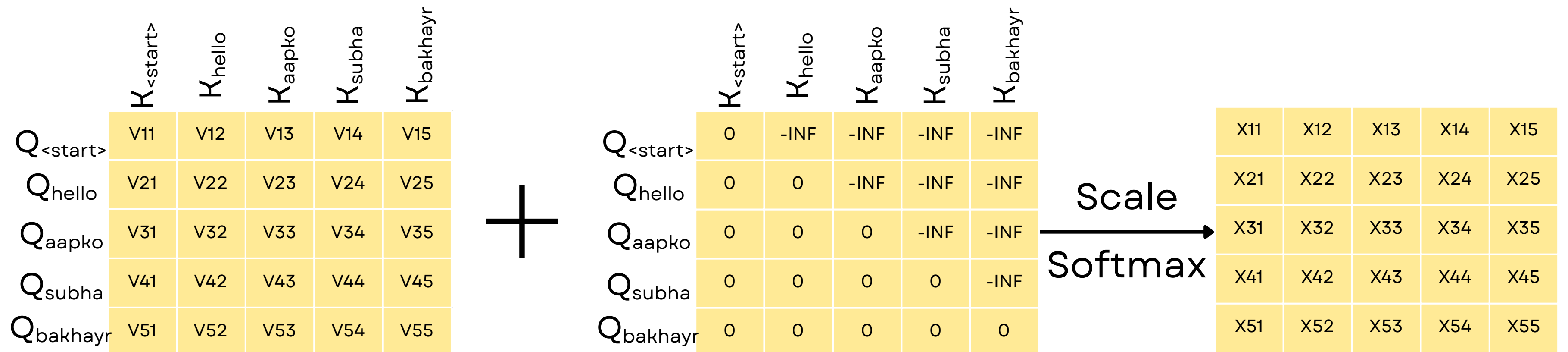
$$\begin{aligned}
 E'_{\text{<start>}} &= 0.9 \times V_{\text{<start>}} + 0.07 \times V_{\text{hello}} + 0.02 \times V_{\text{aapko}} + 0.01 \times V_{\text{subha}} + 0.01 \times V_{\text{bakhayr}} \\
 E'_{\text{hello}} &= 0.9 \times V_{\text{<start>}} + 0.07 \times V_{\text{hello}} + 0.02 \times V_{\text{aapko}} + 0.01 \times V_{\text{subha}} + 0.01 \times V_{\text{bakhayr}} \\
 E'_{\text{aapko}} &= 0.9 \times V_{\text{<start>}} + 0.07 \times V_{\text{hello}} + 0.02 \times V_{\text{aapko}} + 0.01 \times V_{\text{subha}} + 0.01 \times V_{\text{bakhayr}} \\
 E'_{\text{subha}} &= 0.9 \times V_{\text{<start>}} + 0.07 \times V_{\text{hello}} + 0.02 \times V_{\text{aapko}} + 0.01 \times V_{\text{subha}} + 0.01 \times V_{\text{bakhayr}} \\
 E'_{\text{bakhayr}} &= 0.9 \times V_{\text{<start>}} + 0.07 \times V_{\text{hello}} + 0.02 \times V_{\text{aapko}} + 0.01 \times V_{\text{subha}} + 0.01 \times V_{\text{bakhayr}}
 \end{aligned}$$

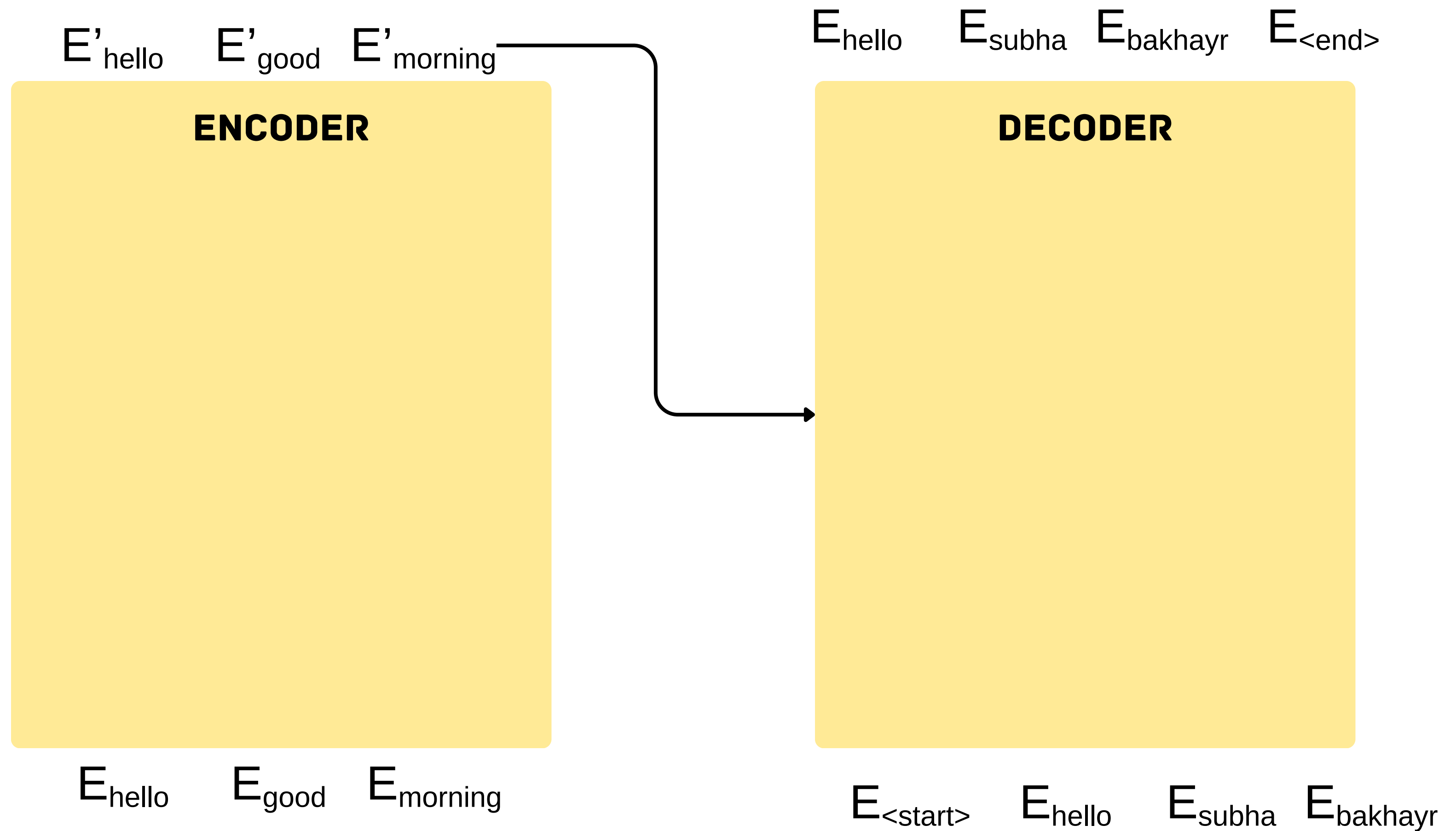
	$K_{\text{<start>}}$	K_{hello}	K_{aapko}	K_{subha}	K_{bakhayr}						
$Q_{\text{<start>}}$	V11	V12	V13	V14	V15	Scale → Softmax	X11	X12	X13	X14	X15
Q_{hello}	V21	V22	V23	V24	V25		X21	X22	X23	X24	X25
Q_{aapko}	V31	V32	V33	V34	V35		X31	X32	X33	X34	X35
Q_{subha}	V41	V42	V43	V44	V45		X41	X42	X43	X44	X45
Q_{bakhayr}	V51	V52	V53	V54	V55		X51	X52	X53	X54	X55

$$\begin{aligned}
 E'_{\text{<start>}} &= 0.9 \times V_{\text{<start>}} + 0.07 \times V_{\text{hello}} + 0.02 \times V_{\text{aapko}} + 0.01 \times V_{\text{subha}} + 0.01 \times V_{\text{bakhayr}} \\
 E'_{\text{hello}} &= 0.9 \times V_{\text{<start>}} + 0.07 \times V_{\text{hello}} + 0.02 \times V_{\text{aapko}} + 0.01 \times V_{\text{subha}} + 0.01 \times V_{\text{bakhayr}} \\
 E'_{\text{aapko}} &= 0.9 \times V_{\text{<start>}} + 0.07 \times V_{\text{hello}} + 0.02 \times V_{\text{aapko}} + 0.01 \times V_{\text{subha}} + 0.01 \times V_{\text{bakhayr}} \\
 E'_{\text{subha}} &= 0.9 \times V_{\text{<start>}} + 0.07 \times V_{\text{hello}} + 0.02 \times V_{\text{aapko}} + 0.01 \times V_{\text{subha}} + 0.01 \times V_{\text{bakhayr}} \\
 E'_{\text{bakhayr}} &= 0.9 \times V_{\text{<start>}} + 0.07 \times V_{\text{hello}} + 0.02 \times V_{\text{aapko}} + 0.01 \times V_{\text{subha}} + 0.01 \times V_{\text{bakhayr}}
 \end{aligned}$$



$$\begin{aligned}
 E'_{\langle \text{start} \rangle} &= 0.9 \times V_{\langle \text{start} \rangle} + 0.07 \times V_{\text{hello}} + 0.02 \times V_{\text{aapko}} + 0.01 \times V_{\text{subha}} + 0.01 \times V_{\text{bakhayr}} \\
 E'_{\text{hello}} &= 0.9 \times V_{\langle \text{start} \rangle} + 0.07 \times V_{\text{hello}} + 0.02 \times V_{\text{aapko}} + 0.01 \times V_{\text{subha}} + 0.01 \times V_{\text{bakhayr}} \\
 E'_{\text{aapko}} &= 0.9 \times V_{\langle \text{start} \rangle} + 0.07 \times V_{\text{hello}} + 0.02 \times V_{\text{aapko}} + 0.01 \times V_{\text{subha}} + 0.01 \times V_{\text{bakhayr}} \\
 E'_{\text{subha}} &= 0.9 \times V_{\langle \text{start} \rangle} + 0.07 \times V_{\text{hello}} + 0.02 \times V_{\text{aapko}} + 0.01 \times V_{\text{subha}} + 0.01 \times V_{\text{bakhayr}} \\
 E'_{\text{bakhayr}} &= 0.9 \times V_{\langle \text{start} \rangle} + 0.07 \times V_{\text{hello}} + 0.02 \times V_{\text{aapko}} + 0.01 \times V_{\text{subha}} + 0.01 \times V_{\text{bakhayr}}
 \end{aligned}$$

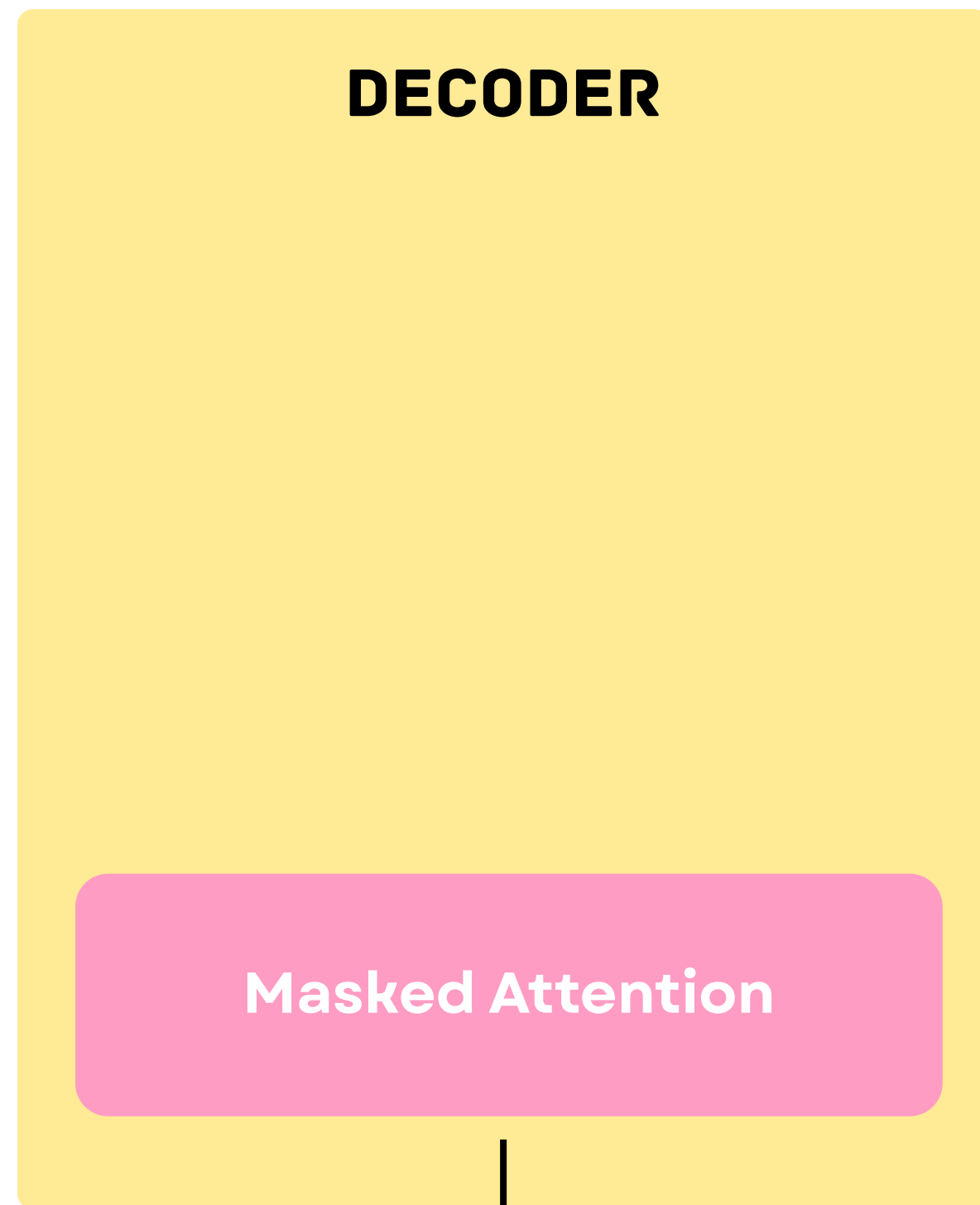


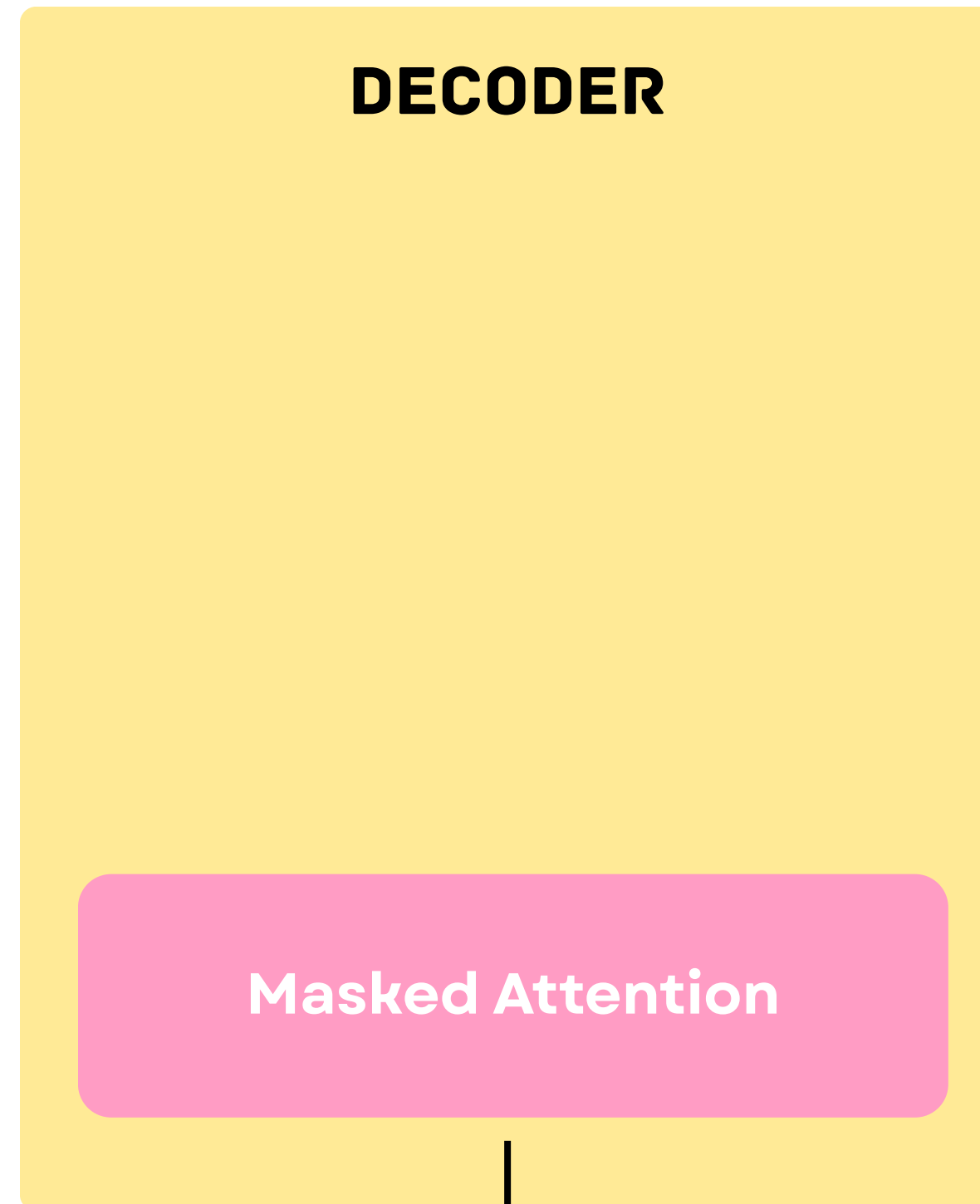


DECODER

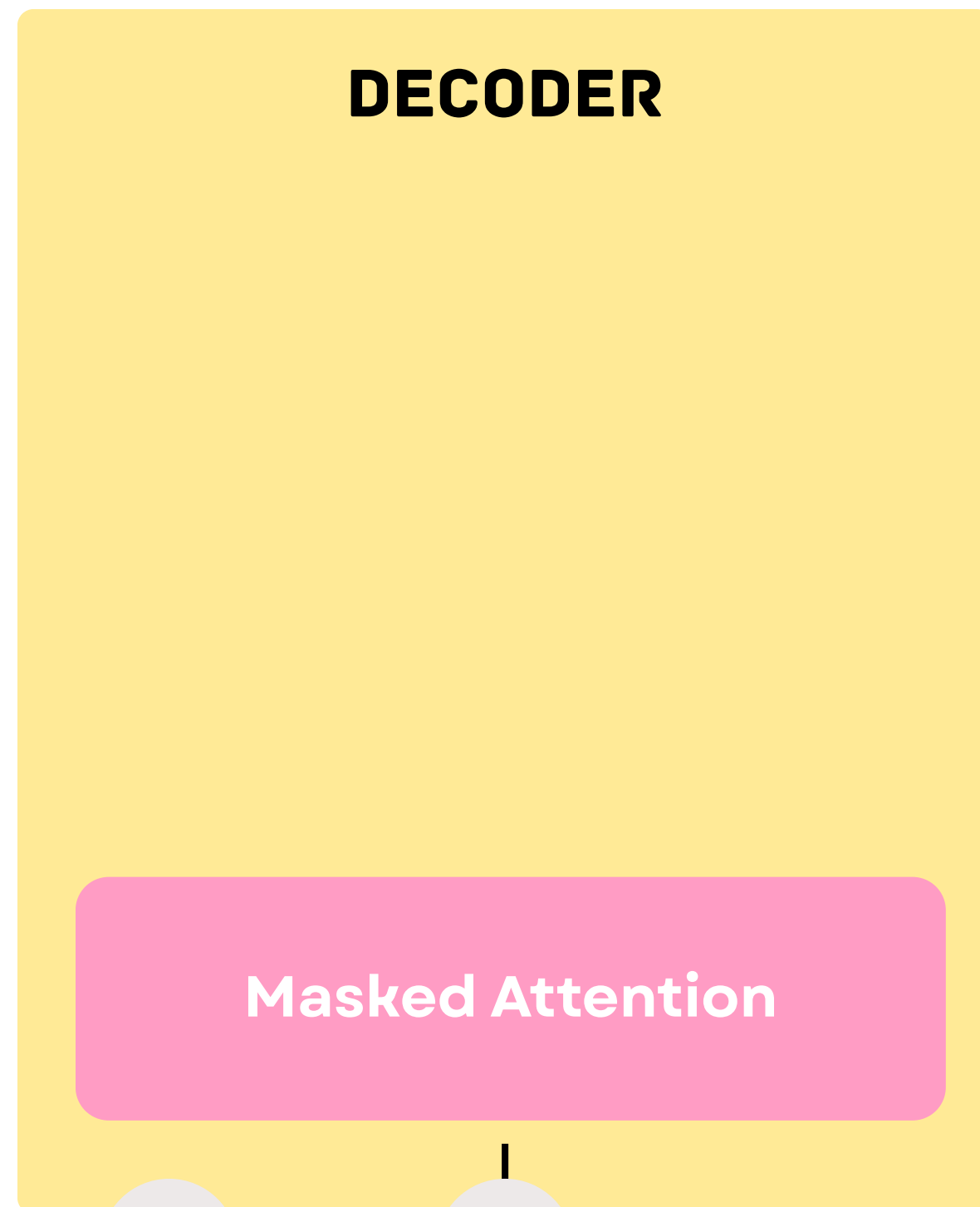
Masked Attention

Input





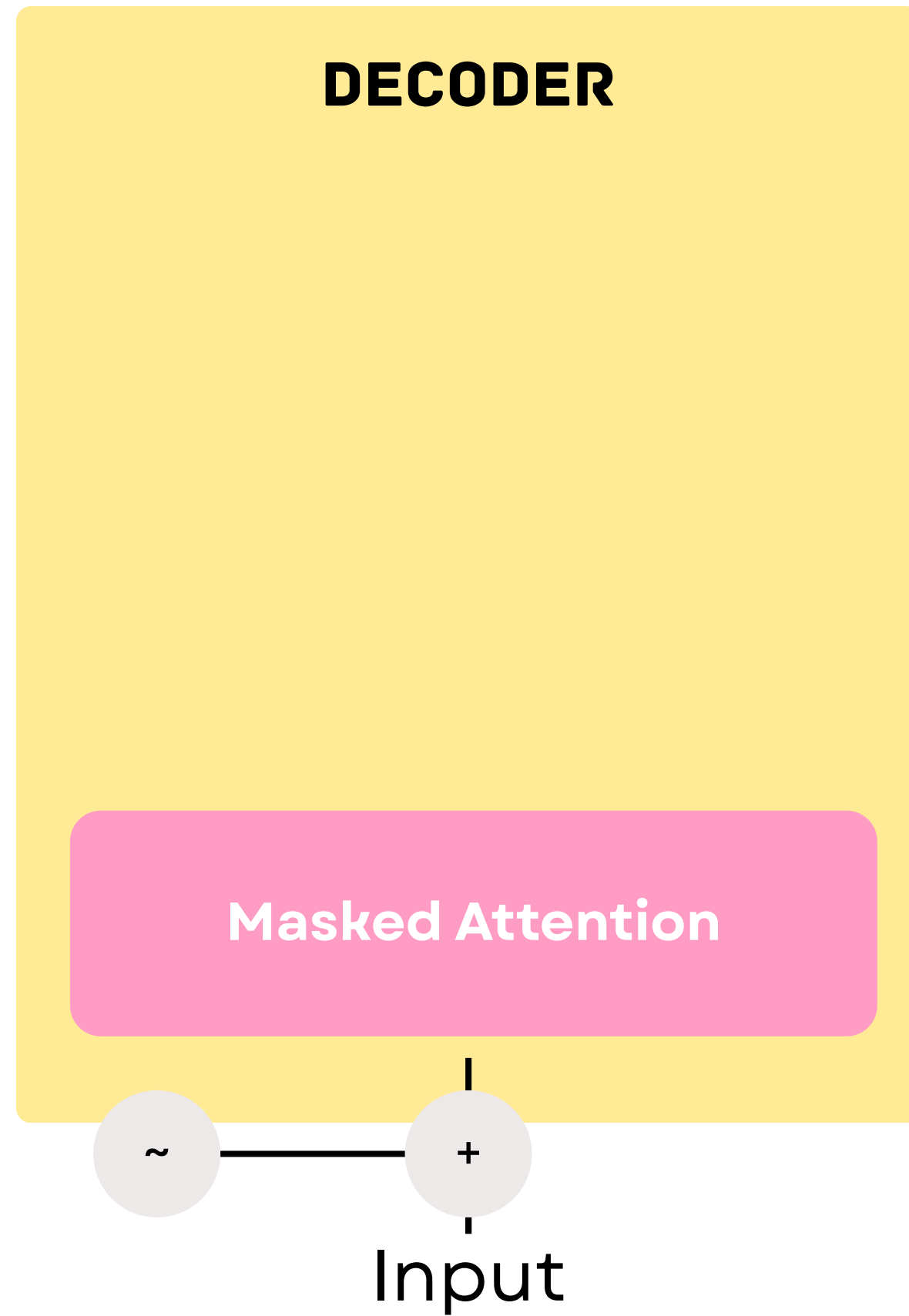
Q: Is there a problem here?



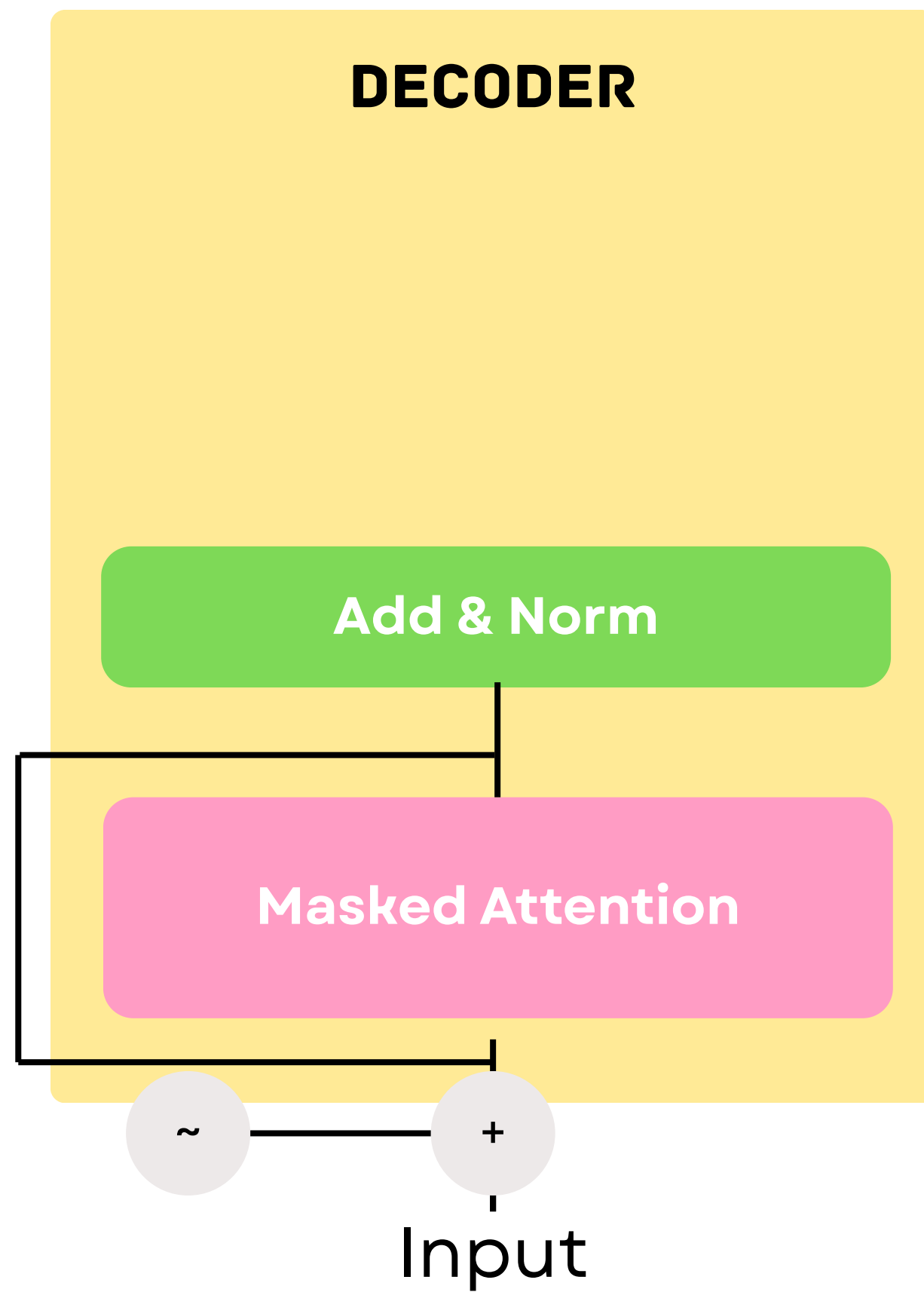
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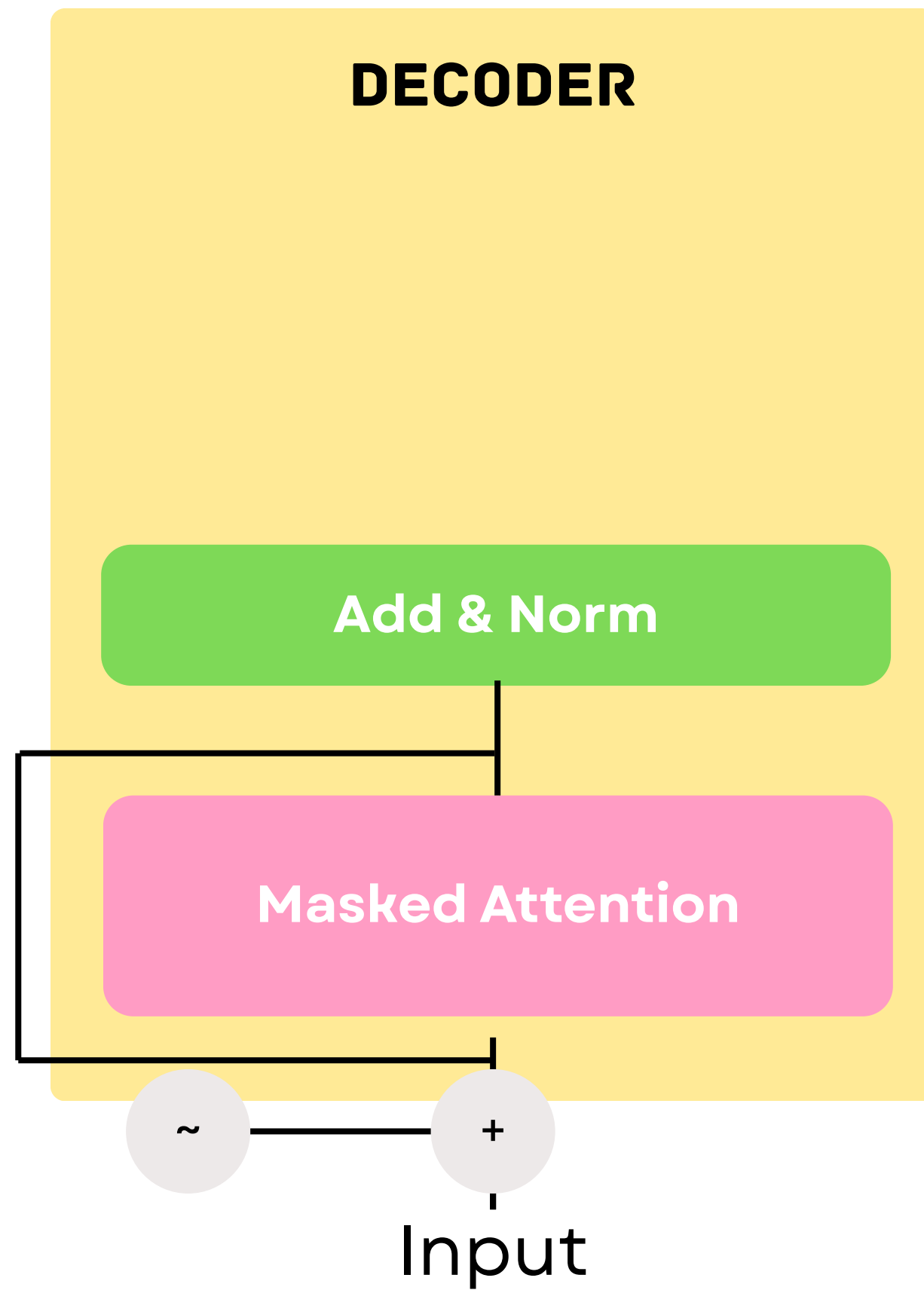
+

Input

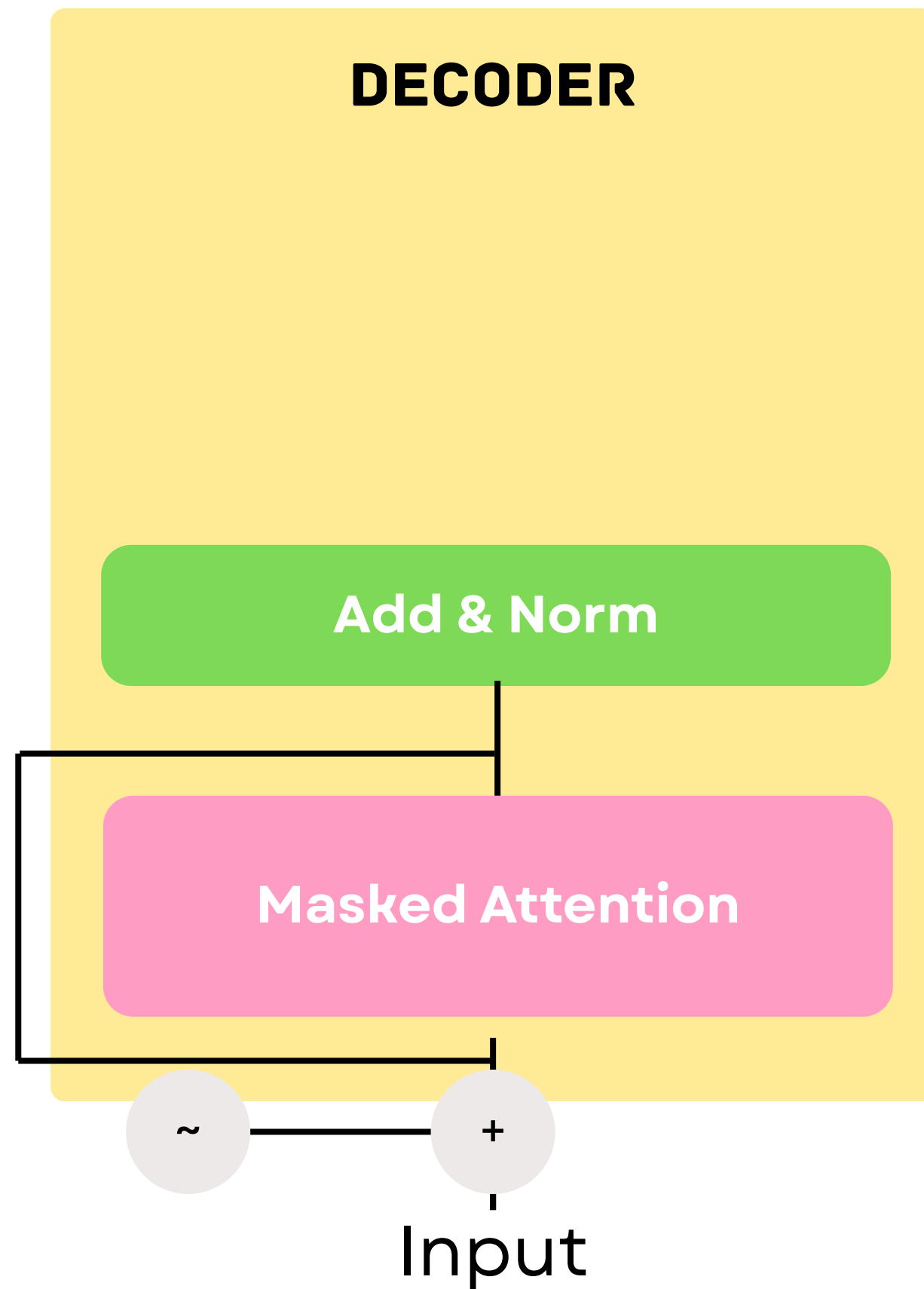


Q: Is there a problem here?

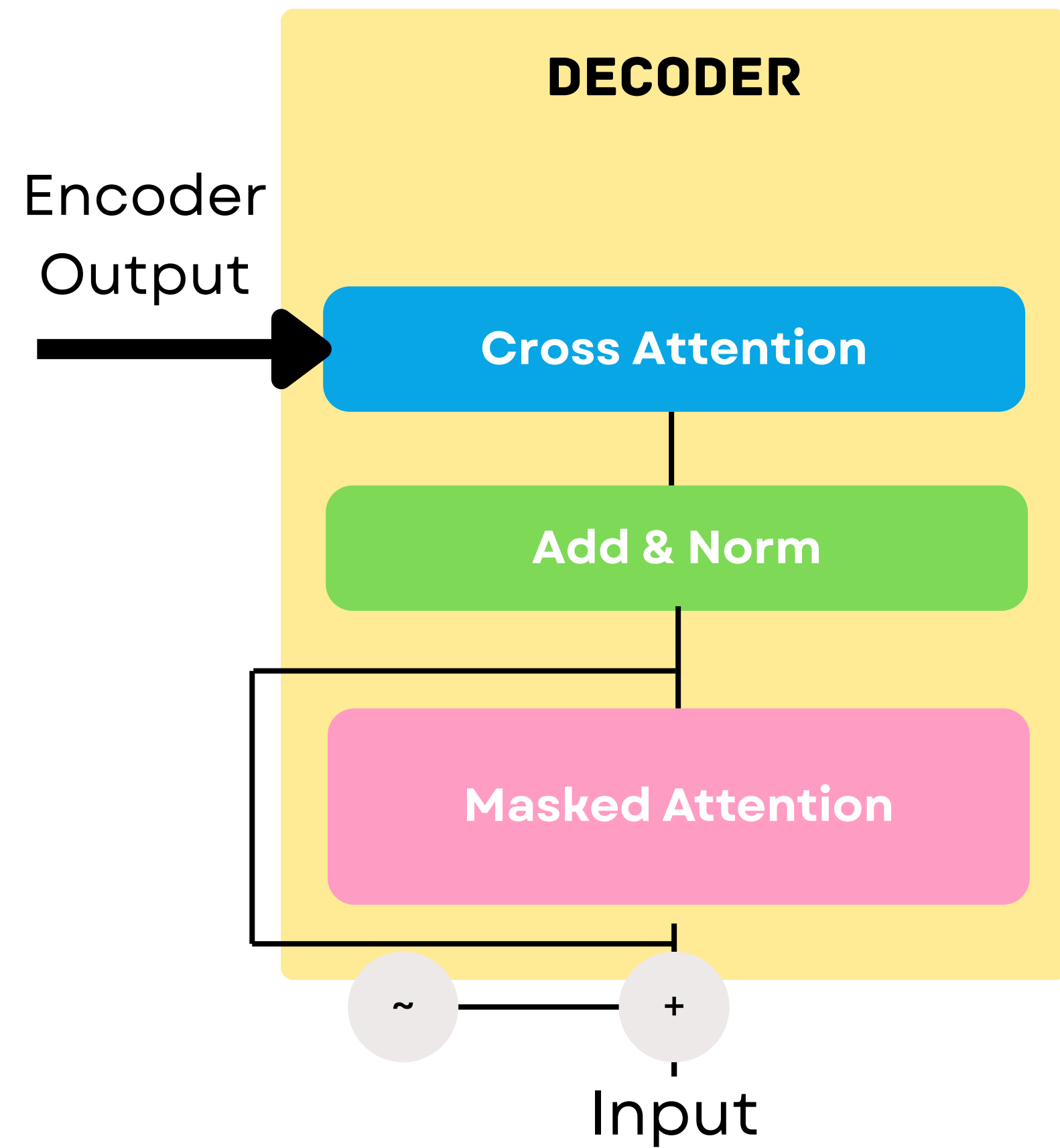


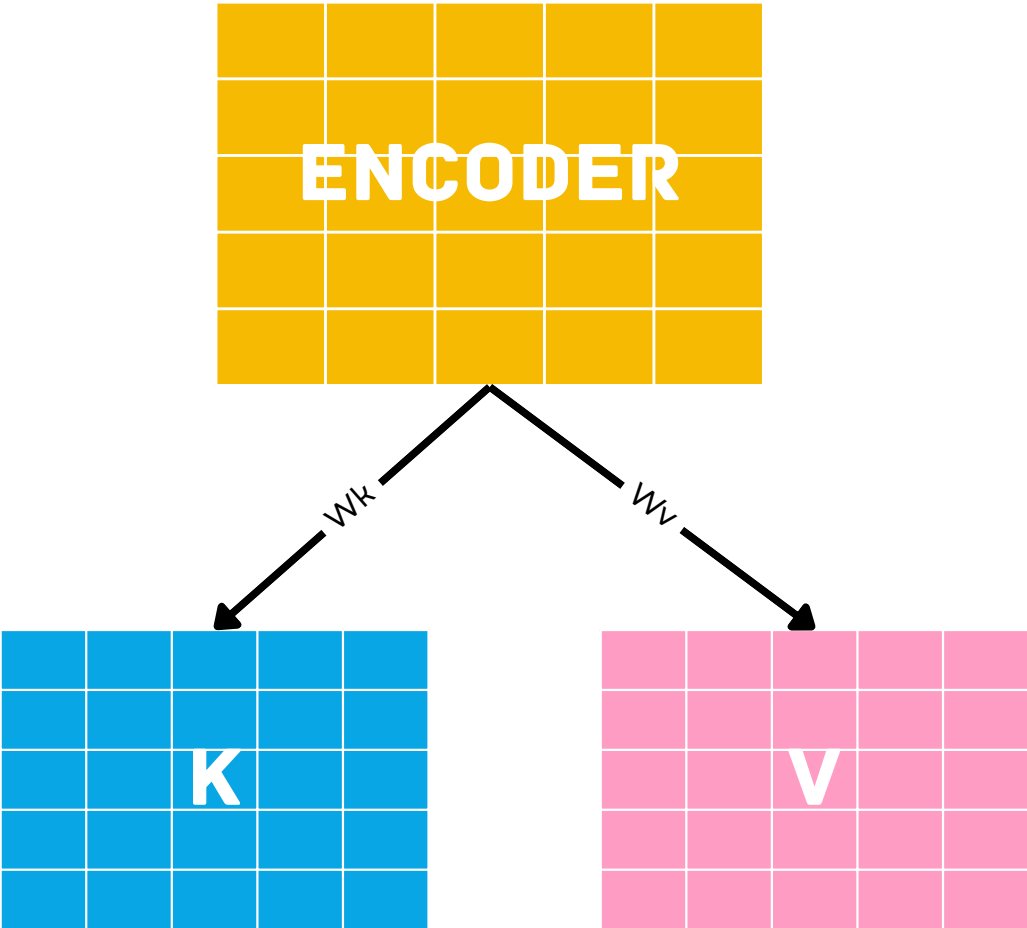
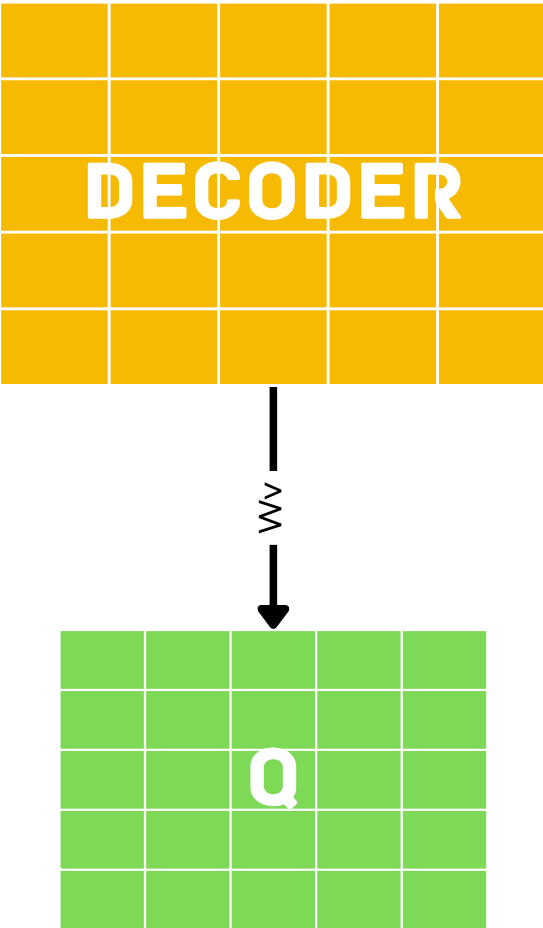


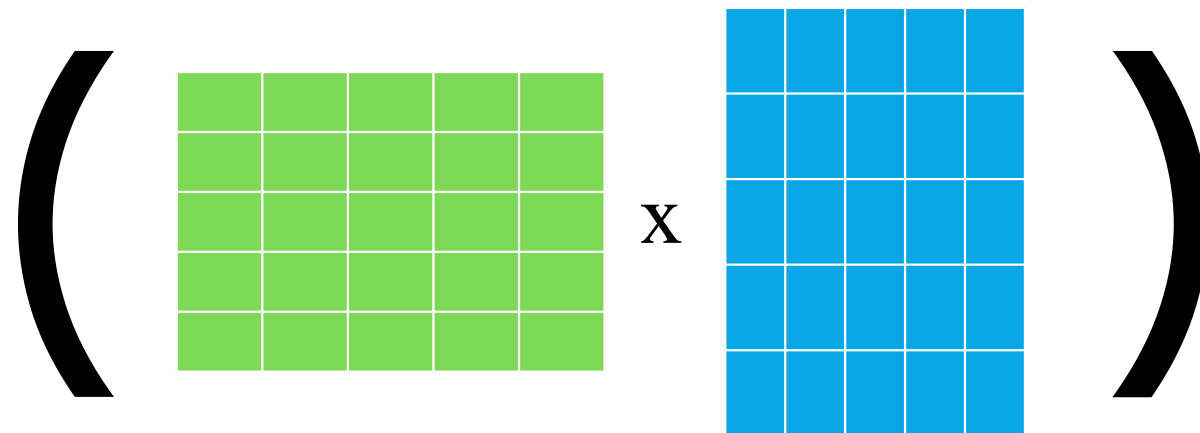
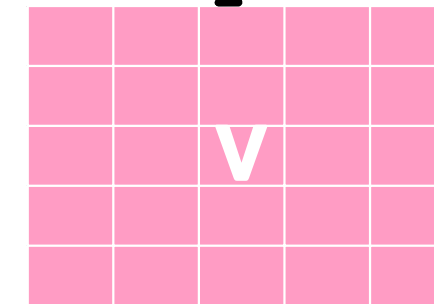
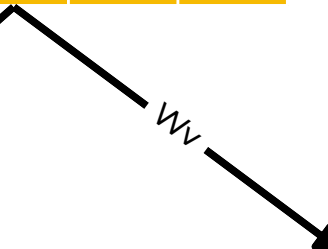
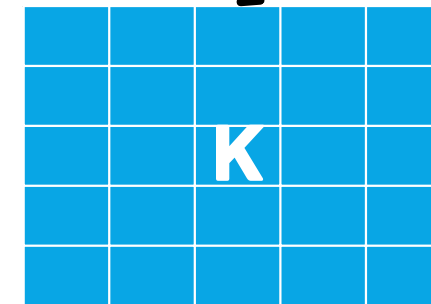
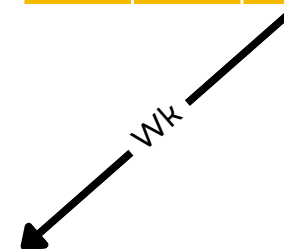
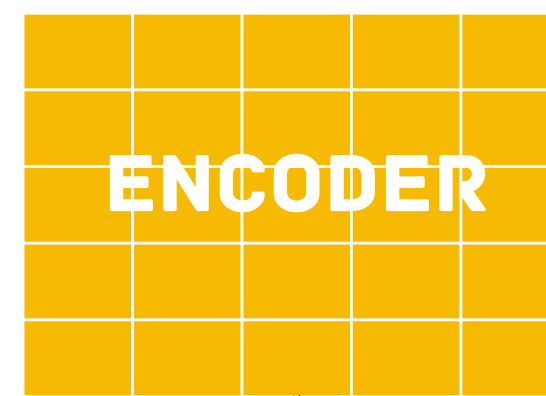
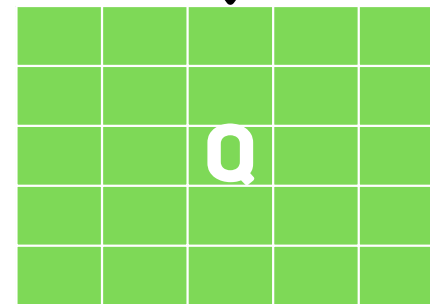
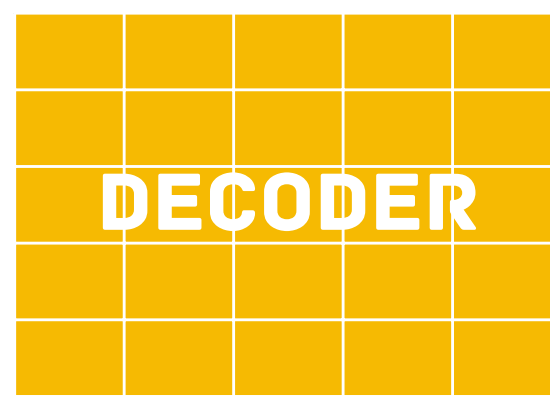
Q: Is there a problem here?



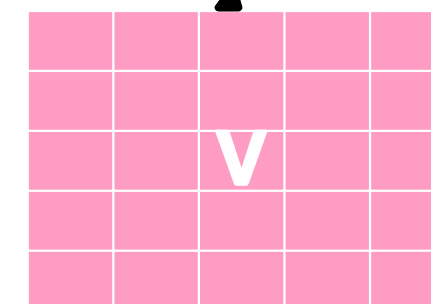
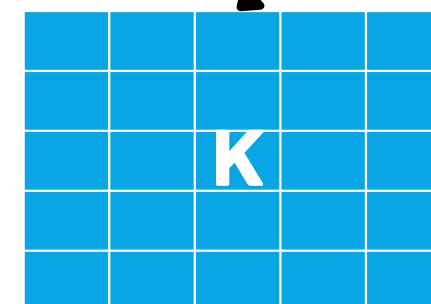
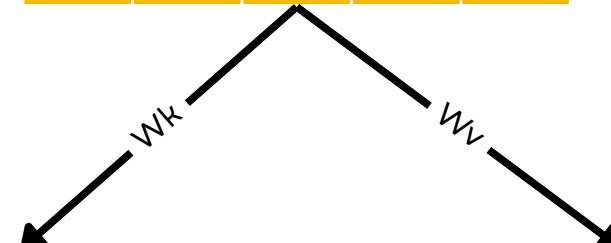
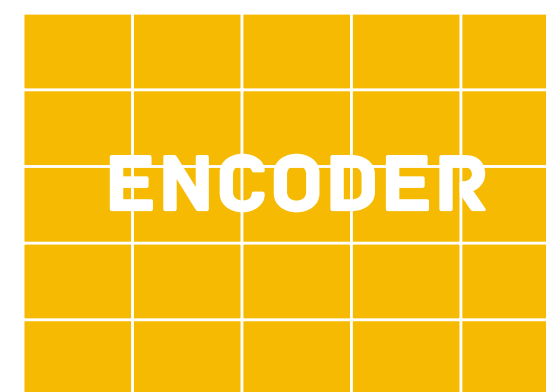
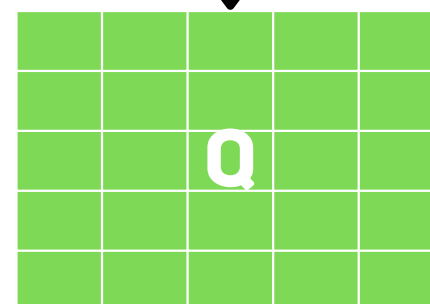
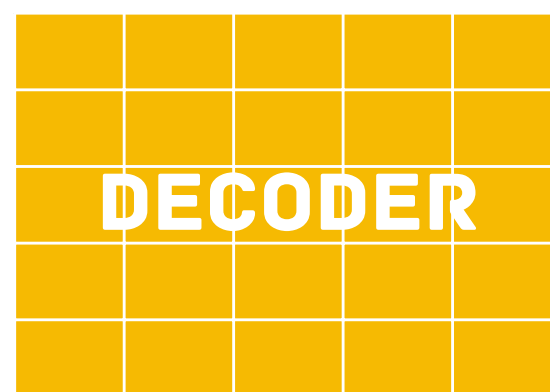
Q: Is there a problem here?
Hint: Brainstorm Full Picture







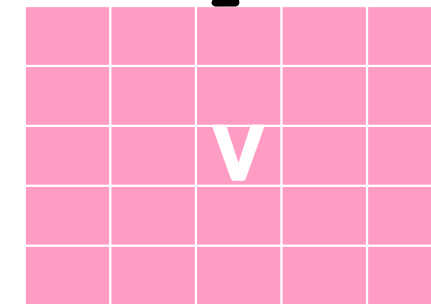
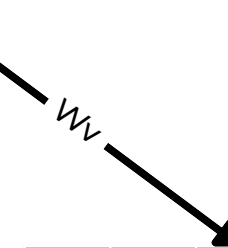
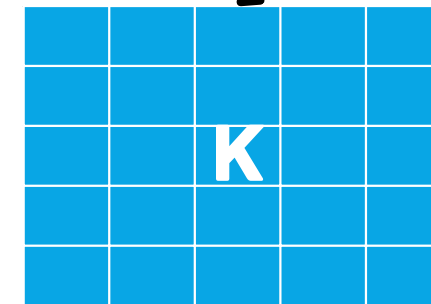
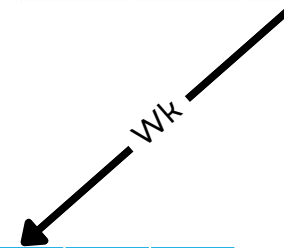
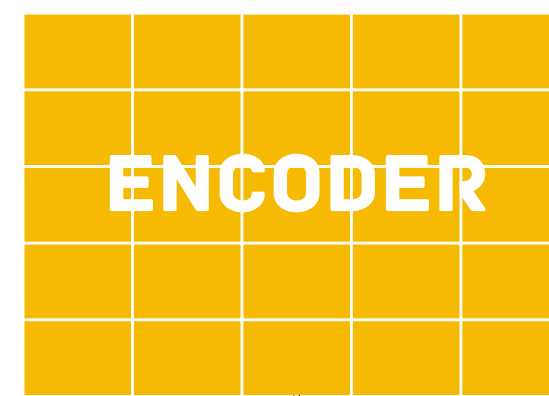
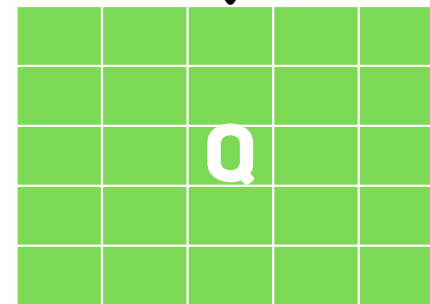
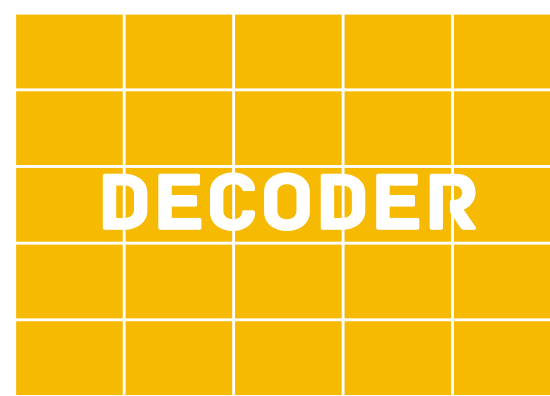
$$Q \times K^T$$



Decoder

$\left(\begin{array}{c} \text{5x5 Green Grid} \\ \text{5x5 Blue Grid} \end{array} \right) \times$

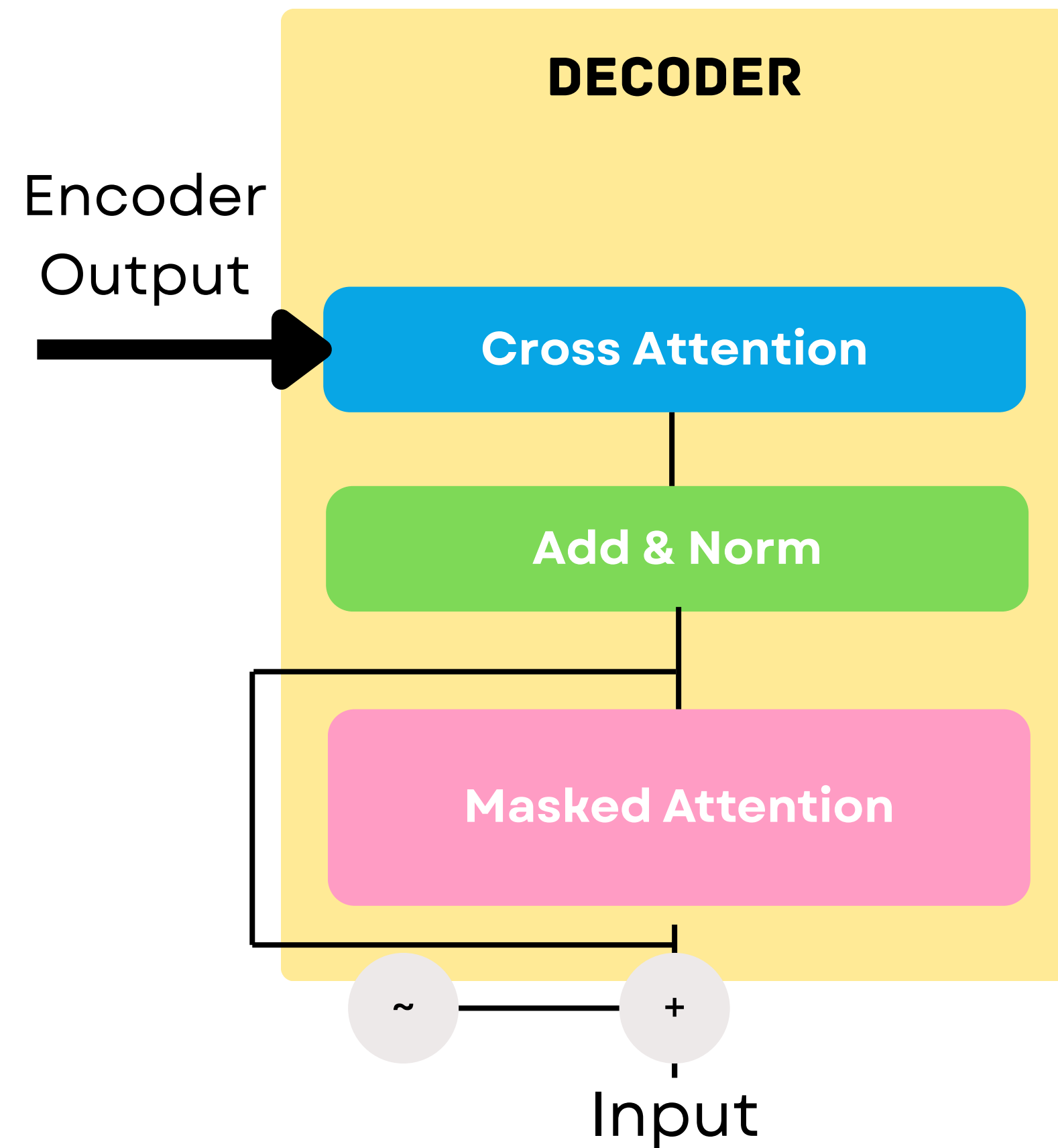
$Q \times K^T$



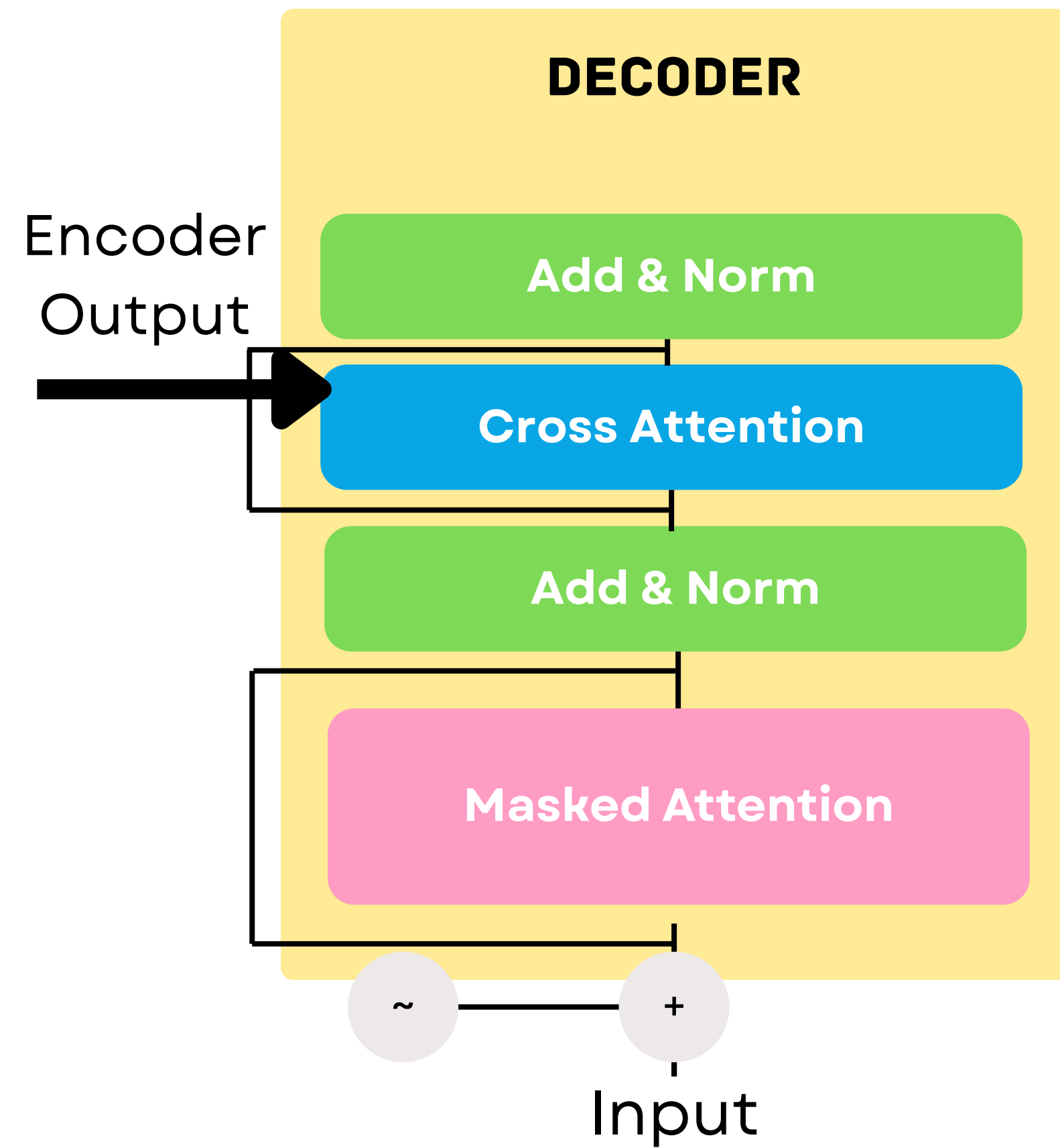
Decoder Encoder

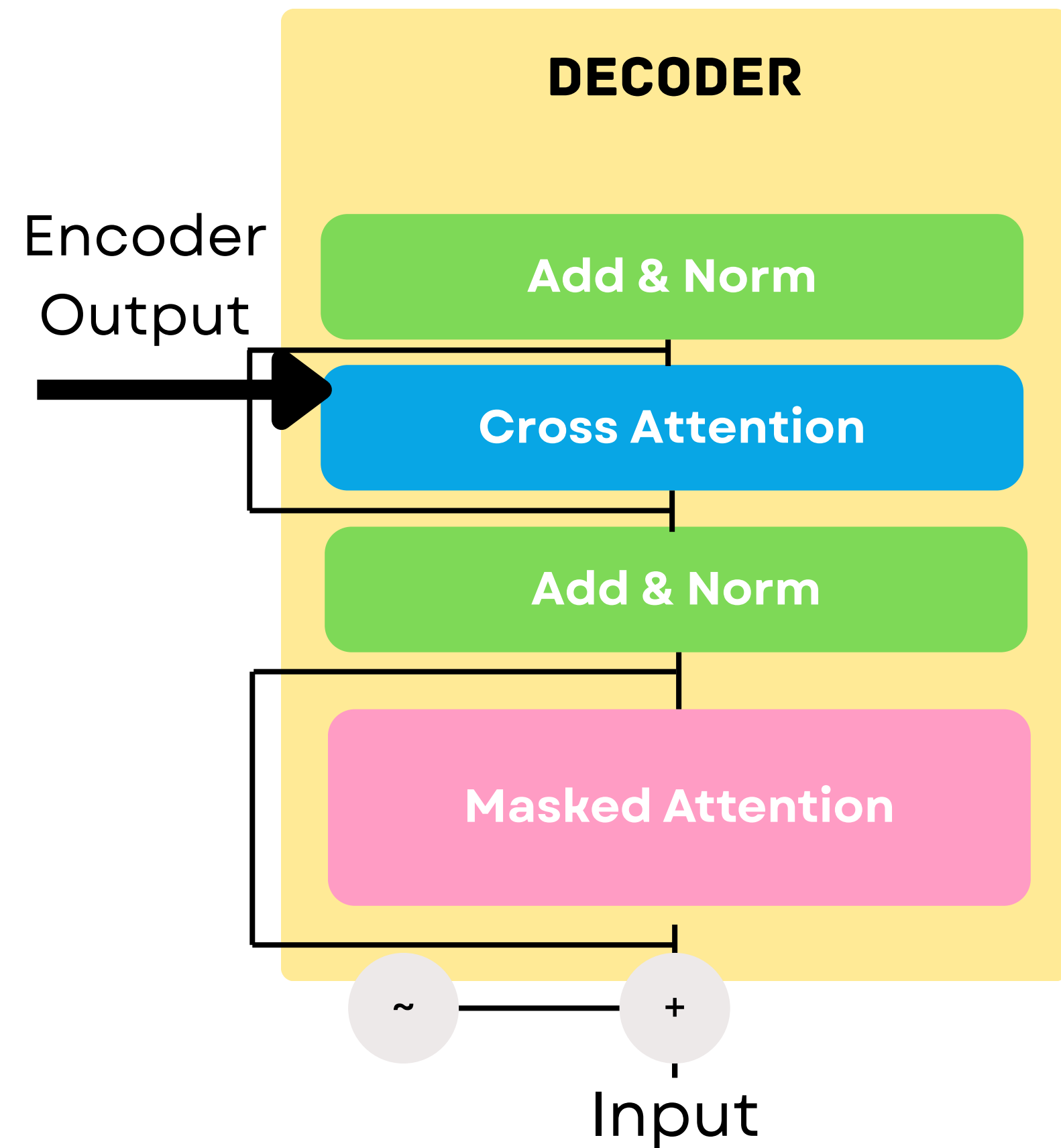
$\left(\begin{array}{c} \text{Decoder} \\ \text{Encoder} \end{array} \right) \times$

$Q \times K^T$



Works with Multi Modal Inputs





Q: Is there a problem here?

